Operator Theory

Lecture notes

(based on the courses of Prof. Simone Warzel)

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1 Banach algebras and their elementary spectral theory

1.1 Banach algebras

Definition 1.1.1 (Banach algebra): A *Banach algebra* is a complex Banach space $\mathcal{A}$ with an associative multiplication

$$\cdot : \mathcal{A} \times \mathcal{A} \to \mathcal{A}, \qquad (x, y) \mapsto x \cdot y,$$

which is submultiplicative with respect to the norm $\|\cdot\|$:

$$\|x \cdot y\| \leq \|x\| \, \|y\| \qquad \text{for all } x, y \in \mathcal{A}.$$

Additional properties $\mathcal{A}$ may have or not:

- *Banach algebra with unit*: there is some $e \in \mathcal{A}$ such that $\|e\| = 1$ and $xe = ex = x$ for all $x \in \mathcal{A}$.
- *Commutative Banach algebra*: $x \cdot y = y \cdot x$ for all $x, y \in \mathcal{A}$.
- C^**-algebra*: Banach algebra with an *involution* $* : \mathcal{A} \to \mathcal{A}$, i.e., for all $x, y \in \mathcal{A}$, $\alpha \in \mathbb{C}$:

  $$(x^*)^* = x, \qquad (x + \alpha y)^* = x^* + \overline{\alpha}\, y^*, \qquad (xy)^* = y^* x^*,$$

  which satisfies the C^**-identity*, i.e., for all $x \in \mathcal{A}$

  $$\|x^* x\| = \|x\|^2 .$$

Example 1.1.2:

(1) For any compact Hausdorff space S,

$$C(S) = \{f : S \to \mathbb{C} \mid f \text{ is continuous}\}$$

with norm $\|f\|_\infty := \sup_{s \in S} |f(s)|$ and involution $f^* := \overline{f}$ is a C^*-algebra. It has unit $f = 1$ and is commutative.

(2) For any Banach space X the bounded linear operators $\mathcal{L}(X)$ form a Banach algebra with unit $\not\Vdash$, where for $a \in \mathcal{L}(X)$

$$\|a\| = \sup_{0 \neq x \in X} \frac{\|ax\|}{\|x\|}.$$

For any Hilbert space $\mathcal{H}$, $\mathcal{L}(\mathcal{H})$ is a C^*-algebra with the adjoint map $* : \mathcal{L}(\mathcal{H}) \to \mathcal{L}(\mathcal{H})$, $a \mapsto a^*$, as involution:

$$\langle av, w \rangle = \langle v, a^* w \rangle , \qquad v, w \in \mathcal{H}.$$

Lemma 1.1.3: For any x in a C^*-algebra: $\|x\| = \|x^*\|$.

Proof. By the C^*-identity and submultiplicativity, $\|x\|^2 = \|x^* x\| \leq \|x^*\| \, \|x\|$, hence $\|x\| \leq \|x^*\|$. Applied to x^* this gives $\|x^*\| \leq \|(x^*)^*\| = \|x\|$. □

Definition 1.1.4 (Invertibility in a Banach algebra with unit e): x is *invertible* if and only if there is $y \in \mathcal{A}$ with $xy = yx = e$.

Remark 1.1.5: The element y is unique and called the *inverse* of x. Notation: $y = x^{-1}$.

Lemma 1.1.6 (Useful algebraic properties):

(1) If x, y are invertible, then xy is invertible with $(xy)^{-1} = y^{-1} x^{-1}$.

(2) If $x_1, x_2, \ldots, x_n$ are mutually commuting, i.e., $x_j x_k = x_k x_j$, then

$$y = x_1 x_2 \cdots x_n \text{ is invertible } \iff \text{ each } x_1, \ldots, x_n \text{ is invertible.}$$

Proof. Exercise. □

1.2 Resolvent and spectrum

Theorem 1.2.1: von Neumann series

Let $\mathcal{A}$ be a Banach algebra with unit, $x \in \mathcal{A}$ invertible, and $y \in \mathcal{A}$ with $\|y\| < 1/\left\|x^{-1}\right\|$. Then $x - y$ is invertible and the series

$$(x-y)^{-1} = \sum_{n=0}^{\infty} (x^{-1}y)^n x^{-1}$$

converges in norm.

Proof. *Case $x = e$, $\|y\| < 1$:*

- $\|y\| < 1$ implies that the partial series is Cauchy. By completeness the geometric series $s := \sum_{n=0}^{\infty} y^n$ converges in norm (with $y^0 := e$).
- $ys = \sum_{n=1}^{\infty} y^n = s - e$ and likewise $sy = s - e$. Hence $s = (e-y)^{-1}$.

Case $x \in \mathcal{A}$, $\|y\| < 1/\left\|x^{-1}\right\|$: We have $x - y = x(e - x^{-1}y)$ with $\left\|x^{-1}y\right\| \leq \left\|x^{-1}\right\| \|y\| < 1$. Therefore

$$(x-y)^{-1} = (e - x^{-1}y)^{-1}x^{-1} = \sum_{n=0}^{\infty} (x^{-1}y)^n x^{-1}. \qquad \square$$

Definition 1.2.1: Let x be an element of a Banach algebra $\mathcal{A}$ with unit. The *resolvent set* of x is

$$\varrho(x) = \{\lambda \in \mathbb{C} \mid \lambda e - x \text{ invertible}\}.$$

The *spectrum* of x is $\sigma(x) = \mathbb{C} \setminus \varrho(x)$. The *resolvent (operator)* of x for $\lambda \in \varrho(x)$ is $(\lambda e - x)^{-1} \equiv (\lambda - x)^{-1}$.

Lemma 1.2.2 (First resolvent formula): For $\lambda, \mu \in \varrho(x)$:

$$(\lambda - x)^{-1} - (\mu - x)^{-1} = (\mu - \lambda)(\lambda - x)^{-1}(\mu - x)^{-1} = (\mu - \lambda)(\mu - x)^{-1}(\lambda - x)^{-1}.$$

Proof. *Exercise.* $\square$

Theorem 1.2.2

(1) $\varrho(x)$ is open and $\varrho(x) \ni \lambda \mapsto (\lambda - x)^{-1} \in \mathcal{A}$ is holomorphic with

$$\left\|(\lambda - x)^{-1}\right\| \leq \frac{1}{|\lambda| - \|x\|} \qquad \text{for } |\lambda| > \|x\|\,.$$

(2) $\sigma(x)$ is closed and bounded.

Proof. From von Neumann's theorem:

(a) If $\lambda \in \varrho(x)$ and $|z| < 1/\left\|(\lambda - x)^{-1}\right\|$, then the power series

$$(\lambda + z - x)^{-1} = \sum_{n=0}^{\infty} (-z)^n (\lambda - x)^{-n-1}$$

is norm convergent. Hence $\varrho(x)$ is open, $\sigma(x) = \mathbb{C} \setminus \varrho(x)$ is closed and the resolvent operator is holomorphic on $\varrho(x)$.

(b) If $|\lambda| > \|x\|$, then

$$\sum_{n=0}^{\infty} \frac{x^n}{\lambda^{n+1}} = \frac{1}{\lambda}\Big(e - \frac{x}{\lambda}\Big)^{-1} = (\lambda - x)^{-1},$$

i.e., $\lambda \in \varrho(x)$ and

$$\big\|(\lambda - x)^{-1}\big\| \le \sum_{n=0}^{\infty} \frac{\|x\|^n}{|\lambda|^{n+1}} = \frac{1}{|\lambda| - \|x\|}.$$

In particular $\sigma(x)$ is bounded. □

Definition 1.2.3: The *spectral radius* of $x \in \mathcal{A}$ is

$$\|\sigma(x)\| := \max_{\lambda \in \sigma(x)} |\lambda|\,.$$

1.3 A first glance at functional calculus

Set-up: Banach algebra $\mathcal{A}$ with unit e. We discuss

(1) polynomial functional calculus,

(2) holomorphic functional calculus (Dunford–Riesz).

Later, in the Hilbert space set-up: measurable functional calculus.

1. Polynomial functional calculus. Let $p : \mathbb{C} \to \mathbb{C}$, $p(z) = \sum_{j=0}^{n} a_j z^j$ be a complex polynomial. For $x \in \mathcal{A}$ set

$$p(x) := \sum_{j=0}^{n} a_j x^j.$$

Remark 1.3.1: This extends to any entire function $f : \mathbb{C} \to \mathbb{C}$: $f(x) := \sum_{j=0}^{\infty} a_j x^j$. Note: for polynomials p, q and $\alpha \in \mathbb{C}$,

$$(p + \alpha q)(x) = p(x) + \alpha q(x), \qquad (p \cdot q)(x) = p(x) \cdot q(x), \qquad p(z) = 1 \;\Rightarrow\; p(x) = e.$$

Theorem 1.3.1: Spectral mapping

For any polynomial p and $x \in \mathcal{A}$:

$$\sigma(p(x)) = p(\sigma(x)).$$

Proof. Factor $p(z) - \lambda = \prod_{j=1}^{n}(z - \mu_j)$ with $\mu_j \in \mathbb{C}$. Since $p(x) - \lambda e = \prod_{j=1}^{n}(x - \mu_j e)$ and $x - \mu_1 e, \dots, x - \mu_n e$ are mutually commuting, the algebraic property (2) implies

$$\lambda \in \sigma(p(x)) \iff \exists j \in \{1, \dots, n\} : \; \mu_j \in \sigma(x) \iff \lambda \in p(\sigma(x)),$$

where we used $p(\mu_j) = \lambda$. □

2. Holomorphic functions $f : G \to \mathbb{C}$ on open $G \subset \mathbb{C}$.

- Algebra $\mathcal{H}(G) = \{f : G \to \mathbb{C} \mid f \text{ holomorphic}\}$ with unit.
- Recall that for any $f \in \mathcal{H}(G)$ and any $\lambda \in G$:

$$f(\lambda) = \oint_{\gamma} (z - \lambda)^{-1} f(z)\, \frac{\mathrm{d}z}{2\pi \mathrm{i}}$$

with a rectifiable Jordan curve γ winding about λ in G anticlockwise once.

Preliminaries for the extension of this formula to $x \in \mathcal{A}$:

(a) Norm convergence of Riemann sums ensures that for any holomorphic $g : G \to \mathcal{A}$ with $G \subset \mathbb{C}$ open and any rectifiable Jordan curve γ in G, $\oint_\gamma g(z)\,\mathrm{d}z$ is well-defined in $\mathcal{A}$.

(b) *Cauchy's theorem in $\mathcal{A}$:* If $g : G \to \mathcal{A}$ is holomorphic and γ is a rectifiable Jordan curve with winding number $n(\gamma, a) = 0$ for all $a \in \mathbb{C} \setminus G$, then $\oint_\gamma g(z)\,\mathrm{d}z = 0$.

Proof idea. Let $\tau \in \mathcal{A}'$. Then $\tau \circ g : G \to \mathbb{C}$ is holomorphic and

$$\tau\Big(\oint_\gamma g(z)\,\mathrm{d}z\Big) = \oint_\gamma \tau(g(z))\,\mathrm{d}z = 0,$$

where the first equality uses that τ is bounded linear and the second is Cauchy's theorem for $\tau \circ g$. By Hahn–Banach the claim follows. □

Example 1.3.2: For any $n \in \mathbb{N}_0$ and $r > \|x\|$:

$$\oint_{|z|=r} (z-x)^{-1} z^n \frac{\mathrm{d}z}{2\pi\mathrm{i}} = \oint_{|z|=r} (e - z^{-1}x)^{-1} z^{n-1} \frac{\mathrm{d}z}{2\pi\mathrm{i}} = \oint_{|z|=r} \sum_{k=0}^{\infty} (z^{-1}x)^k z^{n-1} \frac{\mathrm{d}z}{2\pi\mathrm{i}}$$

$$= \sum_{k=0}^{\infty} x^k \underbrace{\oint_{|z|=r} z^{n-k-1} \frac{\mathrm{d}z}{2\pi\mathrm{i}}}_{=\delta_{n,k}} = x^n,$$

where we used von Neumann's theorem ($\|x\| < |z| = r$) and $\delta_{n,k} = 1$ if $n = k$ and 0 else.

Definition 1.3.3: For $x \in \mathcal{A}$, $s > \|\sigma(x)\|$ and $f \in \mathcal{H}(B(0,s))$ (with $B(0,s)$ the open ball about 0 with radius s), set

$$f(x) := \oint_{|z|=r} (z-x)^{-1} f(z) \frac{\mathrm{d}z}{2\pi\mathrm{i}} \in \mathcal{A}$$

with $\|\sigma(x)\| < r < s$ arbitrary, where the contour is $\gamma : [0, 2\pi] \to \mathbb{C}$, $\gamma(t) = r\mathrm{e}^{\mathrm{i}t}$.

Remark 1.3.4: This is well-defined and independent of the specific $r \in (\|\sigma(x)\|, s)$ by Cauchy's theorem.

Theorem 1.3.2

The map $\Phi : \mathcal{H}(B(0,s)) \to \mathcal{A}$, $\Phi(f) = f(x)$, is an algebra homomorphism, i.e., for $f, g \in \mathcal{H}(B(0,s))$, $\alpha \in \mathbb{C}$:

$$\begin{aligned} \Phi(f + \alpha g) &= \Phi(f) + \alpha\Phi(g) && \text{(linearity)} \\ \Phi(f \cdot g) &= \Phi(f)\,\Phi(g) && \text{(multiplicativity)} \\ \Phi(1) &= e && \text{(preserves unit).} \end{aligned}$$

Proof idea. Linearity follows immediately from that of the integral, and $\Phi(1) = \Phi(z^0) = x^0 = e$ by the example above. Let $f, g \in \mathcal{H}(B(0,s))$ and $\|\sigma(x)\| < r' < r < s$. Then, using the resolvent equation $(z-x)^{-1}(w-x)^{-1} = \big[(w-x)^{-1} - (z-x)^{-1}\big]/(z-w)$,

$$\begin{aligned} \Phi(f)\Phi(g) &= \oint_{|z|=r} \oint_{|w|=r'} (z-x)^{-1}(w-x)^{-1} f(z) g(w) \frac{\mathrm{d}w}{2\pi\mathrm{i}} \frac{\mathrm{d}z}{2\pi\mathrm{i}} \\ &= \oint_{|w|=r'} g(w)(w-x)^{-1} \underbrace{\oint_{|z|=r} (z-w)^{-1} f(z) \frac{\mathrm{d}z}{2\pi\mathrm{i}}}_{=f(w)} \frac{\mathrm{d}w}{2\pi\mathrm{i}} \end{aligned}$$

$$= -\oint_{|z|=r} f(z)(z-x)^{-1} \underbrace{\oint_{|w|=r'} (z-w)^{-1} g(w) \frac{\mathrm{d}w}{2\pi\mathrm{i}}}_{=0} \frac{\mathrm{d}z}{2\pi\mathrm{i}}$$

$$= \Phi(fg). \qquad \square$$

Theorem 1.3.3

(1) $\sigma(x) \neq \emptyset$.

(2) *Spectral radius formula:*

$$\|\sigma(x)\| = \lim_{n\to\infty} \|x^n\|^{1/n} = \inf_{n\in\mathbb{N}} \|x^n\|^{1/n}.$$

Proof. (1) If $\sigma(x) = \emptyset$, then $\mathbb{C} \ni z \mapsto (z-x)^{-1}$ is holomorphic and hence by Cauchy's theorem $\oint_{|z|=r}(z-x)^{-1}\frac{\mathrm{d}z}{2\pi\mathrm{i}} = 0 \neq e$, a contradiction.

(2) Existence of the limit and the second equality is trivial if $\|x^n\| = 0$ for some $n \in \mathbb{N}$. Else it follows from Fekete's lemma (below) with $a_n = \ln\|x^n\|$, using $\|x^{n+m}\| \leq \|x^n\|\,\|x^m\|$ for $m, n \in \mathbb{N}$.

Proof of the first equality:

- If $|\lambda| > \lim_{n\to\infty}\|x^n\|^{1/n}$, then $\sum_{n=0}^{\infty}\frac{x^n}{\lambda^{n+1}}$ is norm convergent and by von Neumann's theorem $\lambda \in \varrho(x)$. Hence $\|\sigma(x)\| \leq \lim_{n\to\infty}\|x^n\|^{1/n}$.
- Let $|\lambda| > \|\sigma(x)\|$. Then $x^n = \oint_{|z|=|\lambda|}(z-x)^{-1}z^n\frac{\mathrm{d}z}{2\pi\mathrm{i}}$, so

$$\|x^n\| \leq \int_0^{2\pi} \left\|(|\lambda|\,\mathrm{e}^{\mathrm{i}t} - x)^{-1}\right\| |\lambda|^{n+1}\frac{\mathrm{d}t}{2\pi} \leq C\,|\lambda|^{n+1}, \qquad C := \max_{t\in[0,2\pi]}\left\|(|\lambda|\,\mathrm{e}^{\mathrm{i}t} - x)^{-1}\right\|.$$

Therefore $\lim_{n\to\infty}\|x^n\|^{1/n} \leq |\lambda|$, i.e., $\lim_{n\to\infty}\|x^n\|^{1/n} \leq \|\sigma(x)\|$. $\square$

Lemma 1.3.5 (Fekete): Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence which is subadditive, i.e., $a_{n+m} \leq a_n + a_m$. Then

$$\lim_{n\to\infty}\frac{a_n}{n} = \inf_{n\in\mathbb{N}}\frac{a_n}{n} \quad \text{exists in } \mathbb{R}\cup\{-\infty\}.$$

Proof. Let $L := \inf_{n\in\mathbb{N}}\frac{a_n}{n}$. Pick $A > L$ arbitrary and $N \in \mathbb{N}$ such that $\frac{a_N}{N} \leq A$. Write $n = p_n N + r_n$ with $r_n \in \{0, 1, \ldots, N-1\}$, $p_n \in \mathbb{N}_0$. By subadditivity,

$$L \leq \frac{a_n}{n} = \frac{a_{p_nN+r_n}}{n} \leq \frac{p_n a_N + a_{r_n}}{n} = \underbrace{\frac{p_nN}{n}}_{\to 1}\frac{a_N}{N} + \underbrace{\frac{a_{r_n}}{n}}_{\to 0} \xrightarrow{n\to\infty} \frac{a_N}{N} \leq A.$$

Hence $\limsup_n a_n/n \leq A$ for every $A > L$, which implies the claim. $\square$

Example 1.3.6 (Application: Volterra integral operators): Let $k : [0,1]^2 \to \mathbb{C}$ be continuous and

$$V_k : C([0,1]) \to C([0,1]), \qquad (V_k f)(t) = \int_0^t k(t,s)f(s)\,\mathrm{d}s, \quad t \in [0,1].$$

Since $|V_k f(t)| \leq \int_0^t |k(t,s)|\,|f(s)|\,\mathrm{d}s$, we have $\|V_k f\|_\infty \leq \|k\|_\infty\|f\|_\infty$ with $\|k\|_\infty = \sup_{t,s\in[0,1]}|k(t,s)|$, i.e., $V_k \in \mathcal{L}(C([0,1]))$. Moreover,

$$(V_k^n f)(t) = \int_{0\leq s_1\leq s_2\leq\cdots\leq s_n\leq t} k(t,s_n)k(s_n,s_{n-1})\cdots k(s_2,s_1)f(s_1)\,\mathrm{d}s_1\,\mathrm{d}s_2\cdots\mathrm{d}s_n,$$

hence

$$\|V_k^n f\|_\infty \le \|k\|_\infty^n \, \|f\|_\infty \sup_{t\in[0,1]} \underbrace{\lambda^n\{(s_1,\dots,s_n) \mid 0 \le s_1 \le \cdots \le s_n \le t\}}_{=t^n/n! \le 1/n!}.$$

Therefore, by the spectral radius formula and Stirling's formula,

$$\|\sigma(V_k)\| = \lim_{n\to\infty} \|V_k^n\|^{1/n} \le \limsup_{n\to\infty} \frac{\|k\|_\infty}{(n!)^{1/n}} = 0, \qquad \text{i.e., } \sigma(V_k) = \{0\}.$$

Example 1.3.7 (ODE problem): Let $g \in C([0,1])$ and consider

$$-u'' + gu = 0, \qquad u(0) = 0, \quad u'(0) = 1. \tag{$*$}$$

- Any solution $u \in C([0,1])$ of

$$u(t) = u_0(t) + \int_0^t (t-s)g(s)u(s)\,\mathrm{d}s \tag{$**$}$$

  with $u_0(t) := t$ also solves $(*)$ and vice versa: indeed $u'(t) = 1 + \int_0^t g(s)u(s)\,\mathrm{d}s$, $u''(t) = g(t)u(t)$, $u(0) = 0$, $u'(0) = 1$.

- $k(t,s) := (t-s)g(s)$ defines a Volterra integral operator V_k on $C([0,1])$ and $(**)$ reads

$$(e - V_k)u = u_0.$$

  Since $\sigma(V_k) = \{0\}$:

$$u = (e - V_k)^{-1}u_0 = u_0 + \sum_{n=1}^{\infty} V_k^n u_0.$$

  (In case $\|V_k\| < 1$ this is norm convergence with respect to $\|\cdot\|_\infty$ in $C([0,1])$.)

1.4 Outlook: Extension of the holomorphic functional calculus

If $G \subset \mathbb{C}$ is open with $\sigma(x) \subset G$ and $f \in \mathcal{H}(G)$, set

$$f(x) := \oint_\Gamma (z-x)^{-1} f(z)\,\frac{\mathrm{d}z}{2\pi\mathrm{i}}$$

with an *admissible contour* $\Gamma = \{\gamma_1,\dots,\gamma_n\} \subset G \setminus \sigma(x)$ consisting of rectifiable Jordan curves bordering the components of $\sigma(x)$ in the mathematically positive sense. This defines an algebra homomorphism $\Phi : \mathcal{H}(G) \to \mathcal{A}$ and:

(1) *Spectral mapping theorem:* $\sigma(f(x)) = f(\sigma(x))$, $f \in \mathcal{H}(G)$.

(2) If $\sigma(x) = \sigma_1 \cup \sigma_2 \cup \cdots \cup \sigma_n$ with $\sigma_j \cap \sigma_k = \emptyset$ for $j \ne k$, then

$$P_j := \oint_{\Gamma_j} (z-x)^{-1}\,\frac{\mathrm{d}z}{2\pi\mathrm{i}},$$

where Γ_j is an admissible contour around σ_j only, define disjoint projections, i.e., $P_j^2 = P_j$, $P_jP_k = 0$ for $j \ne k$, with $e = \sum_{j=1}^n P_j$.

Further reading: B. Simon, *Operator theory*, AMS 2015; P. Lax, *Functional analysis*, Wiley 2002.

2 Gelfand's theory of commutative Banach algebras

Set-up: $\mathcal{A}$ is a commutative Banach algebra with unit e.

2.1 Multiplicative functionals

Definition 2.1.1: An algebra homomorphism $p : \mathcal{A} \to \mathbb{C}$, i.e.,

$$p(x + \alpha y) = p(x) + \alpha p(y), \qquad p(x \cdot y) = p(x)p(y), \qquad p(e) = 1,$$

is called a *multiplicative functional* (or *character*) $p \in \mathcal{A}'$ (the space of continuous linear functionals) on a Banach algebra $\mathcal{A}$ with unit.
Notation: $\Delta(\mathcal{A}) := \{p \in \mathcal{A}' \mid p \text{ multiplicative functional}\}$.

Example 2.1.2: $\mathcal{A} = C(S)$ with S a compact Hausdorff space, $x \in S$:

$$\delta_x : C(S) \to \mathbb{C}, \qquad \delta_x(f) = f(x)$$

lies in $\Delta(\mathcal{A})$.

Lemma 2.1.3:

(1) $x \in \mathcal{A}$ invertible $\Rightarrow p(x) \neq 0$ for all $p \in \Delta(\mathcal{A})$.

(2) $\|p\| = \sup_{0 \neq x \in \mathcal{A}} |p(x)| \,/\, \|x\| = 1$ for all $p \in \Delta(\mathcal{A})$.

Proof. (1) $e = xx^{-1} \Rightarrow 1 = p(e) = p(xx^{-1}) = p(x)p(x^{-1}) \Rightarrow$ claim.
(2) $p(e) = 1 \Rightarrow \|p\| \geq 1$. Suppose there is $0 \neq x \in \mathcal{A}$ with $|p(x)| > \|x\| \neq 0$. Then:

- $e - p(x)^{-1}x$ is invertible by von Neumann's theorem,
- $p(e - p(x)^{-1}x) = \underbrace{p(e)}_{=1} - p(x)^{-1}p(x) = 0$,

a contradiction to (1). □

Theorem 2.1.1: Characterization of invertibility

$$x \in \mathcal{A} \text{ is invertible} \iff p(x) \neq 0 \text{ for all } p \in \Delta(\mathcal{A}).$$

Proof. "⇒": Lemma 2.1.3(1). "⇐": see below. □

2.2 Excursion: Ideals in a commutative Banach algebra with unit

Definition 2.2.1: A (linear) subspace $\mathcal{J}$ of $\mathcal{A}$ for which

(1) $x \in \mathcal{J}$, $y \in \mathcal{A} \Rightarrow xy = yx \in \mathcal{J}$,

(2) $\mathcal{J} \neq \mathcal{A}$,

is called a *proper ideal*. If the proper ideal $\mathcal{J}$ is not contained in any other proper ideal, $\mathcal{J}$ is called a *maximal ideal*.
Notation: $\mathcal{M}(\mathcal{A}) := \{\mathcal{J} \subset \mathcal{A} \mid \mathcal{J} \text{ is a maximal ideal}\}$.

Lemma 2.2.2 (Properties of proper ideals $\mathcal{J}$):

(1) $e \notin \mathcal{J}$.

Proof. Else $\mathcal{J} = \mathcal{A}$ by property (1) of an ideal. □

(2) All $x \in \mathcal{J}$ are not invertible.

Proof. Else $e = xx^{-1} \in \mathcal{J}$. □

(3) The *principal ideal* generated by $x \in \mathcal{A}$,

$$x\mathcal{A} = \{y \in \mathcal{A} \mid \exists z \in \mathcal{A} : \ y = xz\},$$

is a proper ideal if x is not invertible.

Proof. $x\mathcal{A}$ is a subspace of $\mathcal{A}$, and (1) $y \in x\mathcal{A}$, $z \in \mathcal{A} \Rightarrow yz \in x\mathcal{A}$; (2) $e \notin x\mathcal{A}$ and hence $x\mathcal{A} \neq \mathcal{A}$. □

(4) Every proper ideal $\mathcal{J}$ is contained in a maximal ideal.

Proof idea. Proper ideals are partially ordered by "$\subset$" and the union of a chain of proper ideals is a proper ideal. Zorn's lemma then yields the claim. □

(5) Every non-invertible $x \in \mathcal{A}$ belongs to a maximal ideal.

Proof. (3) + (4). □

(6) The closure $\overline{\mathcal{J}}$ of any proper ideal is a proper ideal.

Proof. $\overline{\mathcal{J}}$ is a subspace; continuity of multiplication yields property (1). Moreover

$$\inf_{x \in \mathcal{J}} \|e - x\| = 1 :$$

"$\leq$" holds since $\|e\| = 1$ (take $x = 0$), and "$<$" is ruled out by von Neumann's theorem since otherwise $x = e - (e - x)$ would be invertible. Hence $e \notin \overline{\mathcal{J}}$, i.e., $\overline{\mathcal{J}} \neq \mathcal{A}$. □

(7) Any $\mathcal{J} \in \mathcal{M}(\mathcal{A})$ is closed.

Proof. Else $\mathcal{J} \subsetneq \overline{\mathcal{J}}$, contradicting maximality. □

Lemma 2.2.3 (Characterization of proper ideals): Let $q : \mathcal{A}_1 \to \mathcal{A}_2$ be an algebra homomorphism. Then

$$\ker q = \{x \in \mathcal{A}_1 \mid q(x) = 0\}$$

is a proper ideal in $\mathcal{A}_1$. Every closed proper ideal in $\mathcal{A}_1$ is of the above form.

Proof. $\ker q$ is a subspace of $\mathcal{A}_1$, and

(1) $x \in \ker q$, $y \in \mathcal{A}_1 \Rightarrow q(xy) = q(x)q(y) = 0$, i.e., $xy \in \ker q$;

(2) $q(e_1) = e_2 \Rightarrow e_1 \notin \ker q$, i.e., $\ker q \neq \mathcal{A}_1$.

Conversely, for a closed proper ideal $\mathcal{J}$ the quotient

$$\mathcal{A}_2 := \mathcal{A}_1/\mathcal{J} = \{[x] = x + \mathcal{J} \mid x \in \mathcal{A}_1\}$$

is a commutative Banach algebra with unit $[e_1]$ and norm $\|[x]\| = \inf_{u \in \mathcal{J}} \|x - u\|$. Then $q : \mathcal{A}_1 \to \mathcal{A}_2$, $q(x) := [x]$, is an algebra homomorphism with $\ker q = \mathcal{J}$. □

Lemma 2.2.4: For every $\mathcal{J} \in \mathcal{M}(\mathcal{A})$ the quotient $\mathcal{A}/\mathcal{J}$ is a commutative Banach algebra in which every $0 \neq [x]$ is invertible.

Proof of invertibility. Let $0 \neq x \in \mathcal{A}$, $x \notin \mathcal{J}$, and

$$\tilde{\mathcal{J}} := \{ax + y \mid a \in \mathcal{A}, \ y \in \mathcal{J}\}.$$

Then $\mathcal{J} \subsetneq \tilde{\mathcal{J}}$ and $\tilde{\mathcal{J}}$ is a subspace of $\mathcal{A}$ with property (1). Hence, by maximality, $\tilde{\mathcal{J}} = \mathcal{A}$ and $e \in \tilde{\mathcal{J}}$, i.e., there are $a \in \mathcal{A}$, $y \in \mathcal{J}$ with

$$e = ax + y \quad \Rightarrow \quad [e] = [ax] = [a][x], \quad \text{i.e., } [a] = [x]^{-1}.$$ □

Theorem 2.2.1: Gelfand–Mazur

Any (commutative) Banach algebra $\mathcal{A}$ with unit in which every $x \in \mathcal{A}$ with $x \neq 0$ is invertible is isometrically isomorphic to $\mathbb{C}$ (isomorphism = bijective homomorphism).

Proof. For $x \in \mathcal{A}$ pick $\lambda \in \sigma(x) \neq \emptyset$. Then $\lambda e - x$ is not invertible, hence zero: $x = \lambda e$, hence $\sigma(x) = \{\lambda\}$. Set $p : \mathcal{A} \to \mathbb{C}$, $p(x) = \lambda$. Then p is linear, multiplicative, $p(e) = 1$ and $|p(x)| = |\lambda| = \|x\|$. Hence p is injective. Since $\mathcal{A}$ is a complex vector space, $\mathbb{C}e \subset \mathcal{A}$, and hence p is surjective, hence bijective. □

Lemma 2.2.5 (Characterization of maximal ideals: $\mathcal{M}(\mathcal{A}) = \{\ker p \mid p \in \Delta(\mathcal{A})\}$):

(1) For any $p \in \Delta(\mathcal{A})$: $\ker p \in \mathcal{M}(\mathcal{A})$.

(2) Every $\mathcal{J} \in \mathcal{M}(\mathcal{A})$ is the kernel of some $p \in \Delta(\mathcal{A})$.

Proof. (1) $\ker p$ is a proper ideal. For a proof of maximality, let $x \in \mathcal{A} \setminus \ker p$. Then

$$e = \underbrace{e - \frac{1}{p(x)}x}_{\in \ker p} + \frac{1}{p(x)}x, \qquad \text{i.e., } e \in \operatorname{span}\{\ker p, x\},$$

since $p\big(e - \frac{1}{p(x)}x\big) = p(e) - \frac{1}{p(x)}p(x) = 0$. Since proper ideals cannot contain e, $\ker p$ is maximal.

(2) Let $q : \mathcal{A} \to \mathcal{A}/\mathcal{J}$, $x \mapsto [x]$. Since $\mathcal{A}/\mathcal{J}$ is a commutative Banach algebra in which every $0 \neq [x]$ is invertible, there is an isometric isomorphism $i : \mathcal{A}/\mathcal{J} \to \mathbb{C}$ (Gelfand–Mazur). Set $p := i \circ q$. Then $p \in \Delta(\mathcal{A})$ and for $x \in \mathcal{J}$: $p(x) = i(q(x)) = i([0]) = 0$. □

Proof of the characterization of invertibility "⇐". Equivalent statement: if $x \in \mathcal{A}$ is not invertible, then $p(x) = 0$ for some $p \in \Delta(\mathcal{A})$. Consider the maximal ideal $\mathcal{J} \in \mathcal{M}(\mathcal{A})$ with $x \in \mathcal{J}$ (Lemma 2.2.2(5)). The claim follows from Lemma 2.2.5(2). □

Theorem 2.2.2: Characterization of the spectrum

For all $x \in \mathcal{A}$:

$$\sigma(x) = \{p(x) \mid p \in \Delta(\mathcal{A})\}.$$

Proof.

$$\begin{aligned}\sigma(x) &= \{\lambda \in \mathbb{C} \mid \lambda e - x \text{ not invertible}\} \\ &= \{\lambda \in \mathbb{C} \mid \exists p \in \Delta(\mathcal{A}) : \underbrace{p(\lambda e - x)}_{=\lambda - p(x)} = 0\} = \text{r.h.s.}\end{aligned}$$ □

2.3 The Gelfand transformation

Lemma 2.3.1 (Gelfand topology): $\Delta(\mathcal{A})$ equipped with the weak-$*$-topology is a compact Hausdorff space.

Remark 2.3.2: Recall: the weak-$*$-topology is the coarsest topology which renders $\mathcal{A}' \ni p \mapsto p(x)$ continuous for all $x \in \mathcal{A}$. This topology separates points.

Proof of compactness.

$$\Delta(\mathcal{A}) = \bigcap_{x,y \in \mathcal{A}} \underbrace{\{p \in \mathcal{A}' \mid p(xy) - p(x)p(y) = 0\}}_{\text{weak-}*\text{-closed}} \cap \underbrace{\{p \in \mathcal{A}' \mid p(e) = 1\}}_{\text{weak-}*\text{-closed}}.$$

Hence $\Delta(\mathcal{A})$ is a weak-$*$-closed subset of the unit ball in $\mathcal{A}'$ (recall: $\|p\| = 1$), hence compact by the Banach–Alaoglu theorem. □

Definition 2.3.3 (Gelfand transformation):

$$\Gamma : \mathcal{A} \to C(\Delta(\mathcal{A})), \qquad x \mapsto \Gamma(x) \quad \text{with} \quad \Gamma(x)(p) := p(x).$$

Theorem 2.3.1: Gelfand

(1) Γ is an algebra homomorphism.

(2) $\Gamma(\mathcal{A})$ separates points in $\Delta(\mathcal{A})$ and contains all constant functions.

(3) For all $x \in \mathcal{A}$: $\sigma(x) = \Gamma(x)(\Delta(\mathcal{A}))$ and $\|\sigma(x)\| = \|\Gamma(x)\|_\infty = \sup_{p\in\Delta(\mathcal{A})} |p(x)|$.

Proof. (1) Immediate.

(2) Since $\Gamma(\alpha e)(p) = p(\alpha e) = \alpha p(e) = \alpha$ for any $\alpha \in \mathbb{C}$, $\Gamma(\mathcal{A})$ contains all constant functions. Let $p_1, p_2 \in \Delta(\mathcal{A})$ with $p_1 \neq p_2$. Then there is $x \in \mathcal{A}$ with $p_1(x) \neq p_2(x)$, i.e., $\Gamma(x)(p_1) \neq \Gamma(x)(p_2)$.

(3) See the characterization of the spectrum. □

Remark 2.3.4: In general, Γ is not isometric, cf. Exercise.

Example 2.3.5: Banach algebra $C(S)$, S compact Hausdorff space, with evaluation functionals $\delta_x \in \Delta(C(S))$, $x \in S$.

Theorem 2.3.2

$\Phi : S \to \Delta(C(S))$, $\Phi(x) = \delta_x$, is a homeomorphism (continuous bijection with continuous inverse); the Gelfand transformation $\Gamma : C(S) \to C(\Delta(C(S)))$ is an isometric isomorphism.

Proof. Exercise. □

2.4 Application: Absolutely convergent Fourier series

Let

$$W(S^1) = \left\{ f : S^1 \to \mathbb{C} \;\middle|\; f(\theta) = \sum_{n\in\mathbb{Z}} a_n \mathrm{e}^{\mathrm{i}\theta n} \text{ with some } (a_n) \in \ell^1(\mathbb{Z}) \right\}.$$

- Normed space: $\|f\|_W := \sum_{n\in\mathbb{Z}} |a_n|$.
- Algebra with unit 1 ($= a_0$): $\sum_{n\in\mathbb{Z}} a_n \mathrm{e}^{\mathrm{i}\theta n} \sum_{m\in\mathbb{Z}} b_m \mathrm{e}^{\mathrm{i}\theta m} = \sum_{n,m\in\mathbb{Z}} a_n b_m \mathrm{e}^{\mathrm{i}\theta(n+m)}$.
- Completeness follows from that of $\ell^1(\mathbb{Z})$.

Hence $W(S^1)$ is a commutative Banach algebra with unit. Evaluation functional for $\theta \in S^1$: $\delta_\theta(f) = f(\theta)$.

Lemma 2.4.1: $\Delta(W(S^1)) = \{\delta_\theta \mid \theta \in S^1\}$.

Proof. "$\supset$": Straightforward.

"$\subset$": Let $p \in \Delta(W(S^1))$ and consider $f_\pm(\theta) := \mathrm{e}^{\pm\mathrm{i}\theta}$. Then:

(1) $|p(f_\pm)| \leq \|p\| \, \|f_\pm\|_W = 1$,

(2) $p(f_+)p(f_-) = p(1) = 1$,

hence $|p(f_\pm)| = 1$, i.e., for some $\theta \in S^1$: $p(f_+) = \mathrm{e}^{\mathrm{i}\theta}$. Therefore

$$p(f_+^n) = p(f_+)^n = \mathrm{e}^{\mathrm{i}n\theta} \quad \text{for all } n \in \mathbb{Z} \quad \Rightarrow \quad p\Big(\sum_{n\in F} a_n f_+^n\Big) = \sum_{n\in F} a_n \mathrm{e}^{\mathrm{i}n\theta} \quad \text{for all finite } F \subset \mathbb{Z}.$$

Extension to $F = \mathbb{Z}$ follows by continuity of p and $\sum_{n\in\mathbb{Z}} |a_n| < \infty$. □

Gelfand transform: $\Gamma : W(S^1) \to C(\Delta(W(S^1))) \cong C(S^1)$ (isometrically isomorphic).

Theorem 2.4.1: Wiener

Let $f \in W(S^1)$ with $f(\theta) \neq 0$ for all $\theta \in S^1$. Then $1/f \in W(S^1)$.

Proof. For all $p \in \Delta(W(S^1))$: $p(f) \neq 0$ by the lemma. Hence f is invertible in $W(S^1)$ by the characterization of invertibility. □

3 Spectral theory in C^*-algebras

3.1 Special elements of a C^*-algebra

Definition 3.1.1 (Special elements in a C^*-algebra $\mathcal{A}$): An element $x \in \mathcal{A}$ is called

(1) *normal* if and only if $x^*x = xx^*$,

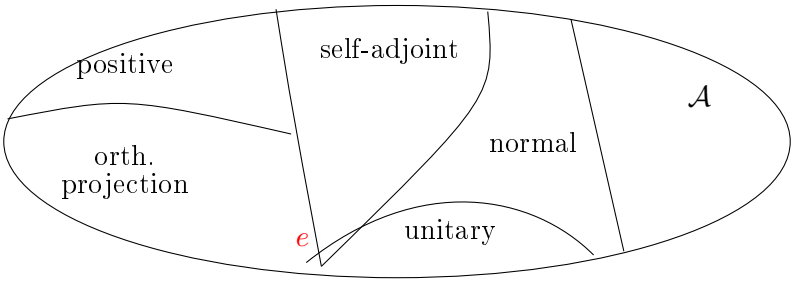
(2) *self-adjoint* if and only if $x = x^*$,

(3) *positive* if and only if $x = y^*y$ for some $y \in \mathcal{A}$,

(4) an *orthogonal projection* if and only if $x^2 = x = x^*$,

(5) *unitary* if and only if $xx^* = x^*x = e$ with unit $e \in \mathcal{A}$.

Notation: x positive $:\iff x \geq 0$. For x, y self-adjoint with $x - y \geq 0$ we write $x \geq y$.



Theorem 3.1.1

If x is normal in a C^*-algebra, then $\|x\| = \|\sigma(x)\|$.

Proof. Exercise. □

Remark 3.1.2: The norm on a C^*-algebra is uniquely determined by the algebraic structure since for $x \in \mathcal{A}$

$$\|x\|^2 = \|x^*x\| = \|\sigma(x^*x)\|\,,$$

where the first equality is the C^*-identity and the second holds since x^*x is positive (in particular normal).

Theorem 3.1.2

For any C^*-algebra $\mathcal{A}$ with unit:

(1) $x \in \mathcal{A}$ is unitary $\Rightarrow \sigma(x) \subset S^1 \subset \mathbb{C}$,

(2) $x \in \mathcal{A}$ is self-adjoint $\Rightarrow \sigma(x) \subset \mathbb{R}$.

Proof. (1) $1 = \|\sigma(e)\| = \|\sigma(x^*x)\| = \|x\|^2 \Rightarrow \sigma(x) \subset \overline{B_1(0)}$. Since $x^{-1} = x^*$, $0 \notin \sigma(x)$. For $\lambda \neq 0$: $\lambda^{-1}e - x^{-1} = \lambda^{-1}x^{-1}(x - \lambda e)$ and hence $0 \neq \lambda \in \sigma(x)$ iff $\lambda^{-1} \in \sigma(x^{-1})$. Since also $x^{-1} = x^*$ is unitary, $\sigma(x^{-1}) \subset \overline{B_1(0)}$. This implies $\sigma(x) \subset S^1$.

(2) Let $\lambda \in \sigma(x)$ and set $y := \sum_{n=1}^{\infty} \frac{\mathrm{i}^n}{n!}(x - \lambda e)^{n-1}$. Then, by the holomorphic functional calculus,

$$\exp(\mathrm{i}x) - \exp(\mathrm{i}\lambda)e = \big[\exp(\mathrm{i}(x - \lambda e)) - e\big] \exp(\mathrm{i}\lambda) = (x - \lambda e)\, y \exp(\mathrm{i}\lambda),$$

where $\exp(\mathrm{i}(x-\lambda e)) = \sum_{n=0}^{\infty} \frac{(\mathrm{i}(x-\lambda e))^n}{n!} = e + \sum_{n=1}^{\infty} \frac{(\mathrm{i}(x-\lambda e))^n}{n!}$, and $\exp(\mathrm{i}x)$ is unitary (holomorphic functional calculus). Since $(x - \lambda e)$ and y commute, $(x - \lambda e)y$ is not invertible, hence $\exp(\mathrm{i}x) - \exp(\mathrm{i}\lambda)e$ is not invertible. Since $\sigma(\exp(\mathrm{i}x)) \subset S^1$ by (1), this implies $|\exp(\mathrm{i}\lambda)| = 1$, i.e., $\lambda \in \mathbb{R}$. □

3.2 Continuous functional calculus

Definition 3.2.1 (C^*-algebra with unit generated by a normal $x \in \mathcal{A}$):

$$C^*(x) := \text{smallest } C^*\text{-algebra containing } x,\ x^* \text{ and } e$$
$$= \Big\{ y \in \mathcal{A} \;\Big|\; \exists \text{ polynomials } q_n(t,s) = \sum_{j,k=0}^{n} a_{jk}^{(n)} t^j s^k : \lim_{n\to\infty} \|y - q_n(x, x^*)\| = 0 \Big\}.$$

Remark 3.2.2 (Properties):

(1) $C^*(x) \subset \mathcal{A}$ is a commutative C^*-algebra with unit.

(2) Any $y \in C^*(x)$ is normal.

Lemma 3.2.3: For any commutative C^*-algebra $\mathcal{A}$ with unit and all $p \in \Delta(\mathcal{A})$, $x \in \mathcal{A}$:

$$p(x^*) = \overline{p(x)}.$$

Proof. Let $\operatorname{Re} x = \frac{1}{2}(x + x^*)$, $\operatorname{Im} x = \frac{1}{2\mathrm{i}}(x - x^*)$. Then $\operatorname{Re} x, \operatorname{Im} x$ are self-adjoint and $x^* = \operatorname{Re} x - \mathrm{i} \operatorname{Im} x$. Since $p(\operatorname{Re} x), p(\operatorname{Im} x) \in \mathbb{R}$ (spectra of self-adjoint elements are real),

$$p(x^*) = p(\operatorname{Re} x) - \mathrm{i}\, p(\operatorname{Im} x) = \overline{p(\operatorname{Re} x) + \mathrm{i}\, p(\operatorname{Im} x)} = \overline{p(x)}.$$ □

Definition 3.2.4: An algebra homomorphism $\varphi : \mathcal{A}_1 \to \mathcal{A}_2$ between C^*-algebras $\mathcal{A}_1, \mathcal{A}_2$ with $\varphi(x^*) = \varphi(x)^*$, $x \in \mathcal{A}_1$, is called a *$*$-homomorphism*. If it is additionally bijective, it is called a *$*$-isomorphism*.

Theorem 3.2.1: Gelfand theorem for C^*-algebras

Let $\mathcal{A}$ be a C^*-algebra with unit and $x \in \mathcal{A}$ normal. Then the Gelfand transformation

$$\Gamma : C^*(x) \to C(\Delta(C^*(x))), \qquad \Gamma(y)(p) = p(y), \quad y \in C^*(x),$$

is an isometric $*$-isomorphism.

Proof.

- From Gelfand's theorem: Γ is a homomorphism.
- From Lemma 3.2.3: $\Gamma(y^*)(p) = p(y^*) = \overline{p(y)} = \overline{\Gamma(y)(p)}$, i.e., Γ is a $*$-homomorphism.
- From the Exercise: Γ is isometric, since every $y \in C^*(x)$ is normal and hence $\|y\| = \|\sigma(y)\| = \|\Gamma(y)\|_\infty$.
- Injectivity is clear by linearity and isometry.
- Surjectivity: Set $\hat{\mathcal{A}} := \Gamma(C^*(x))$. Then
  (a) $\hat{\mathcal{A}}$ is a C^*-subalgebra of $C(\Delta(C^*(x)))$ which is closed,
  (b) $\Gamma(e) = 1$ and $\hat{\mathcal{A}}$ separates points in $\Delta(C^*(x))$,
  (c) $\Gamma(y) \in \hat{\mathcal{A}} \Rightarrow \overline{\Gamma(y)} = \Gamma(y^*) \in \hat{\mathcal{A}}$ (by Lemma 3.2.3).

  The Stone–Weierstrass theorem implies $\hat{\mathcal{A}} = C(\Delta(C^*(x)))$. □

Theorem 3.2.2: Stone–Weierstrass

Let S be a compact Hausdorff space and $\hat{\mathcal{A}}$ a closed subalgebra of $C(S)$ with the properties:

(1) $1 \in \hat{\mathcal{A}}$,

(2) $\hat{\mathcal{A}}$ separates the points in S, i.e., for all $t, s \in S$, $t \neq s$, there is $f \in \hat{\mathcal{A}}$ with $f(t) \neq f(s)$,

(3) for any $f \in \hat{\mathcal{A}}$: $\overline{f} \in \hat{\mathcal{A}}$.

Then $\hat{\mathcal{A}} = C(S)$.

Reference: P. Lax, *Functional analysis*, Wiley 2002.

Theorem 3.2.3: Spectral theorem for C^*-algebras

Let $\mathcal{A}$ be a C^*-algebra with unit and $x \in \mathcal{A}$ normal.

(1) The map $\Gamma(x) : \Delta(C^*(x)) \to \sigma(x)$ is a homeomorphism.

(2) The map $\Psi : C(\sigma(x)) \to C^*(x)$,

$$\Psi(f) := \Gamma^{-1}\big(f \circ \Gamma(x)\big) =: f(x), \qquad (*)$$

is an isometric $*$-isomorphism.

(3) Spectral mapping: for any $f \in C(\sigma(x))$: $\sigma(f(x)) = f(\sigma(x))$.

Remark 3.2.5 (Continuous functional calculus): For $f, g \in C(\sigma(x))$, the operators defined through $(*)$ satisfy

$$\begin{aligned} (f+g)(x) &= f(x) + g(x), & (f \cdot g)(x) &= f(x) \cdot g(x), \\ f(x)^* &= \overline{f}(x), & 1(x) &= e, \qquad \mathrm{id}(x) = x, \\ \|f(x)\| &= \|f\|_\infty, \end{aligned}$$

i.e., Ψ is a $*$-homomorphism. Moreover, if $g \in C(\sigma(x))$ and $f \in C(g(\sigma(x)))$, then

$$(f \circ g)(x) = f(g(x)).$$

Proof. Exercise. Sketch: (i) Notice that for $p \in \Delta(C^*(x))$:

$$p(f(x)) = p\big(\Gamma^{-1}(f\circ\Gamma(x))\big) = \Gamma\big(\Gamma^{-1}(f\circ\Gamma(x))\big)(p) = (f\circ\Gamma(x))(p) = f(\Gamma(x)(p)) = f(p(x)).$$

(ii) By (i), $p\big((f\circ g)(x)\big) = (f\circ g)(p(x)) = f(g(p(x)))$. On the other hand, since $C^*(g(x)) \subset C^*(x)$ and $f(g(x)) \in C^*(x)$,

$$p\big(f(g(x))\big) = p|_{C^*(g(x))}\big(f(g(x))\big) \overset{\text{(i)}}{=} f\big(p|_{C^*(g(x))}(g(x))\big) \overset{\text{(i)}}{=} f(g(p(x))).$$

Since the $p \in \Delta(C^*(x))$ separate points (Gelfand isomorphism), the claim follows. □

Lemma 3.2.6: Let $\mathcal{A}_1 \subset \mathcal{A}_2$ be C^*-algebras with (the same) unit. For all $x \in \mathcal{A}_1$:

$$\sigma_{\mathcal{A}_1}(x) = \sigma_{\mathcal{A}_2}(x).$$

Proof. It is sufficient to show for $x \in \mathcal{A}_1$: x is invertible in $\mathcal{A}_1 \iff x$ is invertible in $\mathcal{A}_2$. "$\Rightarrow$" is obvious.

Proof of "$\Leftarrow$", case $x \in \mathcal{A}_1$ self-adjoint with $0 \notin \sigma_{\mathcal{A}_2}(x)$: Hence $f(t) = \frac{1}{t}$ is continuous on $\sigma_{\mathcal{A}_2}(x) \subset \mathbb{R}$. By the Weierstrass approximation theorem there are polynomials $p_n \in \mathcal{P}(\sigma_{\mathcal{A}_2}(x))$ with $\|f - p_n\|_\infty \to 0$. Since $p_n(x) \in \mathcal{A}_1$ and $(p_n(x))$ is Cauchy, completeness implies that there is $y \in \mathcal{A}_1$ with $\|y - p_n(x)\| \to 0$. Then

$$\|yx - e\| = \lim_{n\to\infty} \|p_n(x)x - e\| = \lim_{n\to\infty} \|\sigma_{\mathcal{A}_2}(p_n(x)x - e)\| = \lim_{n\to\infty} \|q_n(\sigma_{\mathcal{A}_2}(x))\| = 0$$

with $q_n(t) := p_n(t)t - 1$, $t \in \sigma_{\mathcal{A}_2}(x)$, using the polynomial spectral mapping theorem (and that $p_n(x)x - e$ is self-adjoint). Similarly $\|xy - e\| = 0$, i.e., $y = x^{-1} \in \mathcal{A}_1$.

General $x \in \mathcal{A}_1$ with $0 \notin \sigma_{\mathcal{A}_2}(x)$: $y = x^*x \in \mathcal{A}_1$ is self-adjoint with $0 \notin \sigma_{\mathcal{A}_2}(y)$ since $y^{-1} = x^{-1}(x^{-1})^* = x^{-1}(x^*)^{-1}$ exists in $\mathcal{A}_2$. By the first case $y^{-1} \in \mathcal{A}_1$, hence $x^{-1} = y^{-1}x^* \in \mathcal{A}_1$. □

Proof of the spectral theorem for C^*-algebras. (1) Consider $\Gamma(x) : \Delta(C^*(x)) \to \sigma_{C^*(x)}(x) = \sigma(x)$ (Lemma 3.2.6) with $\Gamma(x)(p) = p(x)$.

- Surjectivity is evident from $\sigma_{C^*(x)}(x) = \{p(x) \mid p \in \Delta(C^*(x))\}$.
- Injectivity: Let $p_1, p_2 \in \Delta(C^*(x))$ with $p_1(x) = p_2(x)$; then also $p_1(x^*) = \overline{p_1(x)} = \overline{p_2(x)} = p_2(x^*)$. If $q_n(t,s) = \sum_{j,k} a_{jk}t^j s^k$, then by linearity and multiplicativity $p_1(q_n(x,x^*)) = p_2(q_n(x,x^*))$. By density this extends to all $y \in C^*(x)$: $p_1(y) = p_2(y)$.
- Since $\Delta(C^*(x))$ is equipped with the weak-$*$-topology, $\Gamma(x)$ is a continuous bijective map from the compact set $\Delta(C^*(x))$ to the Hausdorff space $\sigma(x)$, hence a homeomorphism.

(2) Follows immediately from (1) and the Gelfand theorem for C^*-algebras. Note: $\Gamma(x) \in C(\Delta(C^*(x)))$ implies $f\circ\Gamma(x) \in C(\Delta(C^*(x)))$, and since $\Gamma(x)$ is a homeomorphism,

$$\|f\circ\Gamma(x)\|_\infty = \sup_{p\in\Delta(C^*(x))} |f(\Gamma(x)(p))| = \|f\|_\infty .$$

(3) Using the Gelfand theorem, $\Gamma(f(x)) = \Gamma\big(\Gamma^{-1}(f\circ\Gamma(x))\big) = f\circ\Gamma(x)$, i.e., $\Gamma(f(x))(p) = f(p(x))$ for $p \in \Delta(C^*(x))$. Thus

$$\sigma(f(x)) = \Gamma(f(x))\big(\Delta(C^*(x))\big) = \{f(p(x)) \mid p \in \Delta(C^*(x))\} = f(\sigma_{C^*(x)}(x)) = f(\sigma(x)),$$

where the last step uses Lemma 3.2.6. □

Application to $\mathcal{A} = \mathcal{L}(\mathcal{H})$ with a Hilbert space $\mathcal{H}$.

Corollary 3.2.7: Let $A \in \mathcal{L}(\mathcal{H})$ be a normal operator on a complex Hilbert space $\mathcal{H}$. Then there is a (unique) $*$-homomorphism

$$\Psi : C(\sigma(A)) \to \mathcal{L}(\mathcal{H}), \qquad \Psi(f) =: f(A),$$

with the properties

(1) $\Psi(f) = A$ if $f(t) = t$,

(2) $\|\Psi(f)\| = \|f\|_\infty$.

Remark 3.2.8: Uniqueness is a consequence of Weierstrass' approximation theorem asserting that the complex polynomials $q_n(z, \bar{z}) = \sum_{j,k=0}^{n} a_{jk} z^j \bar{z}^k$ are dense in $C(\sigma(A))$. Note: $\Psi(\mathrm{id}) = A$, $\Psi(\overline{\mathrm{id}}) = A^* \Rightarrow \Psi(q_n(\cdot, \bar{\cdot})) = q_n(A, A^*)$.

3.3 Self-adjoint and positive elements in a C^*-algebra with unit

Since $f_\pm : \mathbb{R} \to [0, \infty)$, $f_+(t) = \max\{t, 0\}$, $f_-(t) = \max\{-t, 0\}$ are continuous with $f_+(t) - f_-(t) = t$, the spectral calculus implies for any self-adjoint $x \in \mathcal{A}$:

$$x = x_+ - x_-, \qquad x_+ = f_+(x), \quad x_- = f_-(x).$$

Moreover, $x_\pm \geq 0$ since $f_\pm = \sqrt{f_\pm}\,\sqrt{f_\pm} = \overline{\sqrt{f_\pm}}\,\sqrt{f_\pm}$.

Theorem 3.3.1: Spectral characterization of positivity

The following are equivalent:

(1) $x \geq 0$,

(2) $x = x^* \in \mathcal{A}$ and $\sigma(x) \subset [0, \infty)$.

Remark 3.3.1: For an extension see the Exercises.

Corollary 3.3.2 (Square root): For any $x \geq 0$ there is a unique $a \geq 0$ such that $x = a^2$. Notation: $a = \sqrt{x}$.

Proof. Existence: $a = \sqrt{x}$ from the spectral calculus, using $\sigma(x) \subset [0, \infty)$.

Uniqueness: Suppose $a \geq 0$ satisfies $x = a^2$. Since $\sqrt{x} \in C^*(a)$ we conclude for any $p \in \Delta(C^*(a))$:

$$p(\sqrt{x})^2 = p(x) = p(a)^2 \quad \Rightarrow \quad p(a) = p(\sqrt{x}),$$

since $p(a) \geq 0$ and $p(\sqrt{x}) \geq 0$. Hence $\Gamma(a) = \Gamma(\sqrt{x})$ and since Γ is an isomorphism, $a = \sqrt{x}$. □

Proof of the theorem. (2)⇒(1): $x = x_+ - x_- = x_+$, since $x_- = 0$ as $\sigma(x) \subset [0, \infty)$.

(1)⇒(2): Let $x = y^* y$ with some $y \in \mathcal{A}$. Then $z := y x_-$ satisfies:

- $z^* z = x_- y^* y x_- = x_- x x_- = -x_-^3$,
- $z = \mathrm{Re}\, z + \mathrm{i}\, \mathrm{Im}\, z$ with $\mathrm{Re}\, z, \mathrm{Im}\, z$ self-adjoint, hence

$$\begin{aligned} z z^* &= 2(\mathrm{Re}\, z)^2 + 2(\mathrm{Im}\, z)^2 - z^* z \\ &= 2(\mathrm{Re}\, z)^2 + 2(\mathrm{Im}\, z)^2 + x_-^3. \end{aligned}$$

- Spectral mapping for self-adjoint elements: $\sigma((\mathrm{Re}\, z)^2)$, $\sigma((\mathrm{Im}\, z)^2)$, $\sigma(x_-^3) \subset [0, \infty)$. Lemma 3.3.3(2) below ⇒ $\sigma(z z^*) \subset [0, \infty)$. But $\sigma(z^* z) = \sigma(-x_-^3) \subset (-\infty, 0]$. Lemma 3.3.4 below ⇒ $\sigma(z^* z) = \sigma(z z^*) = \{0\}$, i.e.,

$$z^* z = -x_-^3 = 0 \;\Rightarrow\; x_- = 0 \;\Rightarrow\; \sigma(x) \subset [0, \infty). \qquad \square$$

Lemma 3.3.3: Let $x, y \in \mathcal{A}$ be self-adjoint.

(1) The following are equivalent:

(a) $\sigma(x) \subset [0, \infty)$,
(b) for all $t \geq \|x\|$: $\|te - x\| \leq t$,
(c) for some $t \geq \|x\|$: $\|te - x\| \leq t$.

(2) $\sigma(x), \sigma(y) \subset [0, \infty) \Rightarrow \sigma(x + y) \subset [0, \infty)$.

Proof. (1) Let $f_t(s) := t - s$, $s \in \sigma(x)$, with $t \geq \|x\| = \|\sigma(x)\|$. From the Gelfand isometry:

$$\|f_t(x)\| = \sup_{s \in \sigma(x)} |t - s| = t - \inf \sigma(x).$$

(a)⇒(b): from $\inf \sigma(x) \geq 0$. (b)⇒(c): evident. (c)⇒(a): from $t - \inf \sigma(x) \leq t$.
(2) Let $t = \|x\| + \|y\| \geq \|x + y\|$. Then

$$\|t - (x + y)\| = \|(\|x\| - x) + (\|y\| - y)\| \leq \|\|x\| - x\| + \|\|y\| - y\| \leq \|x\| + \|y\| = t,$$

using (1). Hence $\sigma(x + y) \subset [0, \infty)$ by (1). □

Lemma 3.3.4: For any $x, y \in \mathcal{A}$: $\sigma(xy) \setminus \{0\} = \sigma(yx) \setminus \{0\}$.

Proof. Exercise. □

3.4 Special case: $\mathcal{A} = \mathcal{L}(\mathcal{H})$ with a Hilbert space $\mathcal{H}$

Theorem 3.4.1: Variational bounds on the spectrum

For any self-adjoint $A \in \mathcal{L}(\mathcal{H})$ on some Hilbert space $\mathcal{H}$ let

$$a := \inf_{\|x\|=1} \langle Ax, x \rangle, \qquad b := \sup_{\|x\|=1} \langle Ax, x \rangle.$$

Then $\sigma(A) \subset [a, b]$ and $a, b \in \sigma(A)$.

Proof. Let $\lambda < a$ and $x \in \mathcal{H}$:

$$\langle (A - \lambda)x, x \rangle = \langle Ax, x \rangle - \lambda \|x\|^2 \geq (a - \lambda) \|x\|^2,$$
$$\langle (A - \lambda)x, x \rangle \leq (b + |\lambda|) \|x\|^2.$$

Hence $\mathcal{H} \ni x \mapsto \sqrt{\langle (A - \lambda)x, x \rangle}$ is an equivalent norm on $\mathcal{H}$ with scalar product $\mathcal{H} \times \mathcal{H} \ni (x, y) \mapsto \langle (A - \lambda)x, y \rangle$. By Riesz' theorem, for any $z \in \mathcal{H}$ there is $y = y(z) \in \mathcal{H}$ such that

$$\langle z, x \rangle = \langle (A - \lambda)y, x \rangle \qquad \text{for all } x \in \mathcal{H},$$

i.e., $z = (A - \lambda)y$, and hence $A - \lambda$ is invertible: $\lambda \notin \sigma(A)$. A similar argument implies $\mu \notin \sigma(A)$ for $\mu > b$. Now

$$\max\{|a|, |b|\} \leq \sup_{\|x\|=1} |\langle Ax, x \rangle| = \|A\| = \|\sigma(A)\| \leq \max\{|a|, |b|\},$$

using that A is normal and $\sigma(A) \subset [a, b]$. Hence $\max\{|a|, |b|\} = \|A\| = \|\sigma(A)\|$. Applying this result to $A_c = A + c$ with $c \in \mathbb{R}$ arbitrary,

$$\max\{|a + c|, |b + c|\} = \|\sigma(A_c)\| = \sup_{\lambda \in \sigma(A)} |\lambda + c|$$

(spectral mapping), yields $a, b \in \sigma(A)$. □

Corollary 3.4.1: Let $A \in \mathcal{L}(\mathcal{H})$ be a bounded self-adjoint operator on a Hilbert space $\mathcal{H}$. Then

$$A \text{ is positive } \iff \langle Ax, x\rangle \geq 0 \text{ for all } x \in \mathcal{H}.$$

Proof. "⇒": $A = B^*B$ implies $\langle Ax, x\rangle = \|Bx\|^2 \geq 0$ for all $x \in \mathcal{H}$. "⇐": From the theorem, $\sigma(A) \subset [0, \infty)$. The claim follows from the characterization of positivity in the C^*-algebra $\mathcal{L}(\mathcal{H})$. □

3.5 Distance of spectra of normal elements

The *Hausdorff distance* of sets D_1, D_2 in a metric space is

$$d_H(D_1, D_2) := \max\Big\{ \sup_{\lambda \in D_1} d(\lambda, D_2),\ \sup_{\mu \in D_2} d(\mu, D_1)\Big\} = \inf\{\varepsilon \geq 0 \mid D_2 \subset (D_1)_\varepsilon \ \wedge\ D_1 \subset (D_2)_\varepsilon\},$$

where $(D)_\varepsilon$ denotes the ε-fattening of D.

Theorem 3.5.1

For any normal elements $x, y \in \mathcal{A}$:

$$d_H(\sigma(x), \sigma(y)) \leq \|x - y\|.$$

Proof. Suppose w.l.o.g. (note that $\sigma(x), \sigma(y)$ are compact)

$$\max_{\lambda \in \sigma(x)} \min_{\mu \in \sigma(y)} |\lambda - \mu| > \|x - y\| > 0.$$

Then for some $\lambda \in \sigma(x)$: $\min_{\mu \in \sigma(y)} |\lambda - \mu| > \|x - y\| > 0$. Hence $\lambda \in \varrho(y)$, i.e., $\lambda e - y$ is invertible and by spectral mapping $\sigma((\lambda e - y)^{-1}) = (\lambda - \sigma(y))^{-1}$. Since $(\lambda e - y)^{-1}$ is normal,

$$\big\|(\lambda e - y)^{-1}\big\| = \big\|\sigma((\lambda e - y)^{-1})\big\| = \Big(\min_{\mu \in \sigma(y)} |\lambda - \mu|\Big)^{-1} < \frac{1}{\|x - y\|}.$$

Hence $e + (\lambda e - y)^{-1}(y - x)$ is invertible by von Neumann's theorem, and so is

$$\lambda e - x = \lambda e - y + y - x = (\lambda e - y)\big(e + (\lambda e - y)^{-1}(y - x)\big),$$

a contradiction to $\lambda \in \sigma(x)$. □

4 Spectral theory of bounded operators in Hilbert space

Set-up: bounded linear operators $\mathcal{L}(\mathcal{H})$ on a Hilbert space $\mathcal{H}$.

4.1 Projection-valued measures

Remark 4.1.1 (Recall: spectral theorem for compact normal $A \in \mathcal{K}(\mathcal{H})$): Here $\mathcal{K}(\mathcal{H})$ denotes the compact operators.

- $\mathcal{H} = \ker A \oplus \overline{\operatorname{ran} A}$ with separable $\overline{\operatorname{ran} A}$.
- There is an orthonormal basis (e_n) of eigenvectors in $\overline{\operatorname{ran} A}$ such that

$$Ax = \sum_{n=1}^{\infty} \lambda_n \langle x, e_n\rangle e_n, \qquad x \in \mathcal{H},$$

with (λ_n) the eigenvalues. (The scalar product on $\mathcal{H}$ is linear in the first coordinate!)

Rephrasing: Let P_n be the orthogonal projection onto $\operatorname{span}(e_n)$ and P_0 the orthogonal projection onto $\ker A$. Then

$$Ax = 0 \cdot P_0 x + \sum_{n=1}^{\infty} \lambda_n P_n x, \qquad \mathbb{1}x = \Big(P_0 + \sum_{n=1}^{\infty} P_n\Big)x,$$

with $\mathbb{1}$ the unit in $\mathcal{L}(\mathcal{H})$. The family $P_0, P_1, \dots$ defines via ($\lambda_0 := 0$)

$$E : \Sigma \to \mathcal{L}(\mathcal{H}), \qquad I \mapsto E(I)x := \sum_{n\geq 0} 1[\lambda_n \in I]\, P_n x$$

(with the indicator function $1[\,\cdot\,]$) a PV measure on the Borel-σ-algebra Σ on $\mathbb{C}$, and

$$E_{xx}(I) = \langle E(I)x, x\rangle = \sum_{n\geq 0} 1[\lambda_n \in I]\, \underbrace{\|P_n x\|^2}_{=\langle P_n x, x\rangle}.$$

Definition 4.1.2: A map $E : \Sigma \to \mathcal{L}(\mathcal{H})$, $I \mapsto E(I)$, is called a *projection-valued (PV) measure* if $E(I)$ is an orthogonal projection ($E(I)^2 = E(I) = E(I)^*$) for all $I \in \Sigma$ and

(1) $E(\emptyset) = 0$, $E(\mathbb{C}) = \mathbb{1}$,

(2) $I_1, I_2, \dots \in \Sigma$ mutually disjoint $\Rightarrow E\big(\biguplus_{j\geq 1} I_j\big)x = \sum_{j\geq 1} E(I_j)x$, $x \in \mathcal{H}$.

In case there is a compact $K \subset \mathbb{C}$ such that $E(K) = \mathbb{1}$, E has *compact support*. The smallest such K is the *support*.

Lemma 4.1.3 (Properties):

(a) For any $x, y \in \mathcal{H}$: $E_{xy}(I) := \langle E(I)x, y\rangle$ defines a complex-valued Borel measure on $\mathbb{C}$, i.e., $\mu = \mu_+ - \mu_- + \mathrm{i}(\nu_+ - \nu_-)$ with Borel measures $\mu_\pm, \nu_\pm$. In case $x = y$: E_{xx} is a Borel measure with $E_{xx}(\mathbb{C}) = \|x\|^2$; it is called the *spectral measure associated with* x.

(b) For $I, J \in \Sigma$: $E(I \cap J) = E(I)E(J) = E(J)E(I)$.

Proof of (b). Case $I \subset J$: $E(J) \overset{(2)}{=} E(I) + E(J \setminus I) \Rightarrow \operatorname{ran} E(I) \subset \operatorname{ran} E(J) \Rightarrow E(I)E(J) = E(J)E(I) = E(I)$.

Case $I \cap J = \emptyset$: $E(I \cup J) \overset{(2)}{=} E(I) + E(J)$. Since $I \subset I \cup J$:

$$E(I) = E(I)E(I \cup J) = E(I) + E(I)E(J) = E(I \cup J)E(I) = E(I) + E(J)E(I),$$

hence $E(I)E(J) = E(J)E(I) = 0$.

General case: $E(I) = E(I \setminus J) + E(I \cap J)$. Multiplication by $E(J)$ from the left/right yields $E(J)E(I) = E(I \cap J) = E(I)E(J)$. □

Remark 4.1.4: Property (a) implies (2) above since for any Borel measure and mutually disjoint $I_1, I_2, \dots$

$$E_{x,y}\Big(\biguplus_{j\geq 1} I_j\Big) = \sum_{j\geq 1} E_{x,y}(I_j) \qquad \text{for all } x, y \in \mathcal{H}.$$

This implies (2) by the following lemma, since $(E(I_j)x)_j$ is an orthogonal sequence.

Lemma 4.1.5: Let $(e_n)_{n\in\mathbb{N}}$ be an orthogonal sequence in $\mathcal{H}$. The following statements are equivalent:

(1) $\sum_{n\geq 1} e_n$ converges in $\mathcal{H}$,

(2) $\sum_{n\geq 1} \|e_n\|^2 < \infty$,

(3) $\sum_{n\geq 1} \langle e_n, y\rangle$ converges for all $y \in \mathcal{H}$.

Proof. Due to orthogonality, for $m < n$:

$$\Big\| \sum_{k=m}^{n} e_k \Big\|^2 = \sum_{k=m}^{n}\sum_{l=m}^{n} \langle e_k, e_l\rangle = \sum_{k=m}^{n} \|e_k\|^2 .$$

Hence $\sum_{n\geq 1} e_n$ is Cauchy iff $\sum_{n\geq 1} \|e_n\|^2$ is Cauchy, i.e., (1) $\iff$ (2). (1)⇒(3) follows from continuity of $\langle \cdot, y\rangle$, $y \in \mathcal{H}$. (3)⇒(2): Consider $s_n := \sum_{k=1}^{n} e_k$. Then $(\langle s_n, y\rangle)$ is bounded for all $y \in \mathcal{H}$ and hence, by the uniform boundedness principle, also (s_n) is bounded in $\mathcal{H}$. Hence $\|s_n\|^2 = \sum_{k=1}^{n} \|e_k\|^2$ is a bounded monotonically increasing sequence in $\mathbb{R}$, i.e., the limit exists. □

Remark 4.1.6 (Recall: notions of convergence in a Hilbert space $\mathcal{H}$):

(1) Sequences $(x_n) \subset \mathcal{H}$:

(i) convergence $x_n \to x$: $\lim_{n\to\infty} \|x - x_n\| = 0$;

(ii) weak convergence $x_n \rightharpoonup x$: $\lim_{n\to\infty} \langle x_n, y\rangle = \langle x, y\rangle$ for all $y \in \mathcal{H}$.

We have (i)⇒(ii).

(2) Sequences $(A_n) \subset \mathcal{L}(\mathcal{H})$:

(i) norm convergence to A: $\lim_{n\to\infty} \|A_n - A\| = 0$;

(ii) strong convergence to A: $A_n x \to Ax$ for all $x \in \mathcal{H}$;

(iii) weak convergence to A: $A_n x \rightharpoonup Ax$ for all $x \in \mathcal{H}$.

We have (i)⇒(ii)⇒(iii).

Example 4.1.7: Weakly, but not strongly convergent sequence: for an orthonormal sequence $(e_n) \subset \mathcal{H}$, $e_n \rightharpoonup 0$ but $\|e_n - e_m\|^2 = \|e_n\|^2 + \|e_m\|^2 - 2\,\mathrm{Re}\underbrace{\langle e_n, e_m\rangle}_{=0} = 2$ for any pair $n \neq m$.

Strongly, but not norm convergent sequence (A_n): $A_n x = e_n \langle x, e_n\rangle \to 0$, but $\|A_n\| = 1$.

4.2 Integration with respect to a PV measure

Let $E : \Sigma \to \mathcal{L}(\mathcal{H})$ be a PV measure.

(1) Indicator functions of $I \in \Sigma$: $\int 1_I \,\mathrm{d}E := E(I)$.

(2) Elementary functions $f = \sum_{j=1}^{n} \alpha_j 1_{I_j}$ with $\alpha_j \in \mathbb{C}$, $I_j \in \Sigma$:

$$\int f \,\mathrm{d}E := \sum_{j=1}^{n} \alpha_j E(I_j).$$

Lemma 4.2.1: $\big\|\int f \,\mathrm{d}E\big\| \leq \|f\|_\infty$.

Proof. Let $x \in \mathcal{H}$. W.l.o.g. $I_1, \ldots, I_n$ are mutually disjoint. Then

$$\Big\| \int f \,\mathrm{d}E\, x \Big\|^2 = \Big\| \sum_{j=1}^{n} \alpha_j E(I_j)x \Big\|^2 = \sum_{j,k=1}^{n} \alpha_j \overline{\alpha_k} \underbrace{\langle E(I_j)x, E(I_k)x\rangle}_{=\langle E(I_k)E(I_j)x, x\rangle} = \sum_{j=1}^{n} |\alpha_j|^2 \|E(I_j)x\|^2$$

$$\leq \sup_j |\alpha_j|^2 \underbrace{\sum_{k=1}^{n} \|E(I_k)x\|^2}_{=\left\|E(\biguplus_j I_j)x\right\|^2} = \|f\|_\infty^2 \left\langle E\big(\biguplus_{j=1}^n I_j\big)x, x\right\rangle \leq \|f\|_\infty^2 \|x\|^2,$$

using $\|E(J)\| \leq 1$. □

(3) Bounded measurable $f : K \to \mathbb{C}$, $K \subset \mathbb{C}$ compact:

$$\mathcal{B}_\infty(K) := \{f : K \to \mathbb{C} \mid f \ \Sigma_K\text{-measurable and } \|f\|_\infty < \infty\}.$$

- $\mathcal{B}_\infty(K)$ equipped with $\|\cdot\|_\infty$ is a C^*-algebra.
- Recall from measure theory: for any $f \in \mathcal{B}_\infty(K)$ there is a Cauchy sequence of elementary functions (f_n) such that $\lim_{n\to\infty} \|f - f_n\|_\infty = 0$.

Lemma 4.2.1 implies that $\left(\int f_n \,\mathrm{d}E\right)$ is Cauchy in $\mathcal{L}(\mathcal{H})$. By completeness this allows us to define for $f \in \mathcal{B}_\infty(K)$:

$$\int_K f \,\mathrm{d}E := \lim_{n\to\infty} \int f_n \,\mathrm{d}E \in \mathcal{L}(\mathcal{H}).$$

Lemma 4.2.2 (Borel functional calculus): Suppose $E : \Sigma \to \mathcal{L}(\mathcal{H})$ is a PV measure supported on a compact set $K \subset \mathbb{C}$. Then

$$\hat{\Psi} : \mathcal{B}_\infty(K) \to \mathcal{L}(\mathcal{H}), \qquad \hat{\Psi}(f) := \int_K f \,\mathrm{d}E,$$

is a $*$-homomorphism with $\left\|\hat{\Psi}(f)\right\| \leq \|f\|_\infty$, $f \in \mathcal{B}_\infty(K)$.

Proof. BLT theorem for $\left\|\hat{\Psi}(f)\right\| \leq \|f\|_\infty$. The defining properties of a $*$-homomorphism are clear for elementary functions and extend by linearity and boundedness to all $f \in \mathcal{B}_\infty(K)$. □

Theorem 4.2.1: BLT

Any $T \in \mathcal{L}(X, Y)$ with X normed and Y a Banach space with norm $\|T\|_{\mathcal{L}(X,Y)}$ has a unique extension $\overline{T} \in \mathcal{L}(\overline{X}, Y)$ to the completion $\overline{X}$ with the same norm.

4.3 The spectral theorem

Theorem 4.3.1: Spectral theorem – Borel functional calculus

For every normal $A \in \mathcal{L}(\mathcal{H})$ there is a unique PV measure $E : \Sigma \to \mathcal{L}(\mathcal{H})$ supported on $\sigma(A)$ such that

$$A = \int_{\sigma(A)} t \,\mathrm{d}E.$$

Moreover:

(1) The map $\hat{\Psi} : \mathcal{B}_\infty(\sigma(A)) \to \mathcal{L}(\mathcal{H})$, $\hat{\Psi}(f) := \int_{\sigma(A)} f \,\mathrm{d}E =: f(A)$, is a $*$-homomorphism which extends the inverse Gelfand transformation, and $\|f(A)\| \leq \|f\|_\infty$.

(2) If $C \in \mathcal{L}(\mathcal{H})$ commutes with A and A^*, then $Cf(A) = f(A)C$ for $f \in \mathcal{B}_\infty(\sigma(A))$.

Proof. By the continuous spectral calculus, for any $x, y \in \mathcal{H}$

$$C(\sigma(A)) \to \mathbb{C}, \qquad f \mapsto \langle f(A)x, y\rangle$$

defines a bounded linear functional with norm $\leq \|x\|\,\|y\|$. By the Riesz–Markov representation theorem there is a unique complex-valued Borel measure $\mu_{x,y}$ on $\sigma(A)$ such that

$$\langle f(A)x, y\rangle = \int_{\sigma(A)} f \,\mathrm{d}\mu_{x,y}, \qquad f \in C(\sigma(A)).$$

Properties:

(i) $\mu_{x,y} = \overline{\mu_{y,x}}$, since for $f \in C(\sigma(A))$: $\int f \,\mathrm{d}\mu_{x,y} = \overline{\langle y, f(A)x\rangle} = \overline{\langle f(A)^* y, x\rangle} = \overline{\langle \overline{f}(A)y, x\rangle} = \overline{\int \overline{f}\,\mathrm{d}\mu_{y,x}} = \int f \,\mathrm{d}\overline{\mu_{y,x}}$.

(ii) $\mu_{x+\alpha y,z} = \mu_{x,z} + \alpha\mu_{y,z}$, $x, y, z \in \mathcal{H}$, $\alpha \in \mathbb{C}$.

(iii) $\left|\int f \,\mathrm{d}\mu_{x,y}\right| \leq \|f\|_\infty \|x\|\,\|y\|$ (with $\|f\|_\infty = \|f(A)\|$ for continuous f).

Hence for any $f \in \mathcal{B}_\infty(\sigma(A))$, $\mathcal{H}\times\mathcal{H} \to \mathbb{C}$, $(x, y) \mapsto \int f \,\mathrm{d}\mu_{x,y}$ defines a bounded sesquilinear form with norm $\leq \|f\|_\infty$. By the Fréchet–Riesz representation there is a unique operator $f(A) \in \mathcal{L}(\mathcal{H})$ such that

$$\langle f(A)x, y\rangle = \int f \,\mathrm{d}\mu_{x,y} \qquad \text{for any } x, y \in \mathcal{H}.$$

We thus defined a linear map $\overline{\Psi} : \mathcal{B}_\infty(\sigma(A)) \to \mathcal{L}(\mathcal{H})$, $f \mapsto f(A)$, with $\left\|\overline{\Psi}(f)\right\| \leq \|f\|_\infty$, which extends the continuous functional calculus $\Psi : C(\sigma(A)) \to \mathcal{L}(\mathcal{H})$.

$\overline{\Psi}$ is a $*$-homomorphism, since:

(a) $\overline{\Psi}(1) = \Psi(1) = \mathbb{1}$.

(b) $\overline{\Psi}$ is linear since $f \mapsto \int f \,\mathrm{d}\mu_{x,y}$ is linear.

(c) For $x, y \in \mathcal{H}$: $\langle f(A)^* x, y\rangle = \langle x, f(A)y\rangle = \overline{\langle f(A)y, x\rangle} = \overline{\int f \,\mathrm{d}\mu_{y,x}} \overset{\text{(i)}}{=} \int \overline{f}\,\mathrm{d}\mu_{x,y} = \langle \overline{f}(A)x, y\rangle$. Hence $\overline{\Psi}(f)^* = \overline{\Psi}(\overline{f})$, $f \in \mathcal{B}_\infty(\sigma(A))$.

(d) For $f, g \in C(\sigma(A))$, by the continuous functional calculus ($x, y \in \mathcal{H}$):

$$\int fg \,\mathrm{d}\mu_{x,y} = \langle (fg)(A)x, y\rangle = \langle f(A)g(A)x, y\rangle = \langle g(A)x, \overline{f}(A)y\rangle = \int g \,\mathrm{d}\mu_{x,\overline{f}(A)y}.$$

By a standard mollifier argument$^{(*)}$ replacing g with $\phi_{1/n} * g$, the integral identity extends to all $g \in \mathcal{B}_\infty(\sigma(A))$, $f \in C(\sigma(A))$, such that

$$\int fg \,\mathrm{d}\mu_{x,y} = \langle f(A)g(A)x, y\rangle = \int f \,\mathrm{d}\mu_{g(A)x,y}.$$

This identity now extends (by the same argument in f) to all $f, g \in \mathcal{B}_\infty(\sigma(A))$, i.e.,

$$\langle (fg)(A)x, y\rangle = \langle f(A)g(A)x, y\rangle\,, \qquad x, y \in \mathcal{H}.$$

$^{(*)}$ The proof uses the following approximation result from measure theory.

Proposition 4.3.1: Let μ be a finite Borel measure with compact support $K \subset \mathbb{C}$. Then for every $f \in \mathcal{B}_\infty(K)$ the sequence $f_n := \phi_{1/n} * f$ with Friedrichs mollifiers $0 \leq \phi_{1/n} \in C_c^\infty(\mathbb{C})$ satisfies

$$\int |f - f_n| \,\mathrm{d}\mu \to 0.$$

Proof. Recall $\phi_{1/n}(y) = n^d\phi(ny)$, here $d = 2$. Then

$$\begin{aligned}\int |f - f_n| \,\mathrm{d}\mu &= \int \Big| \int \phi_{1/n}(y)\big(f(x) - f(x-y)\big)\,\mathrm{d}y\Big| \,\mu(\,\mathrm{d}x)\\ &= \int \Big| \int \phi(y)\big(f(x) - f(x-\tfrac{1}{n}y)\big)\,\mathrm{d}y\Big| \,\mu(\,\mathrm{d}x)\\ &\le \int \phi(y) \int \big|f(x) - f(x-\tfrac{1}{n}y)\big| \,\mu(\,\mathrm{d}x)\,\mathrm{d}y.\end{aligned}$$

By the regularity of μ, the translations are L^1-continuous, i.e.,

$$\lim_{n\to\infty} \int \big|f(x) - f(x-\tfrac{1}{n}y)\big| \,\mu(\,\mathrm{d}x) = 0.$$

The result follows from dominated convergence. □

We now define a PV measure $E : \Sigma \to \mathcal{L}(\mathcal{H})$ by

$$E(I) := 1_{I\cap\sigma(A)}(A), \qquad I \in \Sigma.$$

Properties:

(i) $E(I)$ is an orthogonal projection since $\overline{1_{I\cap\sigma(A)}} = 1_{I\cap\sigma(A)} = 1^2_{I\cap\sigma(A)}$.

(ii) $E(\emptyset) = 0$, $E(\mathbb{C}) = E(\sigma(A)) = \mathbb{1}$.

(iii) For $I_1, I_2, \dots \in \Sigma$ mutually disjoint and $x, y \in \mathcal{H}$:

$$\Big\langle E\Big(\biguplus_{j\ge1} I_j\Big)x, y\Big\rangle = \mu_{x,y}\Big(\biguplus_{j\ge1} I_j\Big) = \sum_{j\ge1} \mu_{x,y}(I_j) = \sum_{j\ge1} \langle E(I_j)x, y\rangle\,,$$

since $\mu_{x,y}$ is a complex Borel measure.

Lemma 4.1.5 then implies $E\big(\biguplus_{j\ge1} I_j\big)x = \sum_{j\ge1} E(I_j)x$ for any $x \in \mathcal{H}$. Hence E is a PV measure supported on $\sigma(A)$.

To establish the claimed uniqueness, suppose E' is another spectral measure supported on $\sigma(A)$ with $\int t\,\mathrm{d}E' = A$. Then also $\int \bar{t}\,\mathrm{d}E' = A^*$ and $\int \mathrm{d}E' = \mathbb{1}$. Since by Weierstrass' approximation theorem the functions $f(t) = t, \bar{t}, 1$ generate $C(\sigma(A))$, the identity $\int f\,\mathrm{d}E = \int f\,\mathrm{d}E'$ extends to all $f \in C(\sigma(A))$, i.e.,

$$\int f\,\mathrm{d}E_{x,y} = \int f\,\mathrm{d}E'_{x,y} \qquad \text{for all } f \in C(\sigma(A)),\ x, y \in \mathcal{H},$$

and hence $E_{x,y} = E'_{x,y}$ and $E = E'$.

If C commutes with A, A^*, it commutes with any element in $C^*(A)$. By the spectral theorem for C^*-algebras this implies $Cf(A) = f(A)C$ for any $f \in C(\sigma(A))$ and hence for $x, y \in \mathcal{H}$:

$$\int f\,\mathrm{d}\mu_{x,C^*y} = \langle f(A)x, C^*y\rangle = \langle Cf(A)x, y\rangle = \langle f(A)Cx, y\rangle = \int f\,\mathrm{d}\mu_{Cx,y}.$$

This extends to all $f \in \mathcal{B}_\infty(\sigma(A))$ by the mollifier argument. □

4.4 Case of self-adjoint $A \in \mathcal{L}(\mathcal{H})$

For $A = A^*$ we have $\sigma(A) \subset [-\|A\|, \|A\|] \subset \mathbb{R}$. Spectral measures $E : \Sigma \to \mathcal{L}(\mathcal{H})$ with support in $\mathbb{R}$ can uniquely be characterised in terms of (cf. probability measures and distribution functions)

$$\hat{E} : \mathbb{R} \to \mathcal{L}(\mathcal{H}), \qquad t \mapsto E((-\infty, t]),$$

which satisfies:

(1) $t \leq s \Rightarrow \hat{E}(t) \leq \hat{E}(s)$,

(2) for all $x \in \mathcal{H}$: $\lim_{t\to-\infty} \hat{E}(t)x = 0$ and $\lim_{t\to\infty} \hat{E}(t)x = x$,

(3) for all $x \in \mathcal{H}$: $\lim_{t\downarrow s} \hat{E}(t)x = \hat{E}(s)x$.

Remark 4.4.1: Alternative notions for a PV measure: spectral resolution, spectral measure.

4.5 Multiplication operator form of the spectral theorem

Example 4.5.1 (Multiplication operator by the identity): Let μ be a finite Borel measure on $(\mathbb{C}, \Sigma)$ with compact support $K = \operatorname{supp}\mu \subset \mathbb{C}$ and consider the multiplication operator

$$M_{\mathrm{id}} : L^2(\mathbb{C}, \mu) \to L^2(\mathbb{C}, \mu), \qquad M_{\mathrm{id}}\psi(t) = t\,\psi(t).$$

Then:

(i) $M_{\mathrm{id}} \in \mathcal{L}(L^2(\mathbb{C}, \mu))$ with $\|M_{\mathrm{id}}\| \leq \max_{t\in K} |t|$.

(ii) $M_{\mathrm{id}}^* = M_{\overline{\mathrm{id}}}$, the multiplication operator $M_{\overline{\mathrm{id}}}\psi(t) = \bar{t}\psi(t)$. Hence M_{id} is normal.

(iii) Its PV measure is $E(I) = M_{1_{K\cap I}}$, $I \in \Sigma$, the multiplication operator by the indicator function $1_{K\cap I}$.

Definition 4.5.2: Let $A \in \mathcal{L}(\mathcal{H})$ and $x \in \mathcal{H}$.

(1) The *cyclic subspace* for A generated by x is the smallest closed subspace containing x which is invariant under A and A^*, i.e.,

$$\mathcal{H}_x(A) = \overline{\operatorname{span}\left\{ q(A, A^*)x \;\middle|\; \text{polynomial } q(t,s) = \sum_{j,k=1}^{n} a_{jk} t^j s^k \right\}}.$$

(2) If $\mathcal{H}_x(A) = \mathcal{H}$, then x is a *cyclic vector* for A.

Example 4.5.3: Since the polynomials $q(t, \bar{t})$ are dense in $C(K)$, which in turn is dense in $L^2(K, \mu) \cong L^2(\mathbb{C}, \mu)$, any $x \in L^2(K, \mu)$ with $x \neq 0$ μ-a.e. is cyclic for M_t.

Theorem 4.5.1: Spectral theorem – multiplication operator version I

Let $A \in \mathcal{L}(\mathcal{H})$ be normal and suppose $x \in \mathcal{H}$, $\|x\| = 1$, is cyclic for A. Then there is a Borel probability measure μ_x on $\sigma(A)$ and a unitary $U : \mathcal{H} \to L^2(\mathbb{C}, \mu_x)$ such that

(1) $(UAy)(t) = t\,(Uy)(t)$ and $(UA^*y)(t) = \bar{t}\,(Uy)(t)$, $y \in \mathcal{H}$,

(2) $(Ux)(t) = 1$.

Proof. Since $V = \operatorname{span}\{q(A, A^*)x \mid q \text{ polynomial}\}$ is dense in $\mathcal{H}$ by assumption,

$$\big(U\, q(A, A^*)x\big)(t) := q(t, \bar{t})$$

is a densely defined linear operator $U : V \to L^2(\mathbb{C}, \mu_x)$ with $\mu_x := E_{x,x}$, which satisfies $\mu_x(\mathbb{C}) = \|x\|^2 = 1$ and, by the spectral theorem,

$$\langle q(A, A^*)x, q(A, A^*)x\rangle = \int \big|q(t, \bar{t})\big|^2 \, \mu_x(\,\mathrm{d}t),$$

i.e., U is an isometry. By density of V in $\mathcal{H}$ and of the polynomials in $L^2(\mathbb{C}, \mu_x)$ it uniquely extends to a unitary operator $U : \mathcal{H} \to L^2(\mathbb{C}, \mu_x)$, for which (1) and (2) hold by construction. □

Not every operator possesses a cyclic vector, e.g. $\nVdash : \mathbb{C}^n \to \mathbb{C}^n$. However, the full Hilbert space decomposes into an orthogonal sum of cyclic subspaces. We will prove this in the separable case (general case by Zorn's lemma).

Proposition 4.5.4: Let $A \in \mathcal{L}(\mathcal{H})$ on a separable Hilbert space $\mathcal{H}$. Then there is an orthonormal family $(e_j)_{j\in J}$ with J countable so that

(1) $\mathcal{H}_{e_j}(A) \perp \mathcal{H}_{e_k}(A)$ for all $j \neq k$,

(2) $\mathcal{H} = \bigoplus_{j\in J} \mathcal{H}_{e_j}(A)$.

Proof idea. Consider an orthonormal basis $(f_k)_{k\in\mathbb{N}}$ in $\mathcal{H}$ and set $e_1 := f_1$ and

$$j_n := \min\{j \mid f_j \notin \operatorname{span}\{\mathcal{H}_{e_k}(A) \mid k < n\}\},$$

and set $e_n := \big(f_{j_n} - \sum_{k=1}^{n-1} P_k f_{j_n}\big)/\text{normalisation}$ with P_k the orthogonal projections onto $\mathcal{H}_{e_k}(A)$. □

Theorem 4.5.2: Spectral theorem – multiplication operator version II

Let $A \in \mathcal{L}(\mathcal{H})$ be a normal operator on a separable Hilbert space. Then there are Borel probability measures $(\mu_j)_{j\in J}$ on $\sigma(A)$ and a unitary operator $U : \mathcal{H} \to \bigoplus_{j\in J} L^2(\mathbb{C}, \mu_j)$ with J countable so that for all $y \in \mathcal{H}$

$$(UAy)_j(t) = t\,(Uy)_j(t), \qquad (UA^*y)_j(t) = \bar{t}\,(Uy)_j(t).$$

Corollary 4.5.5: Let $A \in \mathcal{L}(\mathcal{H})$ be normal on a separable Hilbert space. Then there is a finite measure space $(M, \mathcal{A}, \mu)$, a unitary operator $U : \mathcal{H} \to L^2(M, \mu)$ and $a \in L^\infty(M, \mu)$ such that

$$(UAy)(t) = a(t)\,(Uy)(t),$$

i.e., A is unitarily equivalent to the multiplication operator M_a: $M_a U = UA$.

Proof. Set $M := \{1, \dots, |J|\} \times \mathbb{C}$ (with the product topology), $\mathcal{A}$ the σ-algebra of Borel sets, and

$$\mu(\{j\} \times I) := \frac{1}{2^j}\mu_j(I), \qquad j \in \{1, \dots, |J|\},\ I \in \Sigma,$$

with μ_j from the theorem. Then $\mu(M) = \sum_{j=1}^{|J|} \frac{1}{2^j} \underbrace{\mu_j(\mathbb{C})}_{=1} \leq \sum_{j\geq 1} \frac{1}{2^j} < \infty$. Canonical unitary:

$$V : \bigoplus_{j=1}^{|J|} L^2(\mathbb{C}, \mu_j) \to L^2(M, \mu), \qquad (Vf)(j, s) := \sqrt{2^j}\, f_j(s).$$

The claim follows from the previous theorem by composing $U : \mathcal{H} \to \bigoplus_{j=1}^{|J|} L^2(\mathbb{C}, \mu_j)$ with V. □

4.6 Outlook: Uncountable direct sums of Hilbert spaces

Definition 4.6.1: For a (generally uncountable) index set J and a family $(x_j)_{j\in J} \subset X$ in some Banach space X, the series $\sum_{j\in J} x_j$ is said to *converge* to $x \in X$ iff for any $\varepsilon > 0$ there is a finite set $F \subset J$ such that

$$\left\| x - \sum_{j\in K} x_j \right\| < \varepsilon \qquad \text{for all finite } K \subset J,\ F \subset K.$$

$\sum_{j\in J} x_j$ is a *Cauchy sequence* iff for any $\varepsilon > 0$ there is a finite set $F \subset J$ such that for any finite set $K \subset J \setminus F$: $\left\| \sum_{j\in K} x_j \right\| < \varepsilon$.

Remark 4.6.2 (Properties):

(1) $\sum_{j\in J} \|x_j\| = \sup\left\{ \sum_{j\in F} \|x_j\| \mid F \subset J \text{ finite} \right\}$ is finite only if $x_j = 0$ for all but countably many j.

(2) $\sum_{j\in J} x_j$ converges iff it is a Cauchy sequence in X.

Definition 4.6.3: The *direct sum* of a (generally uncountable) family of Hilbert spaces $(\mathcal{H}_j, \langle\cdot,\cdot\rangle_j)$ is the vector space

$$\mathcal{H} := \bigoplus_{j\in J} \mathcal{H}_j := \left\{ x = (x_j)_{j\in J} \;\middle|\; \|x\|^2 := \sum_{j\in J} \|x_j\|_j^2 < \infty \right\}$$

with $x + \alpha y := (x_j + \alpha y_j)_{j\in J}$ for $x, y \in \mathcal{H}$, $\alpha \in \mathbb{C}$.

Remark 4.6.4 (Properties):

(i) $\mathcal{H}$ is a Hilbert space with $\langle x, y\rangle := \sum_{j\in J} \langle x_j, y_j\rangle_j$. For any orthonormal basis $(e_j)_{j\in J}$ of $\mathcal{H}$: $x = \sum_{j\in J} \langle x, e_j\rangle\, e_j$.

(ii) If $A_j \in \mathcal{L}(\mathcal{H}_j)$ for all $j \in J$, then $A := \bigoplus_{j\in J} A_j$ defines via $Ax = (A_j x_j)_{j\in J}$ an operator in $\mathcal{L}(\mathcal{H})$ with $\|A\| = \sup_{j\in J} \|A_j\|$.

The spectral theorem in multiplication operator form (version II) remains valid in case $\mathcal{H}$ is not separable. The same is true for the corollary.

Further reading:

- W. Rudin, *Real and complex analysis*, McGraw-Hill 1987.
- P. Halmos, *A Hilbert space problem book*, Springer 1982.

5 Special classes of operators and their spectra

5.1 Spectra of bounded operators on a Banach space

Let $A \in \mathcal{L}(X)$ be a bounded operator on a Banach space X. Recall:

- $A \in \mathcal{L}(X)$ bijective $\Rightarrow A^{-1} \in \mathcal{L}(X)$ (bounded inverse theorem).
- $A \in \mathcal{L}(X)$ not bijective $\Rightarrow$ (i) A is not injective, or (ii) A is not surjective.

Definition 5.1.1:

(1) $\lambda \in \mathbb{C}$ is an *eigenvalue* of A if $\lambda - A$ is not injective; any $0 \neq x \in \ker(\lambda - A) \neq \{0\}$ is an *eigenvector* of A, i.e., $Ax = \lambda x$. The set of eigenvalues of A (*point spectrum*) is

$$\sigma_p(A) = \{\lambda \in \mathbb{C} \mid \lambda - A \text{ is not injective}\}.$$

(2) *Continuous spectrum* of A:

$$\sigma_c(A) = \{\lambda \in \mathbb{C} \mid \lambda - A \text{ is injective but not surjective and } \overline{\operatorname{ran}(\lambda - A)} = X\}.$$

(3) *Residual spectrum* of A:

$$\sigma_r(A) = \{\lambda \in \mathbb{C} \mid \lambda - A \text{ is injective but not surjective and } \overline{\operatorname{ran}(\lambda - A)} \neq X\}.$$

Remark 5.1.2:

(1) $\sigma(A) = \sigma_p(A) \uplus \sigma_c(A) \uplus \sigma_r(A)$.

(2) For normal $A \in \mathcal{L}(\mathcal{H})$ on a Hilbert space $\mathcal{H}$, $\lambda - A$ is normal and $\ker(\lambda - A) = \big(\overline{\operatorname{ran}(\lambda - A)}\big)^\perp$, i.e., $\sigma_r(A) = \emptyset$. Indeed, $\ker T = (\operatorname{ran} T^*)^\perp = (\overline{\operatorname{ran} T^*})^\perp$ for any T, and $\ker A^* = \ker A$ for normal A since $\|Ax\| = \|A^*x\|$:

$$\langle Ax, Ax\rangle = \langle A^*Ax, x\rangle = \langle AA^*x, x\rangle = \langle A^*x, A^*x\rangle\,.$$

(3) For compact $A \in \mathcal{L}(X)$ on a Banach space X, there is a countable set $\{\lambda_n\} \subset \mathbb{C}$ accumulating at most at 0 such that (Fredholm theory)

$$\sigma(A) = \overline{\{\lambda_n\}}, \qquad \{\lambda_n\} \subset \sigma_p(A) \setminus \{0\}, \quad \dim\ker(\lambda_n - A) < \infty.$$

If $\dim X = \infty$, then $0 \in \sigma(A)$.

Example 5.1.3 (Volterra integral operator): $V_1 : C([0,1]) \to C([0,1])$, $(V_1 f)(t) := \int_0^t f(s)\,\mathrm{d}s$. Recall from Section 1: $\sigma(V_1) = \{0\}$.

- $\sigma_p(V_1) = \emptyset$, since $\ker V_1 = \{0\}$ from calculus.
- $\sigma_c(V_1) = \emptyset$, since $\overline{\operatorname{ran} V_1} \subset \{f \in C([0,1]) \mid f(0) = 0\} \neq C([0,1])$.

Hence $\sigma_r(V_1) = \sigma(V_1) = \{0\}$.

Example 5.1.4 (Shifts on $\ell^2(\mathbb{N}_0)$): Hilbert space $\ell^2(\mathbb{N}_0) = \{(a_n)_{n\geq 0} \mid \sum_{n\geq 0} |a_n|^2 < \infty\}$ with scalar product $\langle a, b\rangle := \sum_{n\geq 0} a_n \overline{b_n}$.

Left shift $(La)_n := a_{n+1}$; $\|L\| \leq 1 \Rightarrow \|\sigma(L)\| \leq 1$.

Eigenvalues of L: $(La)_n = \lambda a_n$ with $|\lambda| \leq 1$ (i.e., $a_1 = \lambda a_0$, $a_2 = \lambda a_1, \dots$) $\iff$ $a = a_0(1, \lambda, \lambda^2, \dots)$, and $|a_0|^2 \sum_{n=0}^\infty |\lambda|^{2n} < \infty$. Since $a \in \ell^2(\mathbb{N}_0)$ iff $|\lambda| < 1$: $\sigma_p(L) = \{\lambda \in \mathbb{C} \mid |\lambda| < 1\}$, hence $\sigma(L) = \overline{B_1(0)}$.

Right shift $(Ra)_n := \begin{cases} 0, & n = 0 \\ a_{n-1}, & n \geq 1. \end{cases}$ $R = L^*$, since $\langle La, b\rangle = \sum_{n\geq 0} a_{n+1}\overline{b_n} = \sum_{n\geq 1} a_n \overline{b_{n-1}} = \langle a, Rb\rangle$. Hence

$$\sigma(R) = \{\overline{\lambda} \mid \lambda \in \sigma(R^*)\} = \sigma(L).$$

$\sigma_p(R) = \emptyset$ since $LR = \not\Vdash$, i.e., $0 \notin \sigma_p(R)$, and for $|\lambda| > 0$: if $\lambda a_n = (Ra)_n$ for all $n \geq 0$ then $a = 0$.

- For any $\lambda \in \mathbb{C}$: $\overline{\operatorname{ran}(\lambda - L)} = \ker(\overline{\lambda} - R)^\perp = \{0\}^\perp = \ell^2(\mathbb{N}_0)$, i.e., $\sigma_r(L) = \emptyset \Rightarrow \sigma_c(L) = S^1$.

- For $|\lambda| = 1$: $\overline{\operatorname{ran}(\lambda - R)} = \ker(\overline{\lambda} - L)^\perp = \ell^2(\mathbb{N}_0)$, so $\sigma_r(R) \subset \{|\lambda| < 1\}$.
- For $|\lambda| < 1$: $\overline{\operatorname{ran}(\lambda - R)} = \ker(\overline{\lambda} - L)^\perp \neq \ell^2(\mathbb{N}_0) \Rightarrow \sigma_r(R) = \{|\lambda| < 1\}$.

Example 5.1.5 (Translations on the torus): Hilbert space $L^2(S^1)$; translations by $\alpha \in S^1$: $(T_\alpha \psi)(x) = \psi((x - \alpha) \bmod 2\pi)$. Note: $T_\alpha T_\beta = T_{\alpha+\beta}$, $T_0 = \mathbb{1}$, $T_\alpha^* = T_{-\alpha} = T_\alpha^{-1}$ (unitary group). Orthonormal basis $e_n(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}nx}$, $n \in \mathbb{Z}$. The Fourier transformation $\mathcal{F} : L^2(S^1) \to \ell^2(\mathbb{Z})$, $(\mathcal{F}\varphi)(n) := \langle \varphi, e_n \rangle$, is a unitary operator, and (Exercise!)

$$\mathcal{F} T_\alpha \mathcal{F}^{-1} : \ell^2(\mathbb{Z}) \to \ell^2(\mathbb{Z}), \qquad (\mathcal{F} T_\alpha \mathcal{F}^{-1} a)_n = \mathrm{e}^{-\mathrm{i}\alpha n} a_n.$$

Hence (spectra are invariant under unitaries!)

$$\sigma_p(T_\alpha) = \sigma_p(\mathcal{F} T_\alpha \mathcal{F}^{-1}) = \{\mathrm{e}^{-\mathrm{i}\alpha n} \mid n \in \mathbb{Z}\}.$$

- Case $\alpha = \frac{p}{q} 2\pi$ with $p, q \in \mathbb{N}$: any $\lambda \notin \sigma_p(T_\alpha)$ has positive distance from $\sigma_p(T_\alpha) \Rightarrow$ (Exercise) $\sigma(T_\alpha) = \sigma_p(T_\alpha)$.
- Case α irrational (multiple of 2π): $\overline{\sigma_p(T_\alpha)} = S^1 = \sigma(T_\alpha)$.

5.2 Polar decomposition and singular values

Set-up in the following: $\mathcal{H}$ Hilbert space. Recall that for any $A \in \mathcal{L}(\mathcal{H})$ the operator A^*A is positive and hence $|A| := \sqrt{A^*A} \geq 0$ is well defined.

Definition 5.2.1: $V \in \mathcal{L}(\mathcal{H})$ is called a *partial isometry* iff $\|Vx\| = \|x\|$ for all $x \in (\ker V)^\perp$.

Lemma 5.2.2 (Polar decomposition): For every $A \in \mathcal{L}(\mathcal{H})$ there is a partial isometry $U \in \mathcal{L}(\mathcal{H})$ such that $A = U|A|$.

Proof. For all $x \in \mathcal{H}$:

$$\|Ax\|^2 = \langle A^*Ax, x \rangle = \||A|\, x\|^2 . \qquad (*)$$

Hence $U : \operatorname{ran}|A| \to \operatorname{ran} A$, $U|A|\, x := Ax$, is an isometry, which uniquely extends to an isometry $U : \overline{\operatorname{ran}|A|} \to \overline{\operatorname{ran} A}$. By setting $Ux = 0$ on $(\operatorname{ran}|A|)^\perp = \ker|A| \overset{(*)}{=} \ker A$, we obtain a partial isometry with the desired property. □

Remark 5.2.3: The partial isometry U is uniquely determined.

Consider now $A \in \mathcal{K}(\mathcal{H})$, the set of compact operators. Then $T := A^*A \geq 0$ is compact and positive. By the spectral theorem there are orthonormal $(\varphi_n) \subset \mathcal{H}$ and $(\nu_n) \subset (0, \infty)$ with

$$Tx = A^*Ax = \sum_n \nu_n \langle x, \varphi_n \rangle \varphi_n. \qquad (*)$$

Definition 5.2.4: For $A \in \mathcal{K}(\mathcal{H})$ we call $s_n := \sqrt{\nu_n}$, with (ν_n) the non-zero eigenvalues of A^*A, the *singular values* of A.

Theorem 5.2.1: Singular value decomposition

For $A \in \mathcal{K}(\mathcal{H})$ there is an orthonormal set $(\varphi_n) \subset \mathcal{H}$ such that for any $x \in \mathcal{H}$:

$$Ax = \sum_n s_n \langle x, \varphi_n \rangle \psi_n,$$

where (s_n) are the singular values of A and $\psi_n := \frac{1}{s_n} A\varphi_n$ forms an orthonormal set. Moreover, $\|A\| = \max_n s_n$.

Proof. Decompose $\mathcal{H} = E \oplus E^\perp$, $E^\perp := \ker A^*A = \ker A$ (since $\|Ax\|^2 = \langle A^*Ax, x\rangle$). By the spectral decomposition $(*)$, (φ_n) is an ONB of E, so for any $x \in \mathcal{H}$: $x = P_{E^\perp}x + \sum_n \langle x, \varphi_n\rangle \varphi_n$, with $P_{E^\perp}$ the orthogonal projection onto $E^\perp$. Hence

$$Ax = \sum_n \langle x, \varphi_n\rangle A\varphi_n = \sum_n s_n \langle x, \varphi_n\rangle \psi_n$$

by definition of $\psi_n = \frac{1}{s_n} A\varphi_n$.

- Orthonormality of (ψ_n): using $A^*A\varphi_n = s_n^2\varphi_n$ by $(*)$,

$$\langle \psi_n, \psi_m\rangle = \frac{1}{s_n s_m}\langle A\varphi_n, A\varphi_m\rangle = \frac{1}{s_n s_m}\langle A^*A\varphi_n, \varphi_m\rangle = \frac{s_n}{s_m}\langle \varphi_n, \varphi_m\rangle = \delta_{n,m}.$$

- $(\max_n s_n)^2 = \max_n s_n^2 = \|\sigma(A^*A)\| = \|A^*A\| = \|A\|^2$ by the C^*-identity. □

Example 5.2.5 (Inverse differential operator): Let

$$L : C_c^2((0,2\pi)) \to L^2((0,2\pi)), \qquad Lu(x) := -u''(x) + q(x)u(x),$$

and assume $q \in C([0,2\pi])$, $q \geq 1$. For any $f \in C([0,2\pi])$ the boundary value problem

$$Lu = f \qquad \text{with} \quad u(0) = u(2\pi) = 0$$

has a unique solution $u =: Af \in C^2([0,2\pi])$.

(a) A extends to a compact operator on $L^2((0,2\pi))$.

Proof. For $u \in C^2([0,2\pi])$ with $u(0) = u(2\pi) = 0$, integrating by parts and using Cauchy–Schwarz,

$$\int_0^{2\pi} |u'|^2 + q|u|^2 \,\mathrm{d}x \overset{\text{P.I.}}{=} \int_0^{2\pi} \overline{u}\, Lu \,\mathrm{d}x = \int_0^{2\pi} \overline{u} f \,\mathrm{d}x$$

$$\leq \Big(\int_0^{2\pi} |u|^2 \,\mathrm{d}x \int_0^{2\pi} |f|^2 \,\mathrm{d}x\Big)^{1/2} \leq \frac{1}{2}\int_0^{2\pi} |u|^2 \,\mathrm{d}x + \frac{1}{2}\int_0^{2\pi} |f|^2 \,\mathrm{d}x.$$

Since $q \geq 1$,

$$\frac{1}{2}\int_0^{2\pi} |u|^2 \,\mathrm{d}x \leq \int_0^{2\pi} |u'|^2 + \frac{1}{2}|u|^2 \,\mathrm{d}x \leq \frac{1}{2}\int_0^{2\pi} |f|^2 \,\mathrm{d}x.$$

Apply this to $u = Af$ to conclude:

- By density of $C_c^\infty((0,2\pi))$ in $L^2((0,2\pi))$ and the BLT theorem, A extends uniquely to a bounded operator $A : L^2((0,2\pi)) \to L^2((0,2\pi))$.
- $A(B_1(0))$ is bounded in $W^{1,2}((0,2\pi)) \Rightarrow A$ is compact by the Rellich criterion. □

(b) $A = A^* \geq 0$.

Proof. For $u, v \in C^2([0,2\pi])$ with $u(0) = v(0) = u(2\pi) = v(2\pi) = 0$:

$$\langle Lu, v\rangle = \int_0^{2\pi} \overline{v}(-u'' + qu)\,\mathrm{d}x \overset{\text{P.I.}}{=} \big[-\overline{v}u'\big]_0^{2\pi} + \int_0^{2\pi} \overline{v}'u' + q\overline{v}u \,\mathrm{d}x \overset{\text{P.I.}}{=} \langle u, Lv\rangle.$$

Apply this to $u = Af$, $v = Ag$ with $f, g \in C([0,2\pi])$:

$$\langle f, Ag\rangle = \langle Af, g\rangle, \qquad \langle Af, f\rangle = \langle u, Lu\rangle \overset{\text{P.I.}}{=} \int_0^{2\pi} |u'|^2 + q|u|^2 \,\mathrm{d}x \geq 0.$$

Since $C_c^\infty((0,2\pi))$ is dense in $L^2((0,2\pi))$ and A is bounded, this extends to all $f, g \in L^2((0,2\pi))$. □

(c) $0 \notin \sigma_p(A)$. Proof: Exercise!

Intermezzo: Sobolev spaces and compactness. For $S \subset \mathbb{R}^n$ open, $p \in [1,\infty]$, $u \in L^p(S)$, $\alpha \in \mathbb{N}_0^n$, one calls $w \in L^p(S)$ the *weak α-th derivative* of u if

$$\forall \varphi \in C_c^\infty(S): \quad \int_S w\varphi \,\mathrm{d}x = (-1)^{|\alpha|} \int_S u\, \partial^\alpha \varphi \,\mathrm{d}x.$$

Notation: $w = \partial^\alpha u$.

Definition 5.2.6: *Sobolev space* for $k \in \mathbb{N}$, $p \in [1,\infty)$:

$$W^{k,p}(S) := \{u \in L^p(S) \mid \partial^\alpha u \in L^p(S)\ \forall\, |\alpha| \le k\}$$

with the *Sobolev norm* $\|u\|_{W^{k,p}} := \left(\sum_{|\alpha|\le k} \|\partial^\alpha u\|_p^p\right)^{1/p}$, and $W_0^{k,p}(S) := \overline{C_c^\infty(S)}^{\|\cdot\|_{W^{k,p}}} \subset W^{k,p}(S)$.

$(W^{k,p}(S), \|\cdot\|_{W^{k,p}})$ is a Banach space.

Theorem 5.2.2: Rellich

For $S \subset \mathbb{R}^n$ an open, bounded Lipschitz domain and $p \in [1,\infty)$, any bounded $F \subset W^{1,p}(S)$ is relatively compact in $L^p(S)$, i.e., $\overline{F}^{\|\cdot\|_p}$ is compact.

Application: $F = A(B_1(0)) \subset W^{1,2}((0,2\pi))$ with $\|u\|_{W^{1,2}}^2 = \|u\|^2 + \|u'\|^2 = \int_0^{2\pi} |u|^2 + |u'|^2 \,\mathrm{d}x$.

By the spectral theorem there are an ONB $(\varphi_n) \subset L^2([0,2\pi])$ and $\lambda_1 > \lambda_2 > \cdots > 0$ with $A\varphi_n = \lambda_n \varphi_n$.
Physics question: quantum particle in a well with potential q with Dirichlet boundary conditions. The energy levels of $L = A^{-1}$ are given by λ_n^{-1}, $n \in \mathbb{N}$.

5.3 Min–max principle

Question: How to approximately calculate the singular values of $A \in \mathcal{K}(\mathcal{H})$?

Theorem 5.3.1: Courant–Fischer min-max variational principle

Let $A \in \mathcal{K}(\mathcal{H})$ and $s_1 \ge s_2 \ge \cdots \ge s_n \ge \ldots$ its ordered sequence of singular values. Then

$$\begin{aligned} s_1 &= \|A\| = \sup\{\|Ax\| \mid \|x\| = 1\}, \\ s_{n+1} &= \inf_{x_1,\ldots,x_n \in \mathcal{H}} \sup\{\|Ax\| \mid \|x\| = 1 \text{ and } x \perp \operatorname{span}\{x_1,\ldots,x_n\}\}. \end{aligned}$$

Proof. Recall $Ax = \sum_n s_n \langle x, \varphi_n\rangle \psi_n$ with (ψ_n) orthonormal. For $x \perp \operatorname{span}\{\varphi_1,\ldots,\varphi_n\}$:

$$\|Ax\|^2 = \sum_k s_k^2 |\langle x,\varphi_k\rangle|^2 \|\psi_k\|^2 = \sum_{k\ge n+1} s_k^2 |\langle x,\varphi_k\rangle|^2 \le s_{n+1}^2 \sum_{k\ge n+1} |\langle x,\varphi_k\rangle|^2 \le s_{n+1}^2 \|x\|^2.$$

Hence $s_{n+1} \ge$ r.h.s. of the claim.

Let $x_1,\ldots,x_n \in \mathcal{H}$ be arbitrary and pick $x \in \operatorname{span}\{\varphi_1,\ldots,\varphi_{n+1}\}$ with $\|x\| = 1$ and $x \perp \operatorname{span}\{x_1,\ldots,x_n\}$:

$$\|Ax\|^2 = \left\| \sum_{k=1}^{n+1} s_k \langle x,\varphi_k\rangle \psi_k \right\|^2 = \sum_{k=1}^{n+1} s_k^2 |\langle x,\varphi_k\rangle|^2 \ge s_{n+1}^2 \sum_{k=1}^{n+1} |\langle x,\varphi_k\rangle|^2 = s_{n+1}^2 \|x\|^2.$$

Hence $s_{n+1} \le$ r.h.s. of the claim. □

Alternative formulation and eigenvalue order:

Corollary 5.3.1:

(1) If $0 \leq A \in \mathcal{K}(\mathcal{H})$, then its ordered sequence $0 < \cdots \leq \alpha_n \leq \cdots \leq \alpha_2 \leq \alpha_1$ of positive eigenvalues satisfies

$$\alpha_1 = \sup\{\|Ax\| \mid \|x\| = 1\},$$
$$\alpha_{n+1} = \inf_{x_1,\ldots,x_n \in \mathcal{H}} \sup\{\|Ax\| \mid \|x\| = 1 \text{ and } x \perp \operatorname{span}\{x_1, \ldots, x_n\}\}.$$

(2) If $A, B \in \mathcal{K}(\mathcal{H})$ are self-adjoint with $A \leq B$, then their ordered sequences of eigenvalues satisfy $\alpha_j \leq \beta_j$ for all j.

Proof. Exercise. □

6 Trace class, trace formula and Hilbert–Schmidt operators

In the following: $\mathcal{K}(\mathcal{H})$ denotes the compact operators on a Hilbert space $\mathcal{H}$.

6.1 Trace class operators

Definition 6.1.1: $A \in \mathcal{K}(\mathcal{H})$ is called *trace class* iff its singular values (s_j) are summable, $\sum_j s_j < \infty$.
Notation: $\mathcal{S}_1(\mathcal{H}) := \{A \in \mathcal{K}(\mathcal{H}) \mid A \text{ trace class}\}$. In this case $\|A\|_1 := \sum_j s_j$ is the *trace norm*.

Theorem 6.1.1: Trace norm properties

Let $A, A' \in \mathcal{S}_1(\mathcal{H})$, $B \in \mathcal{L}(\mathcal{H})$.

(1) $\|A\|_1 = \|A^*\|_1 \geq \|A\|$.

(2) $\|BA\|_1 \leq \|B\| \, \|A\|_1$, $\|AB\|_1 \leq \|A\|_1 \, \|B\|$.

(3) $\|A + A'\|_1 \leq \|A\|_1 + \|A'\|_1$ (triangle inequality).

Proof. (1) $\sigma(A^*A) \setminus \{0\} = \sigma(AA^*) \setminus \{0\}$, cf. Lemma 3.3.4. Hence $s_j(A^*) = s_j(A)$ for all j, and so $\|A^*\|_1 = \|A\|_1 \geq \max_j s_j(A) = \|A\|$ (singular value decomposition).

(2) For all $x \in \mathcal{H}$: $\|BAx\| \leq \|B\| \, \|Ax\|$. From the min-max theorem: $s_j(BA) \leq \|B\| \, s_j(A)$ for all j. Also $s_j(AB) = s_j(B^*A^*) \leq \|B^*\| \, s_j(A^*) = \|B\| \, s_j(A)$.

(3) From the lemma below, since for $(e_j), (f_j)$ orthonormal in $\mathcal{H}$: $|\langle (A + A')e_j, f_j\rangle| \leq |\langle Ae_j, f_j\rangle| + |\langle A'e_j, f_j\rangle|$. □

Lemma 6.1.2: For any $A \in \mathcal{S}_1(\mathcal{H})$:

$$\|A\|_1 = \sup\Big\{ \sum_j |\langle Ae_j, f_j\rangle| \;\Big|\; (e_j), (f_j) \text{ orthonormal in } \mathcal{H} \Big\}.$$

Proof. (1) Pick $e_j = \varphi_j$, $f_j = \psi_j$ from the singular value decomposition of A: $\langle A\varphi_j, \psi_j\rangle = s_j$. Hence $\|A\|_1 \leq$ r.h.s.

(2) For any orthonormal $(e_j), (f_j)$: $\langle Ae_j, f_j\rangle = \sum_k s_k \langle e_j, \varphi_k\rangle \langle \psi_k, f_j\rangle$. Hence, by Cauchy–Schwarz and Bessel,

$$\sum_j |\langle Ae_j, f_j\rangle| \leq \sum_k s_k \sum_j |\langle e_j, \varphi_k\rangle| \, |\langle \psi_k, f_j\rangle|$$
$$\leq \sum_k s_k \Big(\underbrace{\sum_j |\langle e_j, \varphi_k\rangle|^2}_{\leq \|\varphi_k\|^2 = 1} \underbrace{\sum_j |\langle \psi_k, f_j\rangle|^2}_{\leq \|\psi_k\|^2 = 1} \Big)^{1/2} \leq \|A\|_1 .$$ □

Corollary 6.1.3: Let $\mathcal{H}$ be separable, (e_n) an ONB and $A \in \mathcal{S}_1(\mathcal{H})$. Then $\sum_n \langle Ae_n, e_n\rangle$ converges absolutely to a limit which is independent of the chosen ONB.

Proof. Pick $f_j = e_j$ in the lemma to conclude absolute convergence: $\sum_j |\langle Ae_j, e_j\rangle| \le \|A\|_1$. Let (f_n) be some ONB in $\mathcal{H}$. Using the singular value decomposition, Fubini and Parseval,

$$\sum_n \langle Af_n, f_n\rangle = \sum_n \sum_j s_j \langle f_n, \varphi_j\rangle \langle \psi_j, f_n\rangle = \sum_j s_j \sum_n \langle \psi_j, f_n\rangle \langle f_n, \varphi_j\rangle = \sum_j s_j \langle \psi_j, \varphi_j\rangle ,$$

which is clearly basis independent. □

Definition 6.1.4: For any $A \in \mathcal{S}_1(\mathcal{H})$ on a separable Hilbert space $\mathcal{H}$ with ONB (e_n),

$$\operatorname{tr} A := \sum_n \langle Ae_n, e_n\rangle$$

is called the *trace* of A.

Theorem 6.1.2: Properties of the trace

On a separable Hilbert space $\mathcal{H}$, $\mathcal{S}_1(\mathcal{H}) \ni A \mapsto \operatorname{tr} A$ is a linear map with the following properties:

(1) $|\operatorname{tr} A| \le \|A\|_1$,

(2) $\operatorname{tr} A^* = \overline{\operatorname{tr} A}$,

(3) $\operatorname{tr} AB = \operatorname{tr} BA$, $A \in \mathcal{S}_1(\mathcal{H})$, $B \in \mathcal{L}(\mathcal{H})$.

Proof. Exercise. □

Theorem 6.1.3: Lidskii

On any separable Hilbert space

$$\operatorname{tr} A = \sum_{\lambda \in \sigma_p(A)} \lambda \qquad \text{for any } A \in \mathcal{S}_1(\mathcal{H})$$

(eigenvalues counted with algebraic multiplicity).

Proof only for normal A. The spectral theorem guarantees the existence of an ONB (φ_n) and $(\alpha_n) \subset \mathbb{C}$ with $A\varphi_n = \alpha_n \varphi_n$. Hence $\operatorname{tr} A = \sum_n \langle A\varphi_n, \varphi_n\rangle = \sum_n \alpha_n$. □

Remark 6.1.5: See P. Lax, Ch. 30 for the general case.

6.2 Mercer's theorem

Example 6.2.1: Let μ be a finite Borel measure on a compact $S \subset \mathbb{R}^n$, $\mu(S) < \infty$, $k : S \times S \to \mathbb{C}$ a continuous kernel, and

$$K : L^2(S,\mu) \to L^2(S,\mu), \qquad K\psi(t) := \int k(t,s)\psi(s)\,\mu(\mathrm{d}s)$$

the integral operator. K is compact (see e.g. the Hilbert–Schmidt example below) and $KL^2(S,\mu) \subset C(S)$, since for all $\psi \in L^2(S,\mu)$, $t, t' \in S$:

$$\left|K\psi(t) - K\psi(t')\right| \le \int \left|k(t,s) - k(t',s)\right| \, |\psi(s)| \, \mu(\mathrm{d}s).$$

Continuity follows by application of the dominated convergence theorem from that of k.

Special case: $\overline{k(s,t)} = k(t,s) \Rightarrow K = K^*$. From the spectral theorem for compact self-adjoint operators:

$$K\psi = \sum_j \alpha_j \langle \psi, \varphi_j \rangle \varphi_j \qquad \text{with an ONB } (\varphi_j) \subset L^2(S,\mu) \text{ and } (\alpha_j) \subset \mathbb{R}.$$

Since $\alpha_j \varphi_j = K\varphi_j$, each $\varphi_j \in C(S)$ with $\alpha_j \neq 0$.

Theorem 6.2.1: Mercer's expansion and trace formula

In the setting of the example, additionally assume $K \geq 0$. Then

(1) $k(s,t) = \sum_j \alpha_j \varphi_j(s)\overline{\varphi_j(t)}$ for all $s,t \in S$, and the convergence is uniform on $S \times S$;

(2) $K \in \mathcal{S}_1(L^2(S,\mu))$ and $\operatorname{tr} K = \int_S k(s,s)\,\mu(\mathrm{d}s)$.

Proof. (1) Pick $N \in \mathbb{N}$ and set $k_N(s,t) := \sum_{j=1}^N \alpha_j \varphi_j(s)\overline{\varphi_j(t)}$ and $K_N : L^2(S,\mu) \to L^2(S,\mu)$, $(K_N\psi)(s) = \int k_N(s,t)\psi(t)\,\mu(\mathrm{d}t)$. Then

$$\langle (K - K_N)\psi, \psi \rangle = \sum_{j \geq N+1} \alpha_j \left| \langle \varphi_j, \psi \rangle \right|^2 \geq 0.$$

Also $\langle (K-K_N)\psi,\psi\rangle = \iint \big(k(s,t) - k_N(s,t)\big)\psi(t)\overline{\psi(s)}\,\mu(\mathrm{d}t)\,\mu(\mathrm{d}s)$. Concentrating ψ near $s \in S$ hence yields for all $s \in S$:

$$0 \leq k(s,s) - k_N(s,s) = k(s,s) - \sum_{j=1}^N \alpha_j \left|\varphi_j(s)\right|^2.$$

Hence $\lim_{N\to\infty} \sum_{j=1}^N \alpha_j \left|\varphi_j(s)\right|^2 < \infty$ exists, and the convergence is uniform on S by Dini's theorem. By Cauchy–Schwarz,

$$k_\infty(s,t) := \sum_{j \geq 1} \alpha_j \varphi_j(s)\overline{\varphi_j(t)}$$

exists for all $s,t \in S$ and the convergence is uniform on $S \times S$. For all $s \in S$ and k, since (φ_k) is an ONB,

$$\int k_\infty(s,t)\varphi_k(t)\,\mu(\mathrm{d}t) = \sum_{j\geq 1} \alpha_j \varphi_j(s) \int \overline{\varphi_j(t)}\varphi_k(t)\,\mu(\mathrm{d}t) = \alpha_k \varphi_k(s) = \int k(s,t)\varphi_k(t)\,\mu(\mathrm{d}t),$$

hence $\int \big(k_\infty(s,t) - k(s,t)\big)\psi(t)\,\mu(\mathrm{d}t) = 0$ for all $\psi \in L^2(S,\mu)$. Using continuity this implies $k_\infty = k$.

(2) By uniform convergence,

$$\int k(s,s)\,\mu(\mathrm{d}s) = \sum_j \alpha_j \int |\varphi_j(s)|^2\,\mu(\mathrm{d}s) = \sum_j \alpha_j.$$

The singular values of $K \geq 0$ agree with the $\alpha_j > 0$. Hence $K \in \mathcal{S}_1(L^2(S,\mu))$ and $\|K\|_1 = \operatorname{tr} K = \sum_j \alpha_j$. □

6.3 Hilbert–Schmidt operators

Definition 6.3.1: $A \in \mathcal{K}(\mathcal{H})$ is called a *Hilbert–Schmidt operator* iff its singular values (s_j) satisfy $\sum_j s_j^2 < \infty$.
Notation: $\mathcal{S}_2(\mathcal{H}) := \{A \in \mathcal{K}(\mathcal{H}) \mid A \text{ Hilbert–Schmidt}\}$; *Hilbert–Schmidt norm* $\|A\|_2 := \left(\sum_j s_j^2\right)^{1/2}$.

Theorem 6.3.1: HS norm properties

Let $A, A' \in \mathcal{S}_2(\mathcal{H})$, $B \in \mathcal{L}(\mathcal{H})$.

(1) $\|A\|_2 = \|A^*\|_2 \geq \|A\|$.

(2) $\|BA\|_2 \leq \|B\| \, \|A\|_2$, $\|AB\|_2 \leq \|A\|_2 \, \|B\|$.

(3) $\|A + A'\|_2 \leq \|A\|_2 + \|A'\|_2$ (triangle inequality).

Proof. *Exercise.* □

Theorem 6.3.2: Characterisation of HS operators

$A \in \mathcal{L}(\mathcal{H})$ is a Hilbert–Schmidt operator on a separable Hilbert space $\mathcal{H}$ iff for all orthonormal (e_j)

$$\sum_j \|Ae_j\|^2 < \infty.$$

In that case $\|A\|_2^2 = \sup\left\{\sum_j \|Ae_j\|^2 \mid (e_j) \text{ orthonormal}\right\}$.

Proof. "⇒": Singular value decomposition $Ax = \sum_k s_k \langle x, \varphi_k\rangle \, \psi_k$. By Tonelli and Bessel,

$$\sum_j \|Ae_j\|^2 = \sum_j \sum_k s_k^2 \, |\langle e_j, \varphi_k\rangle|^2 = \sum_k s_k^2 \sum_j |\langle e_j, \varphi_k\rangle|^2 \leq \sum_k s_k^2 \, \|\varphi_k\|^2 = \sum_k s_k^2 = \|A\|_2^2 .$$

"⇐": By the lemma below, $A \in \mathcal{K}(\mathcal{H})$. Consider (φ_j) from the singular value decomposition:

$$\sum_j \|A\varphi_j\|^2 = \sum_j s_j^2 \, \|\psi_j\|^2 = \sum_j s_j^2 = \|A\|_2^2 . \qquad \square$$

Lemma 6.3.2 (Recall from Functional Analysis): The following statements are equivalent for $A \in \mathcal{L}(\mathcal{H})$:

(1) $A \in \mathcal{K}(\mathcal{H})$,

(2) $Ae_n \to 0$ for all orthonormal $(e_n) \subset \mathcal{H}$,

(3) $\langle Ae_n, f_n\rangle \to 0$ for all orthonormal $(e_n), (f_n) \subset \mathcal{H}$.

Proof. *Exercise.* □

Theorem 6.3.3

(1) $\mathcal{S}_1(\mathcal{H}) \subset \mathcal{S}_2(\mathcal{H})$.

(2) $A, B \in \mathcal{S}_2(\mathcal{H}) \Rightarrow AB \in \mathcal{S}_1(\mathcal{H})$ with $\|AB\|_1 \leq \|A\|_2 \, \|B\|_2$.

Proof. (1) Follows from $\sum_j s_j^2 \le \big(\sum_j s_j\big)^2$.

(2) Let $(e_j), (f_j)$ be orthonormal in $\mathcal{H}$. By Cauchy–Schwarz,

$$\sum_j |\langle ABe_j, f_j\rangle| = \sum_j |\langle Be_j, A^* f_j\rangle| \le \sum_j \|Be_j\|\ \|A^* f_j\|$$

$$\le \sqrt{\sum_j \|Be_j\|^2}\sqrt{\sum_j \|A^* f_j\|^2} \le \sqrt{\sum_j s_j(B)^2}\sqrt{\sum_j s_j(A^*)^2} = \|B\|_2\,\|A\|_2\,,$$

using $\|A^*\|_2 = \|A\|_2$ and, by Bessel's inequality, $\sum_j \|Be_j\|^2 = \sum_j \sum_k s_k^2\,|\langle e_j, \varphi_k\rangle|^2 \le \sum_k s_k^2\,\|\varphi_k\|^2 = \sum_k s_k^2$. The claim follows from Lemma 6.1.2. □

Example 6.3.3: Let μ be a regular Borel measure on a Borel set $S \subset \mathbb{R}^n$, $k \in L^2(S\times S, \mu\otimes\mu)$ a *Hilbert–Schmidt kernel*, and

$$K : L^2(S,\mu) \to L^2(S,\mu), \qquad K\psi(t) := \int k(t,s)\psi(s)\,\mu(\,\mathrm{d}s).$$

Then $K \in \mathcal{S}_2(L^2(S,\mu))$, since for any ONB (e_j), by Tonelli and Parseval,

$$\sum_j \|Ke_j\|^2 = \sum_j \int \Big|\int k(t,s)e_j(s)\,\mu(\,\mathrm{d}s)\Big|^2 \mu(\,\mathrm{d}t) = \int \sum_j \Big|\Big\langle e_j, \overline{k(t,\cdot)}\Big\rangle\Big|^2 \mu(\,\mathrm{d}t)$$

$$= \int \|k(t,\cdot)\|_2^2\ \mu(\,\mathrm{d}t) = \iint |k(t,s)|^2\ \mu\otimes\mu(\,\mathrm{d}t\ \mathrm{d}s),$$

and hence $\|K\|_2 = \|k\|_2$ by the characterisation of HS operators.

Theorem 6.3.4

(1) $\mathcal{S}_1(\mathcal{H})$ is a Banach space.

(2) For separable $\mathcal{H}$, $\mathcal{S}_2(\mathcal{H})$ is a Hilbert space with the *Hilbert–Schmidt scalar product* $\langle A, B\rangle := \mathrm{tr}(AB^*)$.

Proof. (1) *Exercise.*

(2) $\langle A, A\rangle = \mathrm{tr}\,AA^* = \sum_j s_j(A)^2 = \|A\|_2^2$ (Lidskii, $AA^* \ge 0$). $\langle\cdot,\cdot\rangle$ is evidently linear in the first variable and $\overline{\langle A, B\rangle} = \langle B, A\rangle$, i.e., $\langle\cdot,\cdot\rangle$ is a scalar product for $\|\cdot\|_2$.

For a proof of completeness of $\mathcal{S}_2(\mathcal{H})$, let $(A_n) \subset \mathcal{S}_2(\mathcal{H})$ be a Cauchy sequence, i.e., for all $\varepsilon > 0$ there is $N_\varepsilon \in \mathbb{N}$ such that

$$\|A_n - A_m\|_2 \le \varepsilon \qquad \text{for all } n, m \ge N_\varepsilon. \tag{$*$}$$

Since $\|\cdot\| \le \|\cdot\|_2$, (A_n) is also Cauchy w.r.t. $\|\cdot\|$. By completeness of $\mathcal{L}(\mathcal{H})$ there is $A \in \mathcal{L}(\mathcal{H})$ with $\lim_n \|A - A_n\| = 0$. From the Exercise on Weyl's inequality this implies $A \in \mathcal{K}(\mathcal{H})$ and $\lim_n s_j(A_n) = s_j(A)$ for all j, and hence for all $N \in \mathbb{N}$:

$$\sum_{j=1}^N s_j(A)^2 = \lim_{m\to\infty} \sum_{j=1}^N s_j(A_m)^2 \le \sup_m \|A_m\|_2^2 < \infty.$$

Hence $A \in \mathcal{S}_2(\mathcal{H})$, and for all $M \in \mathbb{N}$ and $n \ge N_\varepsilon$, by $(*)$:

$$\sum_{j=1}^M s_j(A - A_n)^2 = \lim_{m\to\infty} \sum_{j=1}^M s_j(A_m - A_n)^2 \le \varepsilon^2,$$

i.e., $\|A - A_n\|_2 \le \varepsilon$ for $n \ge N_\varepsilon$. □

Remark 6.3.4: The proof of completeness is verbatim the same for the other Schatten classes $\mathcal{S}_p(\mathcal{H})$.

6.4 Outlook: Schatten classes

Definition 6.4.1: $A \in \mathcal{K}(\mathcal{H})$ is called a *Schatten-p operator* with $p \in [1,\infty)$ if $\sum_j s_j^p < \infty$.
Notation: Schatten-p class $\mathcal{S}_p(\mathcal{H}) = \{A \in \mathcal{K}(\mathcal{H}) \mid A \text{ Schatten-}p\}$, Schatten-$p$ norm $\|A\|_p := \left(\sum_j s_j^p\right)^{1/p}$.

Remark 6.4.2 (Properties):

(1) $\|A + A'\|_p \leq \|A\|_p + \|A'\|_p$.

(2) $\|A\|_p \geq \|A\|$.

(3) $p \geq q \Rightarrow \|A\|_p \leq \|A\|_q$, i.e., $\mathcal{S}_q(\mathcal{H}) \subset \mathcal{S}_p(\mathcal{H})$.

(4) $A \in \mathcal{S}_p(\mathcal{H})$, $B \in \mathcal{S}_q(\mathcal{H})$, $\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow AB \in \mathcal{S}_1(\mathcal{H})$ with $\|AB\|_1 \leq \|A\|_p \|B\|_q$ (Hölder inequality).

(5) Dual spaces: $\mathcal{S}_p(\mathcal{H})^* = \mathcal{S}_q(\mathcal{H})$ for $\frac{1}{p} + \frac{1}{q} = 1$, $p \in (1,\infty)$; $\mathcal{K}(\mathcal{H})^* = \mathcal{S}_1(\mathcal{H})$, $\mathcal{S}_1(\mathcal{H})^* = \mathcal{L}(\mathcal{H})$.

Further reading: J. Conway, *A course in functional analysis*, Springer 1990; B. Simon, *Trace ideals and their applications*, AMS 2005.

7 States in a C^*-algebra and the GNS construction

In the following: $\mathcal{A}$ is a C^*-algebra with identity e.

7.1 States

Definition 7.1.1: A linear functional $\omega : \mathcal{A} \to \mathbb{C}$ is called

(1) *hermitian* $:\Longleftrightarrow$ $\omega(x^*) = \overline{\omega(x)}$ for all $x \in \mathcal{A}$,

(2) *positive* $:\Longleftrightarrow$ $\omega(x) \geq 0$ for all $x \geq 0$,

(3) a *state* $:\Longleftrightarrow$ ω is positive and $\omega(e) = 1$.

Example 7.1.2 (Motivation and examples):

(1) *Classical states* on $\mathcal{A} = C(S)$ with S a compact Hausdorff space are given by probability measures μ on S:

$$\omega(f) = \int f \, \mathrm{d}\mu, \qquad f \in C(S).$$

(2) *Quantum states* on $\mathcal{A} = \mathcal{L}(\mathcal{H})$ with $\mathcal{H}$ a separable Hilbert space are given by positive trace class $W \in \mathcal{S}_1(\mathcal{H})$ with $\operatorname{tr} W = 1$:

$$\omega(x) = \operatorname{tr}(Wx), \qquad x \in \mathcal{L}(\mathcal{H}).$$

By the spectral theorem, $W\varphi = \sum_j s_j \psi_j \langle \varphi, \psi_j \rangle = \sum_j s_j P_{\psi_j} \varphi$ with $\sum_j s_j = 1$.

(i) *Pure quantum state* $\psi \in \mathcal{H}$, $\|\psi\| = 1$: $W = P_\psi$ the orthogonal projection onto $\operatorname{span}(\psi)$, $\omega(x) = \langle x\psi, \psi \rangle$.

(ii) *Thermal quantum state* associated to $A = A^*$ with $\mathrm{e}^{-\beta A} \in \mathcal{S}_1(\mathcal{H})$ and inverse temperature $\beta \in (0,\infty)$: $W = \mathrm{e}^{-\beta A}/\operatorname{tr}(\mathrm{e}^{-\beta A})$.

Lemma 7.1.3:

(1) ω is hermitian $\iff$ $\omega(x) \in \mathbb{R}$ for all $x = x^* \in \mathcal{A}$.

(2) ω is positive $\Rightarrow$ ω is hermitian.

(3) ω is hermitian $\Rightarrow$ $\|\omega\| = \sup\{|\omega(x)| \mid x = x^*,\ \|x\| \le 1\}$.

Proof. (1) "$\Rightarrow$": $\operatorname{Im}\omega(x) = \frac{1}{2\mathrm{i}}\big(\omega(x) - \overline{\omega(x)}\big) = \omega\big(\frac{x-x^*}{2\mathrm{i}}\big) = \omega(0) = 0$. "$\Leftarrow$": Write $x = \operatorname{Re}x + \mathrm{i}\operatorname{Im}x$. Then, since $\omega(\operatorname{Re}x), \omega(\operatorname{Im}x) \in \mathbb{R}$,

$$\omega(x^*) = \omega(\operatorname{Re}x - \mathrm{i}\operatorname{Im}x) = \omega(\operatorname{Re}x) - \mathrm{i}\,\omega(\operatorname{Im}x) = \overline{\omega(\operatorname{Re}x) + \mathrm{i}\,\omega(\operatorname{Im}x)} = \overline{\omega(x)}.$$

(2) Let $x = x^* \in \mathcal{A}$. Use the continuous functional calculus to write $x = x_+ - x_-$ with $x_\pm = \frac{1}{2}(|x| \pm x) \ge 0$. Then $\omega(x) = \underbrace{\omega(x_+)}_{\ge 0} - \underbrace{\omega(x_-)}_{\ge 0} \in \mathbb{R}$.

(3) Note $M := \sup\{|\omega(x)| \mid x = x^* \in \mathcal{A},\ \|x\| \le 1\} \le \|\omega\|$. For all $\varepsilon > 0$ there is $x \in \mathcal{A}$, $\|x\| \le 1$, with $\|\omega\| - \varepsilon < |\omega(x)|$. By picking $x_0 \in \mathcal{A}$, $\|x_0\| \le 1$ such that $x_0 = \frac{\overline{\omega(x)}}{|\omega(x)|}x$, we may assume w.l.o.g. $\omega(x) \ge 0$, i.e.,

$$\|\omega\| - \varepsilon < \omega(x) = \omega(x^*) = \omega(\operatorname{Re}x) \le M\,\|\operatorname{Re}x\| \le M,$$

since $\omega(\operatorname{Im}x) = \frac{1}{2\mathrm{i}}(\omega(x) - \omega(x^*)) = 0$ and $\|\operatorname{Re}x\| = \frac{1}{2}\,\|x + x^*\| \le \frac{1}{2}(\|x\| + \|x^*\|) = \|x\|$. □

Theorem 7.1.1: Characterisation of states

ω is a state $\iff$ $\|\omega\| = \omega(e) = 1$.

Proof. "$\Rightarrow$": Let $x = x^*$, $\|x\| \le 1$. Then $e - x \ge 0$ and hence $\omega(e) - \omega(x) = \omega(e - x) \ge 0$. From Lemma 7.1.3(3): $\|\omega\| = \sup\{|\omega(x)| \mid x = x^* \in \mathcal{A},\ \|x\| \le 1\} = \omega(e) = 1$.

"$\Leftarrow$": Let $x \ge 0$. Choose $\varepsilon > 0$ small enough such that, by polynomial spectral mapping, $\sigma(e - \varepsilon x) = 1 - \varepsilon\sigma(x) \subset [0,1]$, using $\sigma(x) \subset [0, \|x\|]$. Then

$$1 - \varepsilon\operatorname{Re}\omega(x) \le |1 - \varepsilon\omega(x)| = |\omega(e - \varepsilon x)| \le \|\omega\|\ \|e - \varepsilon x\| \le 1 \quad \Rightarrow \quad \operatorname{Re}\omega(x) \ge 0.$$

Consider $x_n := x - \operatorname{Re}\omega(x)e + \mathrm{i}n\operatorname{Im}\omega(x)e$, $n \in \mathbb{N}$. Then $\omega(x_n) = (n+1)\mathrm{i}\,\operatorname{Im}\omega(x)$, and

$$\begin{aligned}(n+1)^2(\operatorname{Im}\omega(x))^2 = |\omega(x_n)|^2 &\le \|\omega\|^2\,\|x_n\|^2 = \|x_n\|^2 = \|x_n^* x_n\| \\ &= \big\|(x - \operatorname{Re}\omega(x)e)^2 + n^2(\operatorname{Im}\omega(x))^2 e\big\| \\ &\le \big\|(x - \operatorname{Re}\omega(x)e)^2\big\| + \big\|n^2(\operatorname{Im}\omega(x))^2 e\big\| \\ &= \|x - \operatorname{Re}\omega(x)e\|^2 + n^2(\operatorname{Im}\omega(x))^2.\end{aligned}$$

Hence $(2n+1)(\operatorname{Im}\omega(x))^2 \le \|x - \operatorname{Re}\omega(x)e\|^2$. Since $n \in \mathbb{N}$ is arbitrary, $\operatorname{Im}\omega(x) = 0$. □

Definition 7.1.4: The *state space* is

$$\mathcal{S}(\mathcal{A}) := \{\omega \in \mathcal{A}^* \mid \|\omega\| = \omega(e) = 1\},$$

naturally equipped with the weak-$*$-topology.

Theorem 7.1.2

$\mathcal{S}(\mathcal{A}) \subset \mathcal{A}^*$ is a convex, compact Hausdorff space.

Proof. Convexity: Let $\omega_0, \omega_1 \in \mathcal{S}(\mathcal{A})$, $\lambda \in (0,1)$, $\omega_\lambda := (1-\lambda)\omega_0 + \lambda\omega_1$. Clearly $\omega_\lambda \in \mathcal{A}^*$ with $\omega_\lambda(e) = (1-\lambda)\omega_0(e) + \lambda\omega_1(e) = 1$, and $\|\omega_\lambda\| \leq (1-\lambda)\|\omega_0\| + \lambda\|\omega_1\| = 1$; hence $\|\omega_\lambda\| = 1$.

Compactness: $\mathcal{S}(\mathcal{A})$ is a weak-$*$-closed subset of the unit ball and hence compact by Banach–Alaoglu. □

Recall the Krein–Milman theorem: $\mathcal{S}(\mathcal{A}) = \overline{\mathrm{conv}(\mathrm{ext}\,\mathcal{S}(\mathcal{A}))}$.

Definition 7.1.5: The extremal points $\mathrm{ext}(\mathcal{S}(\mathcal{A})) =: \mathcal{P}(\mathcal{A})$ are called the *pure states* of $\mathcal{A}$.

Theorem 7.1.3: States associated with spectral values

For all $x \in \mathcal{A}$ and $\lambda \in \sigma(x)$ there is $\omega_\lambda \in \mathcal{S}(\mathcal{A})$ with $\omega_\lambda(x) = \lambda$.

Proof. Consider the subspace $V := \{\alpha x + \beta e \mid \alpha, \beta \in \mathbb{C}\} \subset \mathcal{A}$ and let $\omega_0 : V \to \mathbb{C}$, $\omega_0(\alpha x + \beta e) = \alpha\lambda + \beta$ for all $\alpha, \beta \in \mathbb{C}$. Clearly ω_0 is a linear functional with $\omega_0(x) = \lambda$, $\omega_0(e) = 1$. For $\alpha, \beta \in \mathbb{C}$ we have, since $\alpha\lambda + \beta \in \sigma(\alpha x + \beta e)$ and by the spectral radius theorem,

$$|\alpha\lambda + \beta| \leq \|\sigma(\alpha x + \beta e)\| \leq \|\alpha x + \beta e\|,$$

with equality in case $\alpha = 0$. Hence $\sup\{|\omega_0(\alpha x + \beta e)| \mid \|\alpha x + \beta e\| \leq 1\} = 1$. By the Hahn–Banach theorem the linear functional ω_0 extends to a bounded $\omega \in \mathcal{A}^*$ with $\omega(x) = \lambda$, $\omega(e) = 1 = \|\omega\|$; it is a state by the characterisation of states. □

Remark 7.1.6: In case $A = A^* \in \mathcal{L}(\mathcal{H})$ serves as a Hamiltonian of a quantum system on a Hilbert space $\mathcal{H}$, the above state ω_λ with $\lambda \in \sigma(A)$ models a state with fixed energy λ. Special case $\lambda = \inf\sigma(A)$: *ground state.*

7.2 The GNS construction

Any state $\omega \in \mathcal{S}(\mathcal{A})$ on a C^*-algebra with identity e induces an inner product on $\mathcal{A}$:

$$\langle y, x\rangle := \omega(x^*y), \qquad x, y \in \mathcal{A}.$$

Properties:

(i) linearity in y and skew-linearity in x;

(ii) $\overline{\langle y, x\rangle} = \overline{\omega(x^*y)} = \omega((x^*y)^*) = \omega(y^*x) = \langle x, y\rangle$, since ω is hermitian;

(iii) $\langle x, x\rangle = \omega(x^*x) \geq 0$, since ω is positive.

Implication: the Cauchy–Schwarz inequality

$$|\omega(x^*y)| \leq \sqrt{\omega(x^*x)\,\omega(y^*y)} \qquad \forall x, y \in \mathcal{A}.$$

Proof of the Cauchy–Schwarz inequality. Let $x, y \in \mathcal{A}$ and $\lambda \in \mathbb{C}$. By positivity and linearity,

$$0 \leq \omega\big((x+\lambda y)^*(x+\lambda y)\big) = |\lambda|^2\,\omega(y^*y) + 2\,\mathrm{Re}\left[\lambda\,\omega(x^*y)\right] + \omega(x^*x).$$

Pick $\lambda = t\,\overline{\omega(x^*y)}$ with $t \in \mathbb{R}$:

$$0 \leq |\omega(x^*y)|^2\,\omega(y^*y)\,t^2 + 2t\,|\omega(x^*y)|^2 + \omega(x^*x).$$

The quadratic function on the right is non-negative for all $t \in \mathbb{R}$ if and only if its discriminant is non-positive, i.e.,

$$4\,|\omega(x^*y)|^4 \leq 4\,|\omega(x^*y)|^2\,\omega(y^*y)\,\omega(x^*x). \qquad (*)$$

Case $\omega(x^*y) = 0$: CS is trivially satisfied. Case $\omega(x^*y) \neq 0$: divide $(*)$ by $4\,|\omega(x^*y)|^2$ to obtain CS. □

Lemma 7.2.1: For any $\omega \in \mathcal{S}(\mathcal{A})$, the space

$$\mathcal{L}_\omega := \{x \in \mathcal{A} \mid \omega(x^*x) = 0\}$$

is a closed left ideal. Moreover,

$$x \in \mathcal{L}_\omega \ \vee \ y \in \mathcal{L}_\omega \quad \Rightarrow \quad \omega(x^*y) = 0. \tag{$*$}$$

Proof. $(*)$ follows from the Cauchy–Schwarz inequality.

Linearity of $\mathcal{L}_\omega$: for $x, y \in \mathcal{L}_\omega$, $\lambda \in \mathbb{C}$,

$$\omega\big((x+\lambda y)^*(x+\lambda y)\big) = \omega(x^*x) + |\lambda|^2\,\omega(y^*y) + \lambda\,\omega(x^*y) + \overline{\lambda}\,\omega(y^*x) = 0$$

by $(*)$. Closedness follows from continuity of ω.

Left ideal: $x \in \mathcal{L}_\omega$, $y \in \mathcal{A} \Rightarrow \omega\big((yx)^*yx\big) = \omega\big(x^*(y^*yx)\big) \overset{(*)}{=} 0$. □

Consider the quotient space of equivalence classes

$$\mathcal{H}^0_\omega := \mathcal{A}/\mathcal{L}_\omega = \{[x]_\omega = x + \mathcal{L}_\omega \mid x \in \mathcal{A}\}.$$

Then $\mathcal{H}^0_\omega$ is a pre-Hilbert space with

$$\langle [y]_\omega, [x]_\omega \rangle := \omega(x^*y),$$

with completion $\mathcal{H}_\omega := \overline{\mathcal{H}^0_\omega}^{\|\cdot\|_\omega}$. Left-multiplication by $x \in \mathcal{A}$:

$$T^\omega_x : \mathcal{H}^0_\omega \to \mathcal{H}^0_\omega, \qquad T^\omega_x [y]_\omega := [xy]_\omega.$$

Lemma 7.2.2:

(1) $\mathcal{H}^0_\omega = \{T^\omega_x [e]_\omega \mid x \in \mathcal{A}\}$.

(2) $\langle T^\omega_x [e]_\omega, [e]_\omega \rangle = \omega(x)$ for all $x \in \mathcal{A}$.

(3) T^ω_x uniquely extends to a bounded linear operator on $\mathcal{H}_\omega$ with $\|[x]_\omega\| \le \|T^\omega_x\| \le \|x\|$ for all $x \in \mathcal{A}$.

Proof. (1) Evident from $[x]_\omega = T^\omega_x [e]_\omega$.

(2) l.h.s. $= \langle [x]_\omega, [e]_\omega \rangle = \omega(e^*x) =$ r.h.s.

(3) Linearity of $T^\omega_x : \mathcal{H}^0_\omega \to \mathcal{H}^0_\omega$ is clear. Let $x, y \in \mathcal{A}$. Then $x^*x \le \|x\|^2\, e$, i.e., there is $z \in \mathcal{A}$ with $z^*z = \|x\|^2\, e - x^*x$. Since ω is positive,

$$\begin{aligned} 0 \le \omega\big((zy)^*zy\big) = \omega(y^*z^*zy) = \omega\big(y^*(\|x\|^2\, e - x^*x)y\big) &= \|x\|^2\,\omega(y^*y) - \omega\big((xy)^*xy\big) \\ &= \|x\|^2\,\|[y]_\omega\|^2 - \|T^\omega_x [y]_\omega\|^2, \end{aligned}$$

i.e., $\|T^\omega_x [y]_\omega\| \le \|x\|\,\|[y]_\omega\|$, or $\|T^\omega_x\| \le \|x\|$. In particular, T^ω_x extends uniquely to $\mathcal{H}_\omega$ by the BLT theorem. Moreover,

$$\|[x]_\omega\|^2 = \omega(x^*x) = \omega\big((xe)^*xe\big) = \|T^\omega_x [e]_\omega\|^2 \le \|T^\omega_x\|^2 \underbrace{\|[e]_\omega\|^2}_{=1}.$$ □

Lemma 7.2.3: For any $x \in \mathcal{A}$ there is $\omega \in \mathcal{S}(\mathcal{A})$ with $\omega(x^*x) = \|x\|^2$.

Proof. Since x^*x is self-adjoint: $\|\sigma(x^*x)\| = \|x^*x\| = \|x\|^2$. By Theorem 7.1.3, using $\|\sigma(x^*x)\| \in \sigma(x^*x)$, there is $\omega \in \mathcal{S}(\mathcal{A})$ with $\omega(x^*x) = \|\sigma(x^*x)\| = \|x\|^2$. □

Definition 7.2.4: For a C^*-algebra $\mathcal{A}$ with identity e, a pair $(\pi, \mathcal{H})$ of a $*$-homomorphism $\pi : \mathcal{A} \to \mathcal{L}(\mathcal{H})$ with $\pi(e) = \mathbb{1}$ and a Hilbert space $\mathcal{H}$ is called a *representation*.

- $(\pi, \mathcal{H})$ is *faithful* if π is injective.
- Any vector $\Omega \in \mathcal{H}$ such that $\{\pi(x)\Omega \mid x \in \mathcal{A}\}$ is dense in $\mathcal{H}$ is called *cyclic*. In this case, the triple $(\pi, \mathcal{H}, \Omega)$ is called a *cyclic representation*.

Theorem 7.2.1: GNS representation I

For any state $\omega \in \mathcal{S}(\mathcal{A})$ on a C^*-algebra $\mathcal{A}$ with identity e, the map $\pi_\omega : \mathcal{A} \to \mathcal{L}(\mathcal{H}_\omega)$, $\pi_\omega(x) := T^\omega_x$, defines a cyclic representation $(\pi_\omega, \mathcal{H}_\omega, [e]_\omega)$ with the property

$$\omega(x) = \langle \pi_\omega(x)[e]_\omega, [e]_\omega \rangle \qquad \forall x \in \mathcal{A}.$$

This representation is unique up to unitary equivalence, i.e., if $(\pi_j, \mathcal{H}_j, \Omega_j)$, $j = 1, 2$, are cyclic representations such that $\langle \pi_1(x)\Omega_1, \Omega_1 \rangle_{\mathcal{H}_1} = \langle \pi_2(x)\Omega_2, \Omega_2 \rangle_{\mathcal{H}_2}$ for all $x \in \mathcal{A}$, then there is a unitary $U : \mathcal{H}_1 \to \mathcal{H}_2$ such that

$$\Omega_2 = U\Omega_1 \qquad \text{and} \qquad \pi_2(x) = U\pi_1(x)U^* \quad \forall x \in \mathcal{A}.$$

Proof. In view of Lemma 7.2.2, it remains to prove that $\pi_\omega : \mathcal{A} \to \mathcal{L}(\mathcal{H}_\omega)$ is a $*$-homomorphism.

Linearity: for $x_1, x_2, y \in \mathcal{A}$, $\lambda \in \mathbb{C}$:

$$T^\omega_{x_1+\lambda x_2}[y]_\omega = [(x_1 + \lambda x_2)y]_\omega = [x_1 y]_\omega + \lambda [x_2 y]_\omega = T^\omega_{x_1}[y]_\omega + \lambda T^\omega_{x_2}[y]_\omega.$$

Since $\mathcal{H}^0_\omega$ is dense in $\mathcal{H}_\omega$, this implies the claim.

Multiplicativity: for $x_1, x_2, y \in \mathcal{A}$: $T^\omega_{x_1 x_2}[y]_\omega = [x_1 x_2 y]_\omega = T^\omega_{x_1}[x_2 y]_\omega = T^\omega_{x_1} T^\omega_{x_2}[y]_\omega$. Since $\mathcal{H}^0_\omega$ is dense in $\mathcal{H}_\omega$, this implies the claim.

Unit element: $T^\omega_e[y]_\omega = [ey]_\omega = [y]_\omega$.

$*$*-property:* for $x, y_1, y_2 \in \mathcal{A}$:

$$\langle T^\omega_x [y_1]_\omega, [y_2]_\omega \rangle = \omega(y_2^* x y_1) = \omega\big((x^* y_2)^* y_1\big) = \langle [y_1]_\omega, T^\omega_{x^*}[y_2]_\omega \rangle ,$$

hence $\pi_\omega(x)^* = \pi_\omega(x^*)$.

To establish uniqueness up to unitaries, let $U : \mathcal{H}_1 \to \mathcal{H}_2$,

$$U\pi_1(x)\Omega_1 := \pi_2(x)\Omega_2 \qquad \forall x \in \mathcal{A}.$$

Since Ω_1 is cyclic, U is defined on a dense set in $\mathcal{H}_1$. Moreover, for any $x, y \in \mathcal{A}$:

$$\begin{aligned} \langle U\pi_1(x)\Omega_1, U\pi_1(y)\Omega_1 \rangle_{\mathcal{H}_2} &= \langle \pi_2(x)\Omega_2, \pi_2(y)\Omega_2 \rangle_{\mathcal{H}_2} = \langle \Omega_2, \pi_2(x)^*\pi_2(y)\Omega_2 \rangle_{\mathcal{H}_2} \\ &= \langle \Omega_2, \pi_2(x^* y)\Omega_2 \rangle_{\mathcal{H}_2} \\ &= \langle \Omega_1, \pi_1(x^* y)\Omega_1 \rangle_{\mathcal{H}_1} = \langle \Omega_1, \pi_1(x)^*\pi_1(y)\Omega_1 \rangle_{\mathcal{H}_1} \\ &= \langle \pi_1(x)\Omega_1, \pi_1(y)\Omega_1 \rangle_{\mathcal{H}_1} . \end{aligned}$$

In particular, U is well defined and extends to an isometry on $\mathcal{H}_1$. Since Ω_2 is cyclic, $\overline{\operatorname{ran} U} = \mathcal{H}_2$ and hence U is unitary. Moreover:

(i) $U\Omega_1 = U\pi_1(e)\Omega_1 = \pi_2(e)\Omega_2 = \Omega_2$;

(ii) for all $x, y \in \mathcal{A}$: $U\pi_1(x)U^*U\pi_1(y)\Omega_1 = U\pi_1(xy)\Omega_1 = \pi_2(xy)\Omega_2 = \pi_2(x)\pi_2(y)\Omega_2$, hence $U\pi_1(x)U^* = \pi_2(x)$. □

Example 7.2.5: $\mathcal{A} = \mathcal{L}(\mathbb{C}^n)$, $\omega(A) = \frac{1}{n} \operatorname{tr} A$, $A \in \mathcal{L}(\mathbb{C}^n)$.

- Left ideal: $\mathcal{L}_\omega = \{A \in \mathcal{L}(\mathbb{C}^n) \mid \operatorname{tr} A^*A = 0\} = \{0\}$, hence $\mathcal{H}_\omega^0 = \mathcal{L}(\mathbb{C}^n) = \mathcal{H}_\omega$.
- $\langle [A]_\omega, [B]_\omega \rangle = \frac{1}{n} \operatorname{tr}(B^*A)$, i.e., the Hilbert–Schmidt scalar product up to $\frac{1}{n}$. Hence $\mathcal{H}_\omega = \mathcal{S}_2(\mathbb{C}^n)$ with $\|[A]_\omega\| = \frac{1}{\sqrt{n}} \|A\|_2$.
- $\pi_\omega : \mathcal{L}(\mathbb{C}^n) \to \mathcal{L}(\mathcal{S}_2(\mathbb{C}^n))$, $\pi_\omega(A)B = AB$, with

$$\|\pi_\omega(A)\| = \sup_{B \in \mathcal{S}_2(\mathbb{C}^n)} \frac{\|AB\|_2}{\|B\|_2} = \|A\| \qquad \forall A \in \mathcal{L}(\mathbb{C}^n),$$

  i.e., π_ω is isometric and hence $(\pi_\omega, \mathcal{H}_\omega)$ is faithful.

7.3 Outlook

Theorem 7.3.1: GNS representation II

For every C^*-algebra $\mathcal{A}$ with identity e, there is a faithful representation $(\pi, \mathcal{H})$ such that π is isometric.

Proof idea. Set $\mathcal{H} = \bigoplus_{\omega \in \mathcal{S}(\mathcal{A})} \mathcal{H}_\omega$ and $\pi : \mathcal{A} \to \mathcal{L}(\mathcal{H})$, $\pi := \bigoplus_{\omega \in \mathcal{S}(\mathcal{A})} \pi_\omega$.

- By the previous theorem, each π_ω is a $*$-homomorphism and hence π is a $*$-homomorphism.
- By Lemma 7.2.2: $\|\pi_\omega(x)\| \le \|x\|$ for all $x \in \mathcal{A}$.
- By Lemma 7.2.3: there is $\omega \in \mathcal{S}(\mathcal{A})$ with $\|\pi_\omega(x)\| = \|x\|$.

Hence for all $x \in \mathcal{A}$: $\|x\| = \sup_{\omega \in \mathcal{S}(\mathcal{A})} \|\pi_\omega(x)\| = \|\pi(x)\|$, i.e., π is isometric and hence faithful. □

8 Unbounded operators

In the following: X, Y are Banach spaces and $A : \mathcal{D}(A) \to Y$ is a linear operator on some dense subspace $\mathcal{D}(A) \subset X$, referred to as the *domain* of A.

Example 8.0.1 (Maximal multiplication operator): Associated with a Borel measurable $f : \mathbb{R}^d \to \mathbb{C}$ on $L^p(\mathbb{R}^d, \mu)$, $p \in [1, \infty)$, with a regular Borel measure μ:

$$\mathcal{D}(M_f) := \{\psi \in L^p(\mathbb{R}^d, \mu) \mid f\psi \in L^p(\mathbb{R}^d, \mu)\}, \qquad M_f\psi := f\psi, \quad \psi \in \mathcal{D}(M_f).$$

Lemma 8.0.2: $M_f \in \mathcal{L}(L^p(\mathbb{R}^d, \mu)) \iff f \in L^\infty(\mathbb{R}^d, \mu)$.

Proof. "⇐": $\|M_f\psi\|_p = \left(\int |f\psi|^p \, \mathrm{d}\mu\right)^{1/p} \le \|f\|_\infty \|\psi\|_p$.

"⇒": Fix $0 < c < \|f\|_\infty$. By regularity of μ there is $A \in \mathcal{B}(\mathbb{R}^d)$, $A \subset \{|f| > c\}$, with $0 < \mu(A) < \infty$. Then

$$c^p \mu(A) = \int_A c^p \, \mathrm{d}\mu \le \int |f 1_A|^p \, \mathrm{d}\mu = \|M_f 1_A\|_p^p \le \|M_f\|^p \, \mu(A),$$

hence $\|M_f\| \ge c$. Thus $\|M_f\| \ge \|f\|_\infty$. □

Thus M_f is an unbounded operator with $\mathcal{D}(M_f) \ne L^p(\mathbb{R}^d, \mu)$ in case $f \notin L^\infty(\mathbb{R}^d, \mu)$.

Example 8.0.3 (Differentiation operators on $I = (a, b) \subset \mathbb{R}$):

$$D_0 : C_c^\infty(I) \to L^2(I), \quad D_0\psi := -\mathrm{i}\psi'; \qquad D_{\max} := -\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}x} : W^{1,2}(I) \to L^2(I), \quad D_{\max}\psi = -\mathrm{i}\psi'.$$

Since $C_c^\infty(I) \subset W^{1,2}(I) \subset L^2(I)$, $D_{\max}$ is an extension of D_0. Both operators are unbounded.

Definition 8.0.4: $A : \mathcal{D}(A) \to Y$ is an *extension* of $B : \mathcal{D}(B) \to Y$ if $\mathcal{D}(A) \supset \mathcal{D}(B)$ and $Ax = Bx$ for all $x \in \mathcal{D}(B)$. We write $A \supset B$.

8.1 Closed and closable operators

Definition 8.1.1: For $A : \mathcal{D}(A) \to Y$ with $\mathcal{D}(A) \subset X$ we call

(1) $\mathrm{Graph}(A) := \{(x, Ax) \in X \times Y \mid x \in \mathcal{D}(A)\}$ the *graph* of A, and $\|x\|_A := \|(x, Ax)\|_{X\times Y} := \|x\|_X + \|Ax\|_Y$ the *graph norm* of $x \in \mathcal{D}(A)$;

(2) A *closed* iff $\mathrm{Graph}(A)$ is closed w.r.t. the product topology induced by $\|\cdot\|_{X\times Y}$;

(3) A *closable* iff it has a closed extension. The smallest closed extension is denoted by $\overline{A}$ and called the *closure* of A.

Lemma 8.1.2: $G \subset X \times Y$ is the graph of a linear operator $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, if and only if G is a subspace of $X \times Y$ with the property

$$(0, y) \in G \;\Rightarrow\; y = 0.$$

Proof. "⇒": evident, since $A0 = 0$.

"⇐": Let $(x, y), (x, y') \in G$. Since G is a subspace, $(0, y - y') \in G$; hence $y = y'$. Set $\mathcal{D}(A) = \{x \in X \mid \exists y \in Y : (x, y) \in G\}$. Since for any $x \in \mathcal{D}(A)$ there is a unique $y \in Y$ with $(x, y) \in G$, set $Ax = y$. Then $A : \mathcal{D}(A) \to Y$ is linear with $\mathrm{Graph}(A) = G$. □

Lemma 8.1.3: $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, is closable if and only if $\overline{\mathrm{Graph}(A)}$ is the graph of a linear operator.

Proof. "⇐": Let B be the linear operator with $\mathrm{Graph}(B) = \overline{\mathrm{Graph}(A)}$. Then $\mathrm{Graph}(A) \subset \mathrm{Graph}(B)$ and hence $\mathcal{D}(A) \subset \mathcal{D}(B)$, so B is a closed extension.

"⇒": Let $B : \mathcal{D}(B) \to Y$ be a closed extension of A. If $(0, y) \in \overline{\mathrm{Graph}(A)}$, then $(0, y) \in \mathrm{Graph}(B)$ and hence $y = 0$. Since $\overline{\mathrm{Graph}(A)}$ is a subspace, the claim follows from Lemma 8.1.2. □

Theorem 8.1.1: Characterisation of closed operators

The following statements are equivalent for $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$:

(1) A is closed.

(2) $(\mathcal{D}(A), \|\cdot\|_A)$ is complete.

(3) If $(x_n) \subset \mathcal{D}(A)$ with $x_n \to x$ and $Ax_n \to y$, then $x \in \mathcal{D}(A)$ and $Ax = y$.

Proof. (1)⇒(3): $(x_n, Ax_n) \in \mathrm{Graph}(A)$, $\|(x_n, Ax_n) - (x, y)\| \to 0$. Thus $(x, y) \in \mathrm{Graph}(A)$, i.e., $x \in \mathcal{D}(A)$, $Ax = y$.

(3)⇒(2): Let $(x_n) \subset \mathcal{D}(A)$ be a Cauchy sequence w.r.t. $\|\cdot\|_A$. Then (i) (x_n) is Cauchy w.r.t. $\|\cdot\|_X$ and (ii) (Ax_n) is Cauchy w.r.t. $\|\cdot\|_Y$. Completeness of X, Y implies that there are $x \in X$, $y \in Y$ with $\|x_n - x\|_X \to 0$ and $\|Ax_n - y\|_Y \to 0$. Thus $x \in \mathcal{D}(A)$, $Ax = y$ and $\|x - x_n\|_A \to 0$.

(2)⇒(1): Let $(x_n, Ax_n) \subset \mathrm{Graph}(A)$ converge to (x, y). Then $(x_n) \subset \mathcal{D}(A)$ is a Cauchy sequence w.r.t. $\|\cdot\|_A$ and hence, for some $x' \in \mathcal{D}(A)$: $\|x_n - x'\|_A \to 0$, i.e., $\|x_n - x'\|_X \to 0$ and $\|Ax_n - Ax'\|_Y \to 0$. Uniqueness of the limit in X and Y yields $x = x'$ and $y = Ax'$. □

Theorem 8.1.2: Characterisation of closable operators

The following statements are equivalent for $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$:

(1) A is closable.

(2) $\overline{\text{Graph}(A)}$ is the graph of a linear operator.

(3) If $(x_n) \subset \mathcal{D}(A)$ with $\|x_n\|_X \to 0$ and $\|Ax_n - y\|_Y \to 0$, then $y = 0$.

Proof. (1) $\iff$ (2): previous lemma. (2)$\Rightarrow$(3): by assumption $(0, y) \in \overline{\text{Graph}(A)}$ implies $y = 0$. (3)$\Rightarrow$(2): immediate from Lemma 8.1.2. □

Example 8.1.4:

(1) Differentiation operators $D_0/D_{\max}$: $D_{\max}$ is closed since

$$\|\psi\|^2_{D_{\max}} \simeq \|\psi\|^2_2 + \|\psi'\|^2_2 = \|\psi\|^2_{W^{1,2}}$$

and $W^{1,2}$ is a Hilbert space with this norm, generated by the scalar product $\langle \varphi, \psi \rangle_{W^{1,2}} = \langle \varphi, \psi \rangle + \langle \varphi', \psi' \rangle$. D_0 is closable (since $D_{\max}$ is a closed extension).

(2) Evaluation at $0 \in I = (a, b)$ on $L^p(I)$, $p \in [1, \infty)$: $A\psi := \psi(0)$, $\psi \in \mathcal{D}(A) = C(\overline{I})$. A is not closable, since there is $(\psi_n) \subset C(\overline{I})$ with $\psi_n(0) = 1$ and $\|\psi_n\|_p \to 0$. Thus $\psi_n \to 0$ but $A\psi_n = 1 \neq 0$.

Remark 8.1.5:

(1) By the above theorem, the closure $\overline{A}$ of a closable operator $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, is uniquely defined through

$$\mathcal{D}(\overline{A}) = \{x \in X \mid \exists (x_n) \subset \mathcal{D}(A) : \ x_n \to x \text{ and } (Ax_n) \text{ convergent}\} = \overline{\mathcal{D}(A)}^{\|\cdot\|_A},$$

$$\overline{A}x = \lim_{n \to \infty} Ax_n.$$

(2) The closed graph theorem specifies the relation of closedness and boundedness.

Theorem 8.1.3: Closed graph theorem

Let $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, be a closed operator. If $\mathcal{D}(A)$ is closed in X, then A is bounded.

(See Functional Analysis.)

8.2 Adjoint of a densely defined operator on Hilbert space

In the following: $\mathcal{G}, \mathcal{H}$ Hilbert spaces, and $A : \mathcal{D}(A) \to \mathcal{H}$ a linear operator densely defined on $\mathcal{D}(A) \subset \mathcal{G}$.

Recall that the (Hilbert) adjoint of $A \in \mathcal{L}(\mathcal{G}, \mathcal{H})$ is uniquely determined as $A^* \in \mathcal{L}(\mathcal{H}, \mathcal{G})$ through

$$\langle Ax, y \rangle = \langle x, A^* y \rangle \qquad \text{for all } x \in \mathcal{G},\ y \in \mathcal{H},$$

where $A^* y =: y^* \in \mathcal{G}$ exists by Riesz, since $\mathcal{G} \ni x \mapsto \langle Ax, y \rangle$ is a continuous linear functional. For $A : \mathcal{D}(A) \to \mathcal{H}$ densely defined on $\mathcal{D}(A) \subset \mathcal{G}$ consider

$$\mathcal{D}(A) \ni x \mapsto \langle Ax, y \rangle \qquad \text{with } y \in \mathcal{H}.$$

If this linear functional is bounded, it uniquely defines $y^* \in \mathcal{G}$ such that $\langle Ax, y \rangle = \langle x, y^* \rangle$ for all $x \in \mathcal{D}(A)$ by Riesz' theorem. This motivates:

Definition 8.2.1: For $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{G}$, the operator $A^* : \mathcal{D}(A^*) \to \mathcal{G}$, $A^* y = y^*$,

$$\mathcal{D}(A^*) = \{y \in \mathcal{H} \mid \exists y^* \in \mathcal{G} : \ \langle x, y^* \rangle = \langle Ax, y \rangle \ \ \forall x \in \mathcal{D}(A)\},$$

is called the *adjoint* of A.

Remark 8.2.2: $\mathcal{D}(A^*) = \big\{ y \in \mathcal{H} \ \big| \ \sup_{x \in \mathcal{D}(A), \|x\|=1} |\langle Ax, y \rangle| < \infty \big\}$.

Example 8.2.3:

(1) Differentiation operators $D_0 / D_{\max}$: for $\psi \in C_c^\infty(I)$, $\varphi \in W^{1,2}(I)$, integrating by parts,

$$\langle D_0 \psi, \varphi \rangle = \int_a^b \overline{\varphi}(-\mathrm{i}\psi') \, \mathrm{d}x \overset{\text{P.I.}}{=} \int_a^b \overline{(-\mathrm{i}\varphi')} \psi \, \mathrm{d}x = \langle \psi, D_{\max} \varphi \rangle \, .$$

Thus D_0^* is an extension of $D_{\max}$ ($\operatorname{Graph} D_{\max} \subset \operatorname{Graph} D_0^*$). We will see below that D_0^* is closed; in fact $D_0^* = D_{\max}$.

(2) Densely defined operator with adjoint only defined on $\{0\}$: Pick a Cantor bijection $n : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $(k, \ell) \mapsto n(k, \ell)$, and consider

$$A : \mathcal{D}(A) \to \ell^2(\mathbb{N}), \quad \mathcal{D}(A) = \ell_c^2(\mathbb{N}) \text{ (finite sequences in } \ell^2(\mathbb{N})), \qquad (Ax)_k := \sum_{\ell=1}^{\infty} x_{n(k,\ell)}.$$

Then $\mathcal{D}(A^*) = \{0\}$, since for any $y \in \mathcal{D}(A^*)$, $x \in \mathcal{D}(A)$ (all sums are finite!):

$$\sum_k \sum_\ell \overline{y_k} x_{n(k,\ell)} = \langle Ax, y \rangle = \langle x, A^* y \rangle = \sum_n \overline{(A^* y)_n} x_n = \sum_k \sum_\ell \overline{(A^* y)_{n(k,\ell)}} x_{n(k,\ell)}.$$

Hence $(A^* y)_{n(k,\ell)} = y_k$ is independent of ℓ for all k. Since $A^* y \in \ell^2(\mathbb{N})$, this implies $A^* y = 0$, i.e., $y = 0$.

Description of the adjoint through the graph. Let

$$J : \mathcal{G} \times \mathcal{H} \to \mathcal{H} \times \mathcal{G}, \qquad (x, y) \mapsto J(x, y) := (-y, x),$$

where the product spaces carry the scalar product $\langle (y, z), (y', z') \rangle_{\mathcal{H} \times \mathcal{G}} = \langle y, y' \rangle_{\mathcal{H}} + \langle z, z' \rangle_{\mathcal{G}}$.

Lemma 8.2.4: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $B : \mathcal{D}(B) \to \mathcal{H}$, $\mathcal{D}(A), \mathcal{D}(B) \subset \mathcal{G}$, both be densely defined on Hilbert spaces $\mathcal{G}, \mathcal{H}$. Then:

(1) $\operatorname{Graph}(A^*) = \big(J \operatorname{Graph}(A)\big)^\perp = J\big(\operatorname{Graph}(A)^\perp\big)$.

(2) $B \subset A \Rightarrow A^* \subset B^*$.

Proof. (1) By definition of A^*:

$$\begin{aligned}
\operatorname{Graph}(A^*) &= \{(y, z) \in \mathcal{H} \times \mathcal{G} \mid \langle Ax, y \rangle = \langle x, z \rangle \text{ for all } x \in \mathcal{D}(A)\} \\
&= \{(y, z) \in \mathcal{H} \times \mathcal{G} \mid \langle (-Ax, x), (y, z) \rangle_{\mathcal{H} \times \mathcal{G}} = 0 \text{ for all } x \in \mathcal{D}(A)\} \\
&= \{(y, z) \in \mathcal{H} \times \mathcal{G} \mid \langle J(v, w), (y, z) \rangle_{\mathcal{H} \times \mathcal{G}} = 0 \text{ for all } (v, w) \in \operatorname{Graph}(A)\} \\
&= \big(J \operatorname{Graph}(A)\big)^\perp = J\big(\operatorname{Graph}(A)^\perp\big).
\end{aligned}$$

(2) $\operatorname{Graph}(B) \subset \operatorname{Graph}(A) \Rightarrow J \operatorname{Graph}(B) \subset J \operatorname{Graph}(A) \Rightarrow (J \operatorname{Graph}(A))^\perp \subset (J \operatorname{Graph}(B))^\perp$
$\overset{(1)}{\Rightarrow} \operatorname{Graph}(A^*) \subset \operatorname{Graph}(B^*)$. □

Theorem 8.2.1

Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{G}$, be densely defined on Hilbert spaces $\mathcal{G}, \mathcal{H}$. Then:

(1) A^* is closed.

(2) If A admits a closure $\overline{A}$, then $(\overline{A})^* = A^*$.

(3) A^* is densely defined if and only if A is closable.

(4) If A is closable, then its closure is $\overline{A} = (A^*)^*$.

Proof. (1) Since $V^\perp$ is closed for any subspace V of a Hilbert space, $\operatorname{Graph}(A^*)$ is closed by Lemma 8.2.4.

(2) By Lemma 8.2.4: $\operatorname{Graph}(\overline{A}^*) = \big(J \operatorname{Graph}(\overline{A})\big)^\perp = \big(J\overline{\operatorname{Graph}(A)}\big)^\perp = \big(\overline{J \operatorname{Graph}(A)}\big)^\perp = \big(J \operatorname{Graph}(A)\big)^\perp = \operatorname{Graph}(A^*)$.

(3) By Lemma 8.2.4,

$$\begin{aligned}\overline{\operatorname{Graph}(A)} &= \big(\operatorname{Graph}(A)^\perp\big)^\perp = \big(J^{-1} \operatorname{Graph}(A^*)\big)^\perp \\ &= \{(x,y) \in \mathcal{G} \times \mathcal{H} \mid \big\langle (x,y), J^{-1}(z, A^*z)\big\rangle_{\mathcal{G}\times\mathcal{H}} = 0 \text{ for all } z \in \mathcal{D}(A^*)\} \\ &= \{(x,y) \in \mathcal{G} \times \mathcal{H} \mid \langle x, A^*z\rangle = \langle y, z\rangle \text{ for all } z \in \mathcal{D}(A^*)\}.\end{aligned}$$

Thus $(0,y) \in \overline{\operatorname{Graph}(A)} \iff y \in \mathcal{D}(A^*)^\perp$. By Lemma 8.1.2 this implies: $\overline{\operatorname{Graph}(A)}$ is a graph $\iff$ $\mathcal{D}(A^*)$ is dense.

(4) Use (3) to conclude that $(A^*)^*$ is well-defined, and

$$\operatorname{Graph}\big((A^*)^*\big) = \big(J \operatorname{Graph}(A^*)\big)^\perp = \big(JJ \operatorname{Graph}(A)^\perp\big)^\perp = \big(\operatorname{Graph}(A)^\perp\big)^\perp = \overline{\operatorname{Graph}(A)},$$

i.e., $(A^*)^* = \overline{A}$. □

8.3 Self-adjointness

Definition 8.3.1: A densely defined operator $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, on a Hilbert space $\mathcal{H}$ is called

(1) *symmetric* iff $A \subset A^*$,

(2) *self-adjoint* iff $A = A^*$,

(3) *essentially self-adjoint (esa)* iff A^* is self-adjoint.

Remark 8.3.2: A esa $\Rightarrow A \subset (A^*)^* = A^*$, i.e., A is symmetric.

Theorem 8.3.1: Hellinger–Toeplitz

Every symmetric operator A defined on a Hilbert space with $\mathcal{D}(A) = \mathcal{H}$ is bounded.

Proof. By the subsequent theorem, A is closable. Its closure $\overline{A} \supset A$ is bounded by the closed graph theorem. Hence A is bounded. □

Theorem 8.3.2

(1) Every symmetric operator is closable with $\overline{A} = (A^*)^* \subset A^*$.

(2) The following statements are equivalent:

(i) A is esa,

(ii) $\overline{A} = A^*$,

(iii) $\overline{A}$ is self-adjoint.

In this situation $\overline{A}$ is the only self-adjoint extension of A.

Proof. (1) A^* is a closed extension of A.

(2) (i)⇒(ii): $\overline{A} = A^{**} \overset{\text{(i)}}{=} A^*$ (Theorem 8.2.1). (ii)⇒(iii): $\overline{A}^* = A^* \overset{\text{(ii)}}{=} \overline{A}$ (Theorem 8.2.1(2)). (iii)⇒(i): $A^* = \overline{A}^* \overset{\text{(iii)}}{=} \overline{A} = A^{**}$ (Theorem 8.2.1).

If $\tilde{A}$ is a self-adjoint extension of A, then $A \subset \tilde{A}$ and $\tilde{A} = \tilde{A}^* \subset A^* \overset{\text{(ii)}}{=} \overline{A} \subset \tilde{A}$, hence $\tilde{A} = \overline{A}$. □

Example 8.3.3 (Maximal multiplication operator): Associated with a Borel measurable $f : \mathbb{R}^d \to \mathbb{C}$ on $L^2(\mathbb{R}^d, \mu)$ with a regular Borel measure μ: $\mathcal{D}(M_f) = \{\psi \in L^2(\mathbb{R}^d, \mu) \mid f\psi \in L^2(\mathbb{R}^d, \mu)\}$, $M_f\psi := f\psi$. Then

$$M_f^* = M_{\overline{f}} \quad \text{with} \quad \mathcal{D}(M_f^*) = \mathcal{D}(M_{\overline{f}}) = \mathcal{D}(M_f).$$

In particular: $\operatorname{Im} f = 0 \iff M_f$ is self-adjoint.

Proof. Let $(\eta, \varrho) \in \operatorname{Graph}(M_f^*)$, i.e., $\varrho = M_f^*\eta$, $\eta \in \mathcal{D}(M_f^*)$. For all $\psi \in \mathcal{D}(M_f)$: $\left|\int \overline{\eta} f\psi \,\mathrm{d}\mu\right| = |\langle\psi, \varrho\rangle| \leq \|\psi\| \, \|\varrho\|$. Hence $\mathcal{D}(M_f) \ni \psi \mapsto \int \overline{\eta} f\psi \,\mathrm{d}\mu = \langle\psi, \varrho\rangle$ extends uniquely to a bounded linear functional on $L^2(\mathbb{R}^d, \mu)$. Hence $f\overline{\eta} \in L^2(\mathbb{R}^d, \mu)$ and $\overline{\varrho} = f\overline{\eta}$, i.e., $\eta \in \mathcal{D}(M_{\overline{f}})$ and $\varrho = \overline{f}\eta$, i.e., $(\eta, \varrho) \in \operatorname{Graph}(M_{\overline{f}})$.

Conversely, let $(\eta, \varrho) \in \operatorname{Graph}(M_{\overline{f}})$. Then $\overline{\varrho} = f\overline{\eta}$ and for all $\psi \in \mathcal{D}(M_f)$: $\langle\psi, \varrho\rangle = \int f\overline{\eta}\psi \,\mathrm{d}\mu = \langle M_f\psi, \eta\rangle$, i.e., $(\eta, \varrho) \in \operatorname{Graph}(M_f^*)$. □

Example 8.3.4 (Laplace operator): Specifically, let $\mu = \lambda^d$ be the Lebesgue measure. Then $f(k) = |k|^2 = \sum_{j=1}^d k_j^2$ defines a self-adjoint operator M_f. Recall: the Fourier transform $\mathcal{F} : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$,

$$(\mathcal{F}\psi)(k) := \int \mathrm{e}^{-\mathrm{i}k\cdot x}\psi(x)\frac{\mathrm{d}x}{(2\pi)^{d/2}}, \qquad \psi \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d), \quad k \cdot x = \sum_{j=1}^d k_j x_j,$$

extends to a unitary operator on $L^2(\mathbb{R}^d)$. For all $\psi \in \mathcal{D}(M_f)$: $\mathcal{F}^* M_f \mathcal{F}\psi = -\Delta\psi$ with $\Delta\varphi = \sum_{j=1}^d \partial_j^2\varphi$. The Laplace operator $L : \mathcal{D}(\Delta) \to L^2(\mathbb{R}^d)$, $L\varphi = -\Delta\varphi$, with

$$\mathcal{D}(L) = \mathcal{F}^*\mathcal{D}(M_f) = \{\varphi \in L^2(\mathbb{R}^d) \mid \Delta\varphi \in L^2(\mathbb{R}^d)\},$$

is self-adjoint since M_f is self-adjoint.

Remark 8.3.5: Similar reasoning allows us to conclude that all differential operators of the form $\operatorname{Pol}(\nabla)$ are self-adjoint on a suitable domain provided $\operatorname{Pol}(\mathrm{i}k) \in \mathbb{R}$ for all $k \in \mathbb{R}^d$.

Example 8.3.6 (Differentiation operators $D_0/D_{\max}$): From the example above: $D_0 \subset D_{\max} = D_0^*$, i.e., D_0 is symmetric. From the characterisation of the closure:

$$\mathcal{D}(\overline{D_0}) = \overline{C_c^\infty(I)}^{\|\cdot\|_{W^{1,2}}} =: W_0^{1,2}(I) \neq W^{1,2}(I) \quad \text{if } I \neq \mathbb{R}.$$

From PDE, $W^{1,2}(I) \subset C(\overline{I})$ and $W_0^{1,2}(I) = \{\psi \in W^{1,2}(I) \mid \psi(a) = \psi(b) = 0\}$, with $\overline{D_0}\psi = -\mathrm{i}\psi'$. I.e., D_0 is not esa if $I \neq \mathbb{R}$, and $D_{\max}$ is not self-adjoint.

Remark 8.3.7: For self-adjoint extensions of D_0 see Section 9.1 below.

8.4 Bounded invertibility, spectra and resolvents

Definition 8.4.1: A densely defined linear operator $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, on some Banach spaces X, Y is *boundedly invertible* if there is some $B \in \mathcal{L}(Y, X)$ such that

$$AB = \mathrm{id}_Y \qquad \text{and} \qquad BA \subset \mathrm{id}_X.$$

B is called the (bounded) *inverse* of A.

Remark 8.4.2: $BA \in \mathcal{L}(\mathcal{D}(A), X)$.

Theorem 8.4.1: Characterisation of bounded invertibility

(1) A is boundedly invertible if and only if $\ker A = \{0\}$, $A\mathcal{D}(A) = Y$ (i.e., $A : \mathcal{D}(A) \to Y$ is bijective), and A is closed.

(2) If A is boundedly invertible, then its inverse is unique. Notation: $A^{-1} = \frac{1}{A}$.

Proof. (1) "⇒":

(i) If $0 \neq x \in \ker(A)$, then $BAx = 0$, which contradicts $BA \subset \mathrm{id}_X$.

(ii) $AB = \mathrm{id}_Y$ and $BA \subset \mathrm{id}_X \Rightarrow \operatorname{ran} B = \mathcal{D}(A)$ and $A\mathcal{D}(A) = Y$.

(iii) $\mathrm{Graph}(A) = \{(x, Ax) \mid x \in \mathcal{D}(A)\} \overset{\text{(ii)}}{=} \{(By, y) \mid y \in Y\}$. Since B is bounded, this set is closed.

"⇐": Set $B : Y \to X$, $By = x$ if $y = Ax$. This linear operator is well-defined, since A is bijective. Since $\mathrm{Graph}(A)$ is closed, $\mathrm{Graph}(B)$ is closed. By the closed graph theorem, $B \in \mathcal{L}(Y, X)$. Moreover $BA \subset \mathrm{id}_X$, $AB = \mathrm{id}_Y$.

(2) Let $C \in \mathcal{L}(Y, X)$ be another inverse of A. Then for all $y \in Y$: $ACy = y = ABy \Rightarrow Cy = By$, since $\ker A = \{0\}$. □

Definition 8.4.3: For a closed linear operator $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$, on some Banach space X,

$$\varrho(A) := \{z \in \mathbb{C} \mid A - z \text{ is boundedly invertible}\}$$

is the *resolvent set* and $\sigma(A) := \mathbb{C} \setminus \varrho(A)$ the *spectrum*. The *resolvent operator* $(A - z)^{-1} \in \mathcal{L}(X)$ is the inverse of $(A - z)$ for $z \in \varrho(A)$.

Lemma 8.4.4 (Properties):

(1) *First resolvent formula:* for $\lambda, \mu \in \varrho(A)$,

$$(A - \lambda)^{-1} - (A - \mu)^{-1} = (\lambda - \mu)(A - \lambda)^{-1}(A - \mu)^{-1} = (\lambda - \mu)(A - \mu)^{-1}(A - \lambda)^{-1}.$$

(2) *Second resolvent formula:* for $\lambda \in \varrho(A) \cap \varrho(B)$,

$$\begin{aligned}(A-\lambda)^{-1} - (B-\lambda)^{-1} &= (A-\lambda)^{-1}(B-A)(B-\lambda)^{-1} && \text{if } \mathcal{D}(B) \subset \mathcal{D}(A),\\ &= (B-\lambda)^{-1}(B-A)(A-\lambda)^{-1} && \text{if } \mathcal{D}(A) \subset \mathcal{D}(B).\end{aligned}$$

Proof. (1) *Exercise.*

(2) W.l.o.g. $\lambda = 0$, $\mathcal{D}(B) \subset \mathcal{D}(A)$. Since $\operatorname{ran} B^{-1} = \mathcal{D}(B) \subset \mathcal{D}(A)$,

$$A^{-1} = A^{-1}BB^{-1} = A^{-1}(A+B-A)B^{-1} = A^{-1}AB^{-1} + A^{-1}(B-A)B^{-1} = B^{-1} + A^{-1}(B-A)B^{-1}.$$

□

Theorem 8.4.2: von Neumann series – stability of bounded invertibility

Let $A : \mathcal{D}(A) \to Y$, $\mathcal{D}(A) \subset X$, be boundedly invertible and $B : \mathcal{D}(B) \to Y$, $\mathcal{D}(A) \subset \mathcal{D}(B) \subset X$, with $\left\|BA^{-1}\right\| < 1$. Then $A - B : \mathcal{D}(A) \to Y$ is boundedly invertible with

$$(A-B)^{-1} = \sum_{n=0}^{\infty} A^{-1}(BA^{-1})^n = \sum_{n=0}^{\infty} (A^{-1}B)^n A^{-1}.$$

The series converges in norm.

Proof. Exercise. □

Corollary 8.4.5: Let $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$, be closed.

(1) If $z_0 \in \varrho(A)$ and $|z - z_0| < \left\|(A-z_0)^{-1}\right\|^{-1}$, then $z \in \varrho(A)$ and

$$(A-z)^{-1} = \sum_{n=0}^{\infty} (z-z_0)^n \left[(A-z_0)^{-1}\right]^{n+1},$$

i.e., the resolvent is an $\mathcal{L}(X)$-valued analytic function.

(2) $\varrho(A) \subset \mathbb{C}$ is open and $\sigma(A) \subset \mathbb{C}$ is closed.

(3) For all $z \in \varrho(A)$: $\operatorname{dist}(z, \sigma(A)) \geq \left\|(A-z)^{-1}\right\|^{-1}$.

Proof. Exercise. □

Remark 8.4.6 (Remark and warning): There are different possible definitions of the resolvent set:

$$\begin{aligned}\varrho(A) &= \{z \in \mathbb{C} \mid \ker(A-z) = \{0\} \wedge \operatorname{ran}(A-z) = X \wedge \left\|(A-z)^{-1}\right\| < \infty\},\\ \tilde{\varrho}(A) &= \{z \in \mathbb{C} \mid \ker(A-z) = \{0\} \wedge \overline{\operatorname{ran}(A-z)} = X \wedge \left\|(A-z)^{-1}\right\| < \infty\}.\end{aligned}$$

If A is closed, then $\varrho(A) = \tilde{\varrho}(A)$.

Proof. Evidently $\varrho(A) \subset \tilde{\varrho}(A)$. Let $z \in \tilde{\varrho}(A)$ and $(x_n) \subset \mathcal{D}(A)$ with $y_n := (A-z)x_n \to y$. Then $x_n = (A-z)^{-1}y_n \to (A-z)^{-1}y$ and by Theorem 8.1.1, $(A-z)^{-1}y \in \mathcal{D}(A)$ and $y = (A-z)(A-z)^{-1}y \in \operatorname{ran}(A-z)$. □

This is not true in general, e.g. $X = \ell^2(\mathbb{N}) = Y$,

$$R := \operatorname{span}\{(1, \tfrac{1}{2}, 0, \dots), (1, 0, \tfrac{1}{3}, 0, \dots), \dots, (1, 0, \dots, 0, \tfrac{1}{n}, 0, \dots), \dots\} \;\Rightarrow\; \overline{R} = \ell^2(\mathbb{N}),$$

and $B : R \to Y$, $(Bx)_n := x_{n+1}$ (left shift). Then $\|B\| \leq 1$, $\ker B = \{0\}$, and $\overline{\operatorname{ran} B} = Y$ since $\operatorname{ran} B$ contains all e_n's. Let $A = B^{-1} : \operatorname{ran} B \to R$. Then $0 \in \tilde{\varrho}(A)$, but $0 \notin \varrho(A)$; A is not closable since $(e_1, 0) \in \overline{\operatorname{Graph}(B)} \Rightarrow (0, e_1) \in \overline{\operatorname{Graph}(A)}$.

The spectrum of any closed operator $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$, on a Banach space X decomposes into the following components:

(1) the set of eigenvalues or *point spectrum* $\sigma_p(A) := \{z \in \mathbb{C} \mid \ker(A - z) \neq \{0\}\}$; every $x \in \ker(A - z) \setminus \{0\}$ is called an *eigenvector* of A;

(2) the *residual spectrum* $\sigma_r(A) := \{z \in \mathbb{C} \mid \ker(A - z) = \{0\} \wedge \overline{\operatorname{ran}(A - z)} \neq X\}$;

(3) the *continuous spectrum* $\sigma_c(A) := \{z \in \mathbb{C} \mid \ker(A-z) = \{0\} \wedge \operatorname{ran}(A-z) \neq X \wedge \overline{\operatorname{ran}(A - z)} = X\}$.

Theorem 8.4.3

For any closed $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$: $\sigma(A) = \sigma_p(A) \uplus \sigma_r(A) \uplus \sigma_c(A)$.

Proof. Evidently $\sigma_p(A), \sigma_r(A), \sigma_c(A)$ are disjoint sets, and by the characterisation of bounded invertibility

$$\varrho(A) = \{z \in \mathbb{C} \mid \ker(A - z) = \{0\} \wedge \operatorname{ran}(A - z) = X\} = \big(\sigma_p(A) \uplus \sigma_r(A) \uplus \sigma_c(A)\big)^c. \qquad \square$$

Useful criterion for injectivity on Hilbert spaces:

Lemma 8.4.7: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{G}$, be densely defined on Hilbert spaces $\mathcal{G}, \mathcal{H}$. Then:

(1) $\big(A\mathcal{D}(A)\big)^\perp = \ker A^*$.

(2) If A is closed, then $\big(A^*\mathcal{D}(A^*)\big)^\perp = \ker A$.

Proof. (1) Let $y \in (A\mathcal{D}(A))^\perp$, i.e., $\langle Ax, y\rangle = 0 = \langle x, 0\rangle$ for all $x \in \mathcal{D}(A)$. Then $y \in \mathcal{D}(A^*)$ and $A^*y = 0$, i.e., $y \in \ker A^*$. Conversely, let $y \in \mathcal{D}(A^*)$, $A^*y = 0$. Then for all $x \in \mathcal{D}(A)$: $0 = \langle x, A^*y\rangle = \langle Ax, y\rangle$, i.e., $y \in (A\mathcal{D}(A))^\perp$.

(2) If A is closed, then $A = \overline{A} = (A^*)^*$ (Theorem 8.2.1). Thus (2) reduces to (1). $\square$

Lemma 8.4.8: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be closed and densely defined. Then

$$\varrho(A^*) = \varrho(A)^*, \qquad \sigma(A^*) = \sigma(A)^*,$$

where * denotes the complex conjugate of the set.

Proof. It suffices to prove the first assertion. Let $\lambda \in \varrho(A)$.

(i) $\{0\} = \mathcal{H}^\perp = \big((A - \lambda)\mathcal{D}(A)\big)^\perp = \ker(A^* - \overline{\lambda})$ by Lemma 8.4.7 and Theorem 8.2.1.

(ii) $\mathcal{H} = \{0\}^\perp = \ker(A - \lambda)^\perp = \Big[\big((A^* - \overline{\lambda})\mathcal{D}(A^*)\big)^\perp\Big]^\perp = \overline{(A^* - \overline{\lambda})\mathcal{D}(A^*)}$.

Let (y_n) be a sequence in $(A^* - \overline{\lambda})\mathcal{D}(A^*)$, i.e., $y_n = (A^* - \overline{\lambda})x_n$ for some $x_n \in \mathcal{D}(A^*)$, and suppose $y_n \to y \in \mathcal{H}$. Then, since $\lambda \in \varrho(A)$ and $x_n, x_m \in \mathcal{D}(A^*)$,

$$\begin{aligned}
\|x_n - x_m\|^2 &= \big\langle (A - \lambda)(A - \lambda)^{-1}(x_n - x_m), x_n - x_m\big\rangle \\
&= \big\langle (A - \lambda)^{-1}(x_n - x_m), (A^* - \overline{\lambda})(x_n - x_m)\big\rangle \\
&\leq \big\|(A - \lambda)^{-1}\big\| \, \|x_n - x_m\| \, \|y_n - y_m\| \,.
\end{aligned}$$

Hence (x_n) is a Cauchy sequence in $\mathcal{H}$, so $x_n \to x \in \mathcal{H}$. By closedness of $A^* - \overline{\lambda}$, this implies, using Theorem 8.1.1, $x \in \mathcal{D}(A^*)$ and $y = (A^* - \overline{\lambda})x$, i.e., $(A^* - \overline{\lambda})\mathcal{D}(A^*) = \overline{(A^* - \overline{\lambda})\mathcal{D}(A^*)} = \mathcal{H}$. From (i) and (ii): $A^* - \overline{\lambda}$ is boundedly invertible, i.e., $\overline{\lambda} \in \varrho(A^*)$. The converse inclusion follows by applying this to A^* (using $A^{**} = A$). $\square$

Lemma 8.4.9: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be symmetric. Then $\overline{A\mathcal{D}(A)} = \mathcal{H} \Rightarrow \ker(A) = \{0\}$.

Proof. By Lemma 8.4.7: $\ker(A^*) = (A\mathcal{D}(A))^\perp = \overline{A\mathcal{D}(A)}^\perp = \{0\}$. Since $A \subset A^*$, this implies $\ker(A) = \{0\}$. □

Lemma 8.4.10: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be symmetric. Then:

(1) For all $x \in \mathcal{D}(A)$, $z \in \mathbb{C} \setminus \mathbb{R}$: $\|(A - z)x\| \geq |\operatorname{Im} z| \, \|x\|$.

(2) For all $z \in \mathbb{C} \setminus \mathbb{R}$: $\ker(A - z) = \{0\}$ and $(A - z)$ is continuously invertible on $(A - z)\mathcal{D}(A)$ with
$$\left\|(A - z)^{-1}\right\| = \sup_{x \in (A-z)\mathcal{D}(A),\ \|x\|=1} \left\|(A - z)^{-1}x\right\| \leq \frac{1}{|\operatorname{Im} z|}.$$

(3) If A is closed, then for all $z \in \mathbb{C} \setminus \mathbb{R}$: $(A - z)\mathcal{D}(A) = \overline{(A - z)\mathcal{D}(A)}$.

Proof. (1) Writing $A - z = A - \operatorname{Re} z - \mathrm{i} \operatorname{Im} z$,
$$\|(A - z)x\|^2 = \|(A - \operatorname{Re} z)x\|^2 + (\operatorname{Im} z)^2 \|x\|^2 + 2 \operatorname{Re} \underbrace{\langle (A - \operatorname{Re} z)x, -\mathrm{i} \operatorname{Im} z \, x \rangle}_{\in \mathrm{i}\mathbb{R}} \geq (\operatorname{Im} z)^2 \|x\|^2 ,$$
since $\langle (A - \operatorname{Re} z)x, x \rangle = \langle x, (A - \operatorname{Re} z)x \rangle \in \mathbb{R}$.

(2) From (1): $\ker(A - z) = \{0\}$ for all $z \in \mathbb{C} \setminus \mathbb{R}$, i.e., $A - z : \mathcal{D}(A) \to \mathcal{H}$ is injective, thus invertible on $(A - z)\mathcal{D}(A)$. For any $y \in (A - z)\mathcal{D}(A)$, by (1): $\left\|(A - z)^{-1}y\right\| \leq \frac{1}{|\operatorname{Im} z|} \left\|(A - z)(A - z)^{-1}y\right\| = \frac{\|y\|}{|\operatorname{Im} z|}$.

(3) Let $(y_n) \subset (A - z)\mathcal{D}(A)$ with $y_n \to y$. Then $x_n := (A - z)^{-1}y_n \in \mathcal{D}(A)$ converges, since $(A - z)^{-1} : (A - z)\mathcal{D}(A) \to \mathcal{D}(A)$ is continuous by (2). By closedness of A this implies $x_n \to x \in \mathcal{D}(A)$ and $y_n \to (A - z)x = y \in (A - z)\mathcal{D}(A)$. □

Theorem 8.4.4: Spectra and resolvents of self-adjoint operators

Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be self-adjoint. Then:

(1) $\sigma(A) \subset \mathbb{R}$.

(2) $\sigma_r(A) = \emptyset$.

(3) If $z \in \varrho(A)$, then $\overline{z} \in \varrho(A)$ and $\left[(A - z)^{-1}\right]^* = (A - \overline{z})^{-1}$. In particular, $(A - z)^{-1}$ is normal.

(4) If $0 \in \mathbb{C} \setminus \sigma_p(A)$, then $A^{-1} : A\mathcal{D}(A) \to \mathcal{H}$ is self-adjoint.

Proof. (1) Let $z \in \mathbb{C} \setminus \mathbb{R}$. By Lemma 8.4.10: (i) $\ker(A - z) = \{0\}$; (ii) $(A - z)\mathcal{D}(A) = \overline{(A - z)\mathcal{D}(A)} = \ker(A^* - \overline{z})^\perp = \ker(A - \overline{z})^\perp = \mathcal{H}$ (Lemma 8.4.7). Thus $z \in \varrho(A)$.

(2) If $\ker(A - \lambda) = \{0\}$ with $\lambda \in \mathbb{R}$, then by Lemma 8.4.7 $\overline{(A - \lambda)\mathcal{D}(A)} = \ker(A^* - \lambda)^\perp = \ker(A - \lambda)^\perp = \mathcal{H}$. Thus $\sigma_r(A) = \emptyset$.

(3) If $z \in \varrho(A)$, then either $z = \overline{z} \in \varrho(A)$ or $\operatorname{Im} z \neq 0$ and hence $\overline{z} \in \varrho(A)$ by (1). Let $x, y \in \mathcal{H}$. Since $(A - \overline{z})\mathcal{D}(A) = \mathcal{H} = (A - z)\mathcal{D}(A)$, there are $x_0, y_0 \in \mathcal{D}(A)$ with $x = (A - z)x_0$, $y = (A - \overline{z})y_0$, and since $A = A^*$:
$$\left\langle (A - z)^{-1}x, y \right\rangle = \langle x_0, (A - \overline{z})y_0 \rangle = \langle (A - z)x_0, y_0 \rangle = \left\langle x, (A - \overline{z})^{-1}y \right\rangle .$$
Hence $\left[(A - z)^{-1}\right]^* = (A - \overline{z})^{-1}$. Normality is trivial in case $z = \overline{z}$ and follows from the first resolvent formula in case $z \neq \overline{z}$:
$$\frac{1}{z - \overline{z}} \left[(A - z)^{-1} - (A - \overline{z})^{-1}\right] = (A - z)^{-1}(A - \overline{z})^{-1} = (A - \overline{z})^{-1}(A - z)^{-1}.$$

(4) *Exercise.* Note that $A : \mathcal{D}(A) \to A\mathcal{D}(A)$ is invertible, since $\ker A = \{0\}$. □

9 Unbounded self-adjoint operators

9.1 Criteria

Theorem 9.1.1

Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be symmetric.

(1) The following statements are equivalent:

(i) A is self-adjoint;
(ii) $(A - z)\mathcal{D}(A) = \mathcal{H}$ for all $z \in \mathbb{C} \setminus \mathbb{R}$;
(iii) there are $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$: $(A - z_\pm)\mathcal{D}(A) = \mathcal{H}$;
(iv) A is closed and there are $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$: $\ker(A^* - z_\pm) = \{0\}$;
(v) A is closed and $\sigma(A) \subset \mathbb{R}$.

(2) The following statements are equivalent:

(i) A is esa;
(ii) $\overline{(A - z)\mathcal{D}(A)} = \mathcal{H}$ for all $z \in \mathbb{C} \setminus \mathbb{R}$;
(iii) there are $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$: $\overline{(A - z_\pm)\mathcal{D}(A)} = \mathcal{H}$;
(iv) there are $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$: $\ker(A^* - z_\pm) = \{0\}$;
(v) $\sigma(\overline{A}) \subset \mathbb{R}$.

Proof of (1). (i)⇒(iv): A is closed since $A = A^*$ and A^* is closed. Since $\sigma(A) \subset \mathbb{R}$ by Theorem 8.4.4, we have for all $z \in \mathbb{C} \setminus \mathbb{R}$: $\{0\} = \ker(A - z) = \ker(A^* - z)$.

(iv)⇒(iii): By Lemmas 8.4.10 and 8.4.7, $(A - \overline{z_\pm})\mathcal{D}(A) = \overline{(A - \overline{z_\pm})\mathcal{D}(A)} = \ker(A^* - z_\pm)^\perp = \mathcal{H}$; note that $\overline{z_\mp}$ lie in the upper/lower half-plane, respectively.

(iii)⇒(v): By Lemma 8.4.10 one concludes from $(A - z_\pm)\mathcal{D}(A) = \mathcal{H}$ that $z_\pm \in \varrho(A)$. Take $z \in \mathbb{C}$ with $|z - z_\pm| < |\operatorname{Im} z_\pm| \leq \left\|(A - z_\pm)^{-1}\right\|^{-1}$. Then $z \in \varrho(A)$ by Corollary 8.4.5. Repeating this step allows us to reach every $z \in \mathbb{C} \setminus \mathbb{R}$, i.e., $\sigma(A) \subset \mathbb{R}$. Since $A - z_\pm$ is boundedly invertible, A is closed.

(v)⇒(ii): evident.

(ii)⇒(i): Since $A \subset A^*$, it suffices to show $\mathcal{D}(A^*) \subset \mathcal{D}(A)$. Pick $z \in \mathbb{C} \setminus \mathbb{R}$. Then $A - z$ and $A - \overline{z}$ are bijective maps from $\mathcal{D}(A)$ to $\mathcal{H}$, since $(A - z)\mathcal{D}(A) = (A - \overline{z})\mathcal{D}(A) = \mathcal{H}$ and, by Lemma 8.4.10, $\ker(A - z) = \ker(A - \overline{z}) = \{0\}$. Hence $\mathcal{D}\big((A - z)^{-1}\big) = \mathcal{H}$. Pick $x \in \mathcal{D}(A^*)$ and set $x_0 := (A - z)^{-1}(A^* - z)x$. Then:

(a) $x_0 \in \mathcal{D}(A)$ and $Ax_0 = A^*x_0$ since $A \subset A^*$;

(b) $(A - z)x_0 = (A^* - z)x \Rightarrow (A^* - z)(x_0 - x) = (A^* - z)x_0 - (A^* - z)x = 0$, i.e.,

$$x_0 - x \in \ker(A^* - z) = \big[(A - \overline{z})\mathcal{D}(A)\big]^\perp = \mathcal{H}^\perp = \{0\}$$

by Lemma 8.4.7. Hence $x = x_0 \in \mathcal{D}(A)$.

Proof of (2): Exercise. □

Example 9.1.1:

(1) $D_0 : C_c^\infty(\mathbb{R}) \to L^2(\mathbb{R})$, $D_0\psi := -\mathrm{i}\psi'$.

(i) D_0 is symmetric, since $D_0 \subset D_0^* = D_{\max}$ with $D_{\max} : W^{1,2}(\mathbb{R}) \to L^2(\mathbb{R})$, $D_{\max}\psi = -\mathrm{i}\psi'$.

(ii) D_0 is not self-adjoint, since e.g. $\psi(x) = \mathrm{e}^{-x^2}$ is in $\mathcal{D}(D_0^*)$, but not in $\mathcal{D}(D_0) = C_c^\infty(\mathbb{R})$.

(iii) D_0 is esa, since for $\varphi \in \ker(D_0^* \pm \mathrm{i}) = \ker(D_{\max} \pm \mathrm{i}) \subset W^{1,2}(\mathbb{R})$ we have for all $\psi \in C_c^\infty(\mathbb{R})$:

$$\int_{-\infty}^{\infty} \overline{\varphi(x)}\big[-\mathrm{i}\psi'(x) \pm \mathrm{i}\psi(x)\big]\,\mathrm{d}x = 0.$$

By the fundamental theorem of calculus: $\varphi' = \pm\varphi$ a.e. Since $W^{1,2}(\mathbb{R}) \subset C(\mathbb{R})$, this implies $\varphi(x) = C\mathrm{e}^{\pm x}$ for some $C \in \mathbb{C}$. Hence $\varphi \notin L^2(\mathbb{R})$ unless $C = 0$, i.e., $\varphi = 0$.

(2) Let $\beta \in [0, 2\pi)$ and set (β-boundary conditions)

$$D_\beta : \mathcal{D}(D_\beta) \to L^2(0,1), \quad D_\beta\psi := -\mathrm{i}\psi', \qquad \mathcal{D}(D_\beta) := \{\psi \in C^1([0,1]) \mid \psi(0) = \mathrm{e}^{\mathrm{i}\beta}\psi(1)\}.$$

(i) D_β is symmetric, since for $\varphi, \psi \in \mathcal{D}(D_\beta)$:

$$\begin{aligned}\langle D_\beta\psi, \varphi\rangle &= \int_0^1 -\mathrm{i}\psi'\overline{\varphi}\,\mathrm{d}x \overset{\text{P.I.}}{=} \big[-\mathrm{i}\psi\overline{\varphi}\big]_0^1 + \int_0^1 \psi\overline{(-\mathrm{i}\varphi')}\,\mathrm{d}x \\ &= (-\mathrm{i})\underbrace{\big[\psi(1)\overline{\varphi(1)} - \psi(0)\overline{\varphi(0)}\big]}_{=0} + \langle\psi, D_\beta\varphi\rangle\,.\end{aligned}$$

(ii) D_β is esa, since there are $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$ and $\overline{(D_\beta - z_\pm)\mathcal{D}(D_\beta)} = L^2(0,1)$.

Proof of 2(ii). It suffices to show that $C([0,1]) \subset (D_\beta - z_\pm)\mathcal{D}(D_\beta)$ for some $z_\pm \in \mathbb{C}$ with $\operatorname{Im} z_+ > 0$, $\operatorname{Im} z_- < 0$. Let $f \in C([0,1])$ and $z \in \mathbb{C} \setminus \mathbb{R}$. We search for a solution of

$$-\mathrm{i}\psi' - z\psi = f \quad \text{with} \quad \psi(0) = \mathrm{e}^{\mathrm{i}\beta}\psi(1).$$

Equivalently, $\mathrm{e}^{\mathrm{i}xz}\frac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{e}^{-\mathrm{i}xz}\psi(x)\big) = \mathrm{i}f(x)$, i.e.,

$$\psi(x) = \mathrm{e}^{\mathrm{i}zx}\psi(0) + \mathrm{i}\mathrm{e}^{\mathrm{i}zx}\int_0^x \mathrm{e}^{-\mathrm{i}zy}f(y)\,\mathrm{d}y, \qquad (*)$$

and $\psi(0) = \mathrm{e}^{\mathrm{i}\beta}\psi(1) = \mathrm{e}^{\mathrm{i}(z+\beta)}\big[\psi(0) + \mathrm{i}\int_0^1 \mathrm{e}^{-\mathrm{i}zy}f(y)\,\mathrm{d}y\big]$, i.e.,

$$\big(1 - \mathrm{e}^{\mathrm{i}(\beta+z)}\big)\psi(0) = \mathrm{i}\mathrm{e}^{\mathrm{i}(\beta+z)}\int_0^1 \mathrm{e}^{-\mathrm{i}zy}f(y)\,\mathrm{d}y.$$

Pick $z \in \mathbb{C} \setminus \mathbb{R}$ such that $\mathrm{e}^{\mathrm{i}(\beta+z)} \neq 1$. Then ψ from $(*)$ with $\psi(0) = \big(\mathrm{e}^{-\mathrm{i}(\beta+z)} - 1\big)^{-1}\mathrm{i}\int_0^1 \mathrm{e}^{-\mathrm{i}zy}f(y)\,\mathrm{d}y$ satisfies $(D_\beta - z)\psi = f$. □

Outlook.

Definition 9.1.2: For a symmetric operator $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, and $z \in \mathbb{C} \setminus \mathbb{R}$, one calls $d(z) := \dim\ker(A^* - \overline{z})$ a *deficiency index*.

Remark 9.1.3 (Properties):

(1) A is esa if and only if there are $z_\pm \in \mathbb{C}^\pm$ with $d(z_\pm) = 0$ (Theorem 9.1.1(2)(iv)).

(2) The deficiency index $d(z)$ is constant, $n_\pm$, in the upper/lower complex half-plane. One may show (*von Neumann's criterion*): A has self-adjoint extensions if and only if $n_+ = n_-$.

Reference: P. Lax, *Functional analysis*, Wiley 2002.

9.2 Stability of self-adjointness and closedness

Definition 9.2.1: Let $A : \mathcal{D}(A) \to Y$, $B : \mathcal{D}(B) \to Y$, $\mathcal{D}(A), \mathcal{D}(B) \subset X$, be densely defined on Banach spaces X, Y. The operator B is called *A-bounded* if and only if

(i) $\mathcal{D}(A) \subset \mathcal{D}(B)$,

(ii) there are $a, b \in [0, \infty)$ such that for all $x \in \mathcal{D}(A)$: $\|Bx\| \leq a \|Ax\| + b \|x\|$.

The infimum over all such a's is called the *A-bound* of B.

Theorem 9.2.1: Kato–Rellich

Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be self-adjoint and $B : \mathcal{D}(B) \to \mathcal{H}$ be symmetric and A-bounded with A-bound < 1. Then $A + B : \mathcal{D}(A) \to \mathcal{H}$ is self-adjoint.

Proof. $A + B : \mathcal{D}(A) \to \mathcal{H}$ is symmetric, since for all $x, y \in \mathcal{D}(A)$, using $A = A^*$ and $B \subset B^*$:

$$\langle (A+B)x, y\rangle = \langle Ax, y\rangle + \langle Bx, y\rangle = \langle x, Ay\rangle + \langle x, By\rangle = \langle x, (A+B)y\rangle\,.$$

Let $x \in \mathcal{D}(A)$, $\eta \in \mathbb{R} \setminus \{0\}$. Then $\|(A + \mathrm{i}\eta)x\|^2 = \|Ax\|^2 + \eta^2 \|x\|^2$. Thus $x = (A + \mathrm{i}\eta)^{-1} y$ with $y = (A + \mathrm{i}\eta)x$ satisfies

$$\left\|A(A+\mathrm{i}\eta)^{-1}y\right\| \leq \|y\| \qquad \text{and} \qquad \left\|(A+\mathrm{i}\eta)^{-1}y\right\| \leq \|y\| / |\eta|\,.$$

Hence

$$\left\|B(A+\mathrm{i}\eta)^{-1}y\right\| \leq a\left\|A(A+\mathrm{i}\eta)^{-1}y\right\| + b\left\|(A+\mathrm{i}\eta)^{-1}y\right\| \leq a\|y\| + \frac{b}{|\eta|}\|y\|\,.$$

Pick $|\eta|$ large enough such that $a + \frac{b}{|\eta|} < 1$. By von Neumann's theorem (Theorem 8.4.2), $C := \mathbb{1} + B(A + \mathrm{i}\eta)^{-1}$ is a bounded invertible operator from $\mathcal{H}$ to $\mathcal{H}$. Since $\mathcal{H} = (A + \mathrm{i}\eta)\mathcal{D}(A)$, we have

$$\mathcal{H} = C(A+\mathrm{i}\eta)\mathcal{D}(A) = (A + B + \mathrm{i}\eta)\mathcal{D}(A)$$

for both signs of η. Thus $A + B$ is self-adjoint by Theorem 9.1.1. □

Remark 9.2.2: One may also show that if $A : \mathcal{D}(A) \to \mathcal{H}$ is esa and $B : \mathcal{D}(A) \to \mathcal{H}$ is symmetric with A-bound < 1, then $A + B : \mathcal{D}(A) \to \mathcal{H}$ is esa with $\overline{A + B} = \overline{A} + \overline{B}$.

Theorem 9.2.2: Stability of closedness

Let $A : \mathcal{D}(A) \to Y$ and $B : \mathcal{D}(B) \to Y$, $\mathcal{D}(A), \mathcal{D}(B) \subset X$, be densely defined on Banach spaces X, Y, and assume that B is A-bounded with A-bound < 1. Then the following statements are equivalent:

(1) A is closed,

(2) $A + B : \mathcal{D}(A) \to Y$ is closed.

Proof. Exercise. □

9.3 Quadratic forms and Friedrichs' extension

Definition 9.3.1: An (unbounded) *quadratic form* is a map

$$q : Q(q) \times Q(q) \to \mathbb{C}$$

with $Q(q)$ a dense subspace of a Hilbert space $\mathcal{H}$, which is called the *form domain*, such that the map $Q(q) \ni x \mapsto q(x, y)$ is linear for all $y \in Q(q)$ and $Q(q) \ni y \mapsto q(x, y)$ is skew-linear for all $x \in Q(q)$. A form q is called

(1) *hermitian* if $q(x,y) = \overline{q(y,x)}$ for all $x, y \in Q(q)$,

(2) *semibounded* if there is $c \in \mathbb{R}$ such that $q(x,x) \geq c\,\|x\|^2$ for all $x \in Q(q)$.

Example 9.3.2: Any symmetric $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, gives rise to a hermitian quadratic form

$$q_A(x,y) = \langle Ax, y\rangle\,, \qquad Q(q_A) = \mathcal{D}(A).$$

Any semibounded hermitian quadratic form defines a scalar product

$$\langle x, y\rangle_q := q(x,y) + (1-c)\,\langle x, y\rangle\,, \qquad x, y \in Q(q),$$

(note $\langle x, x\rangle_q \geq \|x\|^2$) and renders $Q(q)$ a pre-Hilbert space. Its completion

$$\mathcal{H}_q := \overline{Q(q)}^{\|\cdot\|_q}, \qquad \|\cdot\|_q := \sqrt{\langle\cdot,\cdot\rangle_q},$$

is a Hilbert space.

Definition 9.3.3: A semibounded hermitian quadratic form q is called *closed* if $\mathcal{H}_q = Q(q)$. It is called *closable* if $\mathcal{H}_q$ is a subspace of $\mathcal{H}$.

Remark 9.3.4:

(1) Closedness requires to check whether any Cauchy sequence $(x_n) \subset Q(q)$ w.r.t. $\|\cdot\|_q$ converges to some $x \in Q(q)$. Since $\|\cdot\| \leq \|\cdot\|_q$ this also implies $\lim_{n\to\infty} \|x_n - x\| = 0$: $0 = \lim_n \|x - x_n\|_q \geq \lim_n \|x - x_n\|$.

(2) The embedding $\iota : (Q(q), \|\cdot\|_q) \to (\mathcal{H}, \|\cdot\|)$, $\iota(x) = x$, is always continuous with norm ≤ 1. It can hence be uniquely extended to a continuous map $\hat\iota : (\mathcal{H}_q, \|\cdot\|_q) \to (\mathcal{H}, \|\cdot\|)$, $\hat\iota(x) = x$. Closability requires $\hat\iota$ to be injective.

Theorem 9.3.1

Any semibounded, hermitian, closed quadratic form $q : Q(q)\times Q(q) \to \mathbb{C}$ defines a self-adjoint operator $A : \mathcal{D}(A) \to \mathcal{H}$ with

$$\mathcal{D}(A) = \{x \in Q(q) \mid \text{there is } x' \in \mathcal{H} \text{ s.t. for all } y \in Q(q) :\ q(x,y) = \langle x', y\rangle\},$$
$$Ax = x',$$

and

$$\langle Ax, y\rangle = q(x,y) \qquad \text{for all } x \in \mathcal{D}(A),\ y \in Q(q). \tag{$*$}$$

Moreover, A is the only self-adjoint operator with property $(*)$ and $\mathcal{D}(A) \subset Q(q)$.

Proof. W.l.o.g. $c = 1$. Let $x \in \mathcal{H}$. The functional

$$j_x : \mathcal{H}_q \to \mathbb{C}, \qquad j_x(y) := \langle y, x\rangle\,,$$

is continuous since $|j_x(y)| \leq \|x\|\,\|y\| \leq \|x\|\,\|y\|_q$. By Riesz' theorem there is a unique $x_0 \in \mathcal{H}_q$ such that for all $y \in \mathcal{H}_q$: $\langle y, x\rangle = \langle y, x_0\rangle_q = q(y, x_0)$. Since x_0 depends linearly on x, this defines a linear operator

$$B : \mathcal{H} \to \mathcal{H}_q, \qquad Bx = x_0.$$

(i) B is symmetric, since for $x, y \in \mathcal{H}$, using that q is hermitian: $\langle By, x\rangle = q(By, Bx) = \overline{q(Bx, By)} = \overline{\langle Bx, y\rangle} = \langle y, Bx\rangle$. Since $\mathcal{D}(B) = \mathcal{H}$, B is bounded and self-adjoint (Hellinger–Toeplitz).

(ii) B is injective, since $Bx = 0$ implies for all $y \in Q(q)$: $0 = q(y, Bx) = \langle y, x\rangle$, i.e., $x \in Q(q)^\perp = \mathcal{H}^\perp = \{0\}$, as $Q(q)$ is dense in $\mathcal{H}$.

From Lemma 8.4.7 one concludes $\mathcal{H} = (\ker B)^\perp = \big((\operatorname{ran} B)^\perp\big)^\perp = \overline{\operatorname{ran} B}$, i.e., $\operatorname{ran} B$ is dense in $\mathcal{H}$ and thus

$$A : \operatorname{ran} B \to \mathcal{H}, \qquad A = B^{-1},$$

is densely defined and well-defined. Note: $\operatorname{ran} B = \mathcal{D}(A)$. Since B is self-adjoint, $B^{-1} = A$ is self-adjoint (Theorem 8.4.4(4)). Moreover, for all $x \in \mathcal{D}(A)$, $y \in Q(q)$:

$$q(x, y) = \overline{q(y, x)} = \overline{q(y, BAx)} = \overline{\langle y, Ax\rangle} = \langle Ax, y\rangle\,.$$

Uniqueness: Suppose $C : \mathcal{D}(C) \to \mathcal{H}$ is another self-adjoint operator with $\langle Cx, y\rangle = q(x, y)$, $x \in \mathcal{D}(C)$, $y \in Q(q)$. Then for $x \in \mathcal{D}(C)$, $y \in \mathcal{D}(A)$:

$$\langle Ay, x\rangle = q(y, x) = \overline{q(x, y)} = \overline{\langle Cx, y\rangle} = \langle y, Cx\rangle\,.$$

Hence $x \in \mathcal{D}(A^*)$ and $A^*x = Cx$, i.e., $C \subset A^* = A \Rightarrow A = A^* \subset C^* = C$, i.e., $A = C$. □

Example 9.3.5 (Laplace quadratic form):

$$q : W^{1,2}(\mathbb{R}^d) \times W^{1,2}(\mathbb{R}^d) \to \mathbb{C}, \qquad q(\varphi, \psi) := \int \nabla\varphi \cdot \overline{\nabla\psi}\,\mathrm{d}x = \int \sum_{j=1}^{d} \partial_j\varphi\,\overline{\partial_j\psi}\,\mathrm{d}x.$$

- $q \geq 0$, q hermitian;
- q is closed since $W^{1,2}(\mathbb{R}^d)$ is complete w.r.t. the Sobolev norm

$$\|\psi\|_{W^{1,2}} := \sqrt{q(\psi, \psi) + \|\psi\|^2} = \|\psi\|_q\,.$$

Corollary 9.3.6: Let $\mathcal{G}, \mathcal{H}$ be Hilbert spaces and $A : \mathcal{D}(A) \to \mathcal{G}$ be densely defined on $\mathcal{D}(A) \subset \mathcal{H}$ and closed. Then $A^*A : \mathcal{D}(A^*A) \to \mathcal{H}$ is self-adjoint on

$$\mathcal{D}(A^*A) = \{x \in \mathcal{D}(A) \mid Ax \in \mathcal{D}(A^*)\}.$$

Proof. Consider $q : \mathcal{D}(A) \times \mathcal{D}(A) \to \mathbb{C}$, $q(x, y) = \langle Ax, Ay\rangle$.

- $q \geq 0$, q hermitian.
- q is closed, since $\|x\|_q = \sqrt{\|Ax\|^2 + \|x\|^2}$ is equivalent to the graph norm $\|x\|_A$ and $\overline{\mathcal{D}(A)}^{\|\cdot\|_A} = \mathcal{D}(A)$ by closedness of A (Theorem 8.1.1).

Hence there is a unique self-adjoint operator $B : \mathcal{D}(B) \to \mathcal{H}$ with

$$\begin{aligned}\mathcal{D}(B) &= \{x \in \mathcal{D}(A) \mid \text{there is } x' \in \mathcal{H} \text{ s.t. for all } y \in \mathcal{D}(A) : \ \langle x', y\rangle = q(x, y) = \langle Ax, Ay\rangle\} \\ &= \{x \in \mathcal{D}(A) \mid Ax \in \mathcal{D}(A^*)\} = \mathcal{D}(A^*A)\end{aligned}$$

and $Bx = x'$, i.e., $B = A^*A$. □

Theorem 9.3.2: Friedrichs' extension

Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be a symmetric operator which is positive: $\langle Ax, x\rangle \geq 0$ for all $x \in \mathcal{D}(A)$. Then:

(1) the quadratic form $q_A(x, y) := \langle Ax, y\rangle$, $x, y \in \mathcal{D}(A)$, is closable with closure $\hat{q}_A$;

(2) $\hat{q}_A$ is the quadratic form of a unique self-adjoint operator $\hat{A} : \mathcal{D}(\hat{A}) \to \mathcal{H}$ which obeys (i) $A \subset \hat{A}$ and (ii) $\hat{A}$ is positive.

Remark 9.3.7: $\hat{A}$ is called the *Friedrichs extension* of A. A similar result applies if for some $c \in \mathbb{R}$: $\langle Ax, x\rangle \geq c\,\|x\|^2$, $x \in \mathcal{D}(A)$.

Proof. (1) Let $\mathcal{H}_{q_A} := \overline{\mathcal{D}(A)}^{\|\cdot\|_{q_A}}$, $\|x\|_{q_A} := \sqrt{q_A(x,x) + \|x\|^2}$, and consider $x \in \mathcal{H}_{q_A}$ with $\hat{\imath}(x) = 0$. There is a Cauchy sequence $(x_n) \subset \mathcal{D}(A)$ such that

$$\|\hat{\imath}(x_n)\| = \|\hat{\imath}(x_n - x)\| \leq \|x_n - x\|_{q_A} \to 0.$$

In particular $(\|x_n\|_{q_A})$ is a sequence bounded by some $M < \infty$. Using $\langle\cdot,\cdot\rangle_{q_A} = q_A(\cdot,\cdot) + \langle\cdot,\cdot\rangle = \langle(\mathbb{1} + A)\cdot,\cdot\rangle$,

$$\begin{aligned}\|x_n\|_{q_A}^2 &= \langle x_n, x_n\rangle_{q_A} = \langle x_n - x_m, x_n\rangle_{q_A} + \langle x_m, x_n\rangle_{q_A} = \langle x_n - x_m, x_n\rangle_{q_A} + \langle x_m, (\mathbb{1} + A)x_n\rangle \\ &\leq \|x_n - x_m\|_{q_A} \underbrace{\|x_n\|_{q_A}}_{\leq M} + \underbrace{\|x_m\|}_{\to 0} \|(\mathbb{1} + A)x_n\|\,.\end{aligned}$$

Thus $\|x_n\|_{q_A}^2 \leq M \liminf_{m\to\infty} \|x_n - x_m\|_{q_A}$, and since (x_n) is a Cauchy sequence w.r.t. $\|\cdot\|_{q_A}$ we have $\lim_{n\to\infty} \|x_n\|_{q_A} = 0$, i.e., $x = 0$. Hence $\hat{\imath}$ is injective, i.e., q_A is closable.

(2) Thanks to Theorem 9.3.1 it remains to show (i) and (ii). (i) For $x \in \mathcal{D}(A) \subset Q(\hat{q}_A)$, $y \in \mathcal{D}(\hat{A}) \subset Q(\hat{q}_A)$: $\langle Ax, y\rangle = \hat{q}_A(x,y) = \left\langle x, \hat{A}y\right\rangle$. Hence $\hat{A} \subset A^* \Rightarrow A \subset \overline{A} = (A^*)^* \subset \hat{A}^* = \hat{A}$ (Theorem 8.3.2). (ii) For $x \in \mathcal{D}(\hat{A})$: $\left\langle \hat{A}x, x\right\rangle = \hat{q}_A(x,x) \geq 0$ since $Q(\hat{q}_A) = \overline{\mathcal{D}(A)}^{\|\cdot\|_{q_A}}$ and $q_A \geq 0$. □

Example 9.3.8 (Dirichlet Laplacian on $\Lambda := (a_1, b_1) \times \cdots \times (a_d, b_d)$): Symmetric operator

$$L_0 : C_c^\infty(\Lambda) \to L^2(\Lambda), \qquad L_0\psi := -\sum_{j=1}^d \partial_j^2 \psi.$$

Closable form $q(\varphi,\psi) := \langle L_0\varphi, \psi\rangle \overset{\text{P.I.}}{=} \int_\Lambda \sum_{j=1}^d \partial_j\varphi\,\overline{\partial_j\psi}\,\mathrm{d}x$ with closure domain $\overline{C_c^\infty(\Lambda)}^{\|\cdot\|_q} =: W_0^{1,2}(\Lambda)$. The self-adjoint operator $-\Delta_D$ associated with the closure of q is called the (negative) *Dirichlet Laplacian*.

Theorem 9.3.3: KLMN

Let $q : Q(q) \times Q(q) \to \mathbb{C}$ be a positive, hermitian, closed quadratic form. Let $p : Q(q) \times Q(q) \to \mathbb{C}$ be a hermitian quadratic form such that for some $a \in [0,1)$ and $b \in \mathbb{R}$:

$$|p(x,x)| \leq a\,q(x,x) + b\,\|x\|^2\,, \qquad x \in Q(q).$$

Then the form $q + p : Q(q) \times Q(q) \to \mathbb{C}$ is semibounded, hermitian and closed.

Proof. Exercise. □

Example 9.3.9:

(1) $-\partial_x^2 + c\,\delta_0$ on $L^2(\mathbb{R})$ with $c \in \mathbb{R}$:

(i) $q(\psi,\psi) := \int |\psi'|^2\,\mathrm{d}x$, $\psi \in W^{1,2}(\mathbb{R})$; $q \geq 0$, hermitian, closed.

(ii) $p(\psi,\psi) := c\,|\psi(0)|^2$, $\psi \in W^{1,2}(\mathbb{R}) \subset C(\mathbb{R})$, i.e., $p(\varphi,\psi) = c\,\varphi(0)\overline{\psi(0)}$. p is hermitian with, for $\varepsilon > 0$ arbitrary,

$$|p(\psi,\psi)| \leq |c|\,\|\psi\|_\infty^2 \leq 2\,|c|\,\|\psi'\|\,\|\psi\| \leq \varepsilon^2\,\|\psi'\|^2 + \frac{|c|^2}{\varepsilon^2}\,\|\psi\|^2\,,$$

using $2|ab| \le |a|^2 + |b|^2$ and

$$|\psi(x)|^2 = \int_{-\infty}^{x} \frac{\mathrm{d}}{\mathrm{d}y}\big(\psi(y)\overline{\psi(y)}\big)\,\mathrm{d}y = \int_{-\infty}^{x} \big[\psi'(y)\overline{\psi(y)} + \psi(y)\overline{\psi'(y)}\big]\,\mathrm{d}y \le 2\,\big\|\psi'\big\|\,\|\psi\|\,.$$

Hence $(p+q)(\psi,\psi) = \int |\psi'|^2\,\mathrm{d}x + c\,|\psi(0)|^2$ gives rise to a unique self-adjoint operator.

(2) *Consequence of Hardy's inequality* ($d \ge 3$):

$$\int \frac{|\psi(x)|^2}{|x|^2}\,\mathrm{d}x \le \frac{4}{(d-2)^2}\int |\nabla\psi(x)|^2\,\mathrm{d}x, \qquad \psi \in W^{1,2}(\mathbb{R}^d).$$

(Ref.: J. Weidmann, *Linear operators in Hilbert spaces*, Springer 1980.) Any operator of the form $-\Delta + \frac{c}{|x|^\alpha}$ with $c \in \mathbb{R}$ and $\alpha \in [0,2)$ can be rendered self-adjoint in $L^2(\mathbb{R}^d)$, since Hardy's inequality implies for $\psi \in W^{1,2}(\mathbb{R}^d)$, using $\frac{1}{|x|^\alpha} = |x|^{2-\alpha}\,\frac{1}{|x|^2}$:

$$\begin{aligned}
|c|\int_{\mathbb{R}^d} \frac{|\psi(x)|^2}{|x|^\alpha}\,\mathrm{d}x &= |c|\int_{|x|\le R} \frac{|\psi(x)|^2}{|x|^\alpha}\,\mathrm{d}x + |c|\int_{|x|>R} \frac{|\psi(x)|^2}{|x|^\alpha}\,\mathrm{d}x \\
&\le |c|\,R^{2-\alpha}\int \frac{|\psi(x)|^2}{|x|^2}\,\mathrm{d}x + \frac{|c|}{R^\alpha}\int |\psi(x)|^2\,\mathrm{d}x \\
&\le \underbrace{|c|\,R^{2-\alpha}\frac{4}{(d-2)^2}}_{<1 \text{ if } R \text{ is small enough}}\int |\nabla\psi(x)|^2\,\mathrm{d}x + \frac{|c|}{R^\alpha}\,\|\psi\|^2\,.
\end{aligned}$$

Hence we may apply the KLMN theorem to

$$q(\psi,\psi) = \int |\nabla\psi|^2\,\mathrm{d}x,\ \psi \in W^{1,2}(\mathbb{R}^d), \qquad p(\varphi,\psi) = c\int \frac{\varphi(x)\overline{\psi(x)}}{|x|^\alpha}\,\mathrm{d}x.$$

10 Semigroups of operators

Set-up in the following: X Banach space.

10.1 Strongly continuous semigroups

Definition 10.1.1: A family $Q(t)$, $t \ge 0$, of bounded operators on some Banach space X is called a (one-parameter) *semigroup of operators* if

(1) $Q(0) = \mathrm{id}_X$,

(2) $Q(t+s) = Q(t)Q(s)$ for all $t, s \ge 0$.

It is called *strongly continuous* or a C_0-*semigroup* if additionally for all $x \in X$

$$\lim_{t\downarrow 0} \|Q(t)x - x\| = 0.$$

For a C_0-semigroup we call

$$\mathcal{D}(A) := \Big\{x \in X \;\Big|\; \lim_{t\downarrow 0} \underbrace{\tfrac{1}{t}\big(Q(t)x - x\big)}_{=:A_t x} \text{ exists}\Big\}, \qquad Ax := \lim_{t\downarrow 0} A_t x,$$

its *infinitesimal generator*.

Theorem 10.1.1

Let $Q(t)$, $t \geq 0$, be a C_0-semigroup on X.

(1) For some $C, \gamma < \infty$ and all $t \geq 0$: $\|Q(t)\| \leq C\mathrm{e}^{\gamma t}$.

(2) For all $x \in X$: $[0,\infty) \ni t \mapsto Q(t)x$ is continuous.

(3) The infinitesimal generator A is densely defined and closed.

(4) The limit $Q'(t)x := \lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}\big(Q(t+\varepsilon) - Q(t)\big)x$ exists for all $x \in \mathcal{D}(A)$ and all $t \geq 0$, and

$$Q'(t)x = AQ(t)x = Q(t)Ax.$$

Proof. (1) *Step 1: For some $\delta > 0$, $C < \infty$ and all $t \in [0,\delta]$: $\|Q(t)\| \leq C$.* Proof by contradiction: Assume there is $(t_n) \subset [0,\infty)$ with $\lim_n t_n = 0$ and $\lim_n \|Q(t_n)\| = \infty$. Then there is $x \in X$ with $\lim_n \|Q(t_n)x\| = \infty$, since otherwise by the uniform boundedness principle $\sup_n \|Q(t_n)\| < \infty$. This contradicts $\lim_n \|Q(t_n)x - x\| = 0$.

Step 2: For some $C, \gamma < \infty$ and all $t \geq 0$: $\|Q(t)\| \leq C\mathrm{e}^{\gamma t}$. For $t > 0$ and $\delta > 0$ from Step 1 there is $n_\delta \in \mathbb{N}$ with $(n_\delta - 1)\delta \leq t \leq n_\delta \delta$. Hence $\|Q(t/n_\delta)\| \leq C$ by Step 1. From the semigroup property:

$$\|Q(t)\| = \|Q(t/n_\delta)^{n_\delta}\| \leq \|Q(t/n_\delta)\|^{n_\delta} \leq C^{n_\delta} \leq C\, C^{t/\delta}.$$

(2) Consider $s \leq t \leq T$, $x \in X$:

$$\|Q(t)x - Q(s)x\| = \big\|Q(s)\big(Q(t-s)x - x\big)\big\| \leq \|Q(s)\|\, \|Q(t-s)x - x\| \leq C\mathrm{e}^{\gamma s} \underbrace{\|Q(t-s)x - x\|}_{\to 0 \text{ for } t-s \to 0}.$$

(3) For $x \in X$, $t > 0$ set $M_t x := \frac{1}{t}\int_0^t Q(s)x\,\mathrm{d}s$. By (1), $\|M_t x\| \leq \frac{1}{t}\int_0^t \|Q(s)x\|\,\mathrm{d}s \leq C\mathrm{e}^{\gamma t}\|x\|$, i.e., $M_t \in \mathcal{L}(X)$. By (2):

$$\lim_{t \to 0} M_t x = x. \tag{$*$}$$

For $A_s = \frac{1}{s}\big(Q(s) - \mathrm{id}_X\big)$, $s > 0$, we claim:

(i) $A_s M_t = A_t M_s = M_t A_s$ for all $t, s > 0$;

(ii) for all $x \in X$: $M_t x \in \mathcal{D}(A)$, $AM_t x = A_t x$.

Proof of (i): For all $t, s > 0$, $x \in X$:

$$\begin{aligned}\int_s^{s+t} Q(\tau)x\,\mathrm{d}\tau - \int_0^t Q(\tau)x\,\mathrm{d}\tau &= \int_0^t \big(\underbrace{Q(s+\tau)}_{=Q(s)Q(\tau)=Q(\tau)Q(s)} x - Q(\tau)x\big)\,\mathrm{d}\tau \\ &= \big(Q(s) - \mathrm{id}_X\big)\int_0^t Q(\tau)x\,\mathrm{d}\tau = st\, A_s M_t x.\end{aligned} \tag{$**$}$$

Analogously, $\int_t^{t+s} Q(\tau)x\,\mathrm{d}\tau - \int_0^s Q(\tau)x\,\mathrm{d}\tau = st\, A_t M_s x$. Equality of the left-hand sides yields the first equality. Moreover

$$st\, A_s M_t x \overset{(**)}{=} \int_0^t \big(Q(s+\tau)x - Q(\tau)x\big)\,\mathrm{d}\tau = \int_0^t Q(\tau)\big(Q(s) - \mathrm{id}_X\big)x\,\mathrm{d}\tau = st\, M_t A_s x.$$

Proof of (ii): For any $x \in X$, $t > 0$:

$$\lim_{s \to 0} A_s M_t x \overset{\text{(i)}}{=} \lim_{s \to 0} A_t M_s x \overset{(*)}{=} A_t x, \qquad \text{i.e., } M_t x \in \mathcal{D}(A).$$

Now, since $\lim_{t\downarrow 0} M_t x = x$, property (ii) implies that $\mathcal{D}(A)$ is dense in X. Let $(x_n) \subset \mathcal{D}(A)$ with $x_n \to x$ and $Ax_n \to y$. Then for all $t > 0$:

$$A_t x \leftarrow A_t x_n \overset{\text{(ii)}}{=} A M_t x_n \overset{\text{(i)}}{=} M_t A x_n \to M_t y.$$

Since $\lim_{t\downarrow 0} M_t y = y$, we conclude $x \in \mathcal{D}(A)$ and $Ax = y$.

(4) For $t > 0$, $x \in \mathcal{D}(A)$:

$$Q(t)Ax = Q(t) \lim_{\varepsilon\downarrow 0} \tfrac{1}{\varepsilon}\big(Q(\varepsilon) - \mathrm{id}_X\big)x = \lim_{\varepsilon\downarrow 0} \tfrac{1}{\varepsilon}\big(Q(t+\varepsilon) - Q(t)\big)x = Q'(t)x.$$

Differentiating $A\int_0^t Q(\tau)x\,\mathrm{d}\tau \overset{\text{(ii)}}{=} Q(t)x - x$, we get $AQ(t)x = Q'(t)x$. □

Corollary 10.1.2: Let $Q(t)$, $t \geq 0$, and $S(t)$, $t \geq 0$, be C_0-semigroups and A and B their infinitesimal generators, respectively. Then

$$A = B \quad \Rightarrow \quad Q(t) = S(t) \text{ for all } t \geq 0.$$

Proof. Let $x \in \mathcal{D}(A) = \mathcal{D}(B)$, $t > 0$. From Theorem 10.1.1(4), $(0,t) \ni s \mapsto Q(t-s)S(s)x$ is differentiable with

$$\frac{\mathrm{d}}{\mathrm{d}s} Q(t-s)S(s)x = -AQ(t-s)S(s)x + Q(t-s)BS(s)x = Q(t-s)\big[B - A\big]S(s)x = 0.$$

Thus this function is constant, so $Q(t)x = S(t)x$ (using $Q(0)x = x = S(0)x$). Since $\mathcal{D}(A)$ is dense in X, we conclude the assertion. □

Definition 10.1.3: A semigroup $Q(t)$, $t \geq 0$, is *uniformly continuous* on X if

$$\lim_{t\downarrow 0} \|Q(t) - \mathrm{id}_X\| = 0.$$

Theorem 10.1.2

Let $Q(t)$, $t \geq 0$, be a C_0-semigroup. Then its infinitesimal generator A is bounded if and only if $Q(t)$, $t \geq 0$, is uniformly continuous. In this case

$$Q(t) = \exp(tA) = \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n,$$

$$Q'(t) := \lim_{s\to t} \frac{Q(t) - Q(s)}{t - s} = AQ(t) = Q(t)A.$$

Proof. Exercise. □

10.2 Contractions and the Hille–Yosida theorem

Definition 10.2.1: A semigroup $Q(t)$, $t \geq 0$, is called a *contraction semigroup* if $\|Q(t)\| \leq 1$ for all $t \geq 0$.

Remark 10.2.2: If for some $\gamma > 0$ and all $t \geq 0$: $\|Q(t)\| \leq \mathrm{e}^{\gamma t}$, then $S(t) := \mathrm{e}^{-\gamma t}Q(t)$ is a contraction semigroup.

Example 10.2.3: If $A \in \mathcal{L}(X)$ satisfies $A \leq \gamma\,\mathrm{id}_X$ for some $\gamma \in \mathbb{R}$, then $Q(t) = \exp(tA)$ obeys $\|Q(t)\| \leq \mathrm{e}^{\gamma t}$.

Theorem 10.2.1: Hille–Yosida

$A : \mathcal{D}(A) \to X$ is the infinitesimal generator of a C_0-semigroup $Q(t)$, $t \geq 0$, of contractions if and only if:

(1) A is densely defined and closed, and

(2) $(0, \infty) \subset \varrho(A)$ and $\left\|(A - \lambda \operatorname{id}_X)^{-1}\right\| \leq \frac{1}{\lambda}$ for all $\lambda > 0$.

Remark 10.2.4:

(1) The proof below can be modified to show: if A is the infinitesimal generator of a C_0-semigroup of contractions, then $\{\lambda \in \mathbb{C} \mid \operatorname{Re}\lambda > 0\} \subset \varrho(A)$ and $\left\|(A - \lambda)^{-1}\right\| \leq \frac{1}{\operatorname{Re}\lambda}$ for all $\operatorname{Re}\lambda > 0$.

(2) Condition (1) is satisfied for any generator of a C_0-semigroup by Theorem 10.1.1.

Proof. "⇒": For $\lambda > 0$, $x \in X$ set

$$R(\lambda)x := \int_0^\infty \mathrm{e}^{-\lambda t} Q(t)x \, \mathrm{d}t.$$

- $R(\lambda)$ is well-defined since $t \mapsto Q(t)x$ is continuous and $\|R(\lambda)x\| \leq \int_0^\infty \mathrm{e}^{-\lambda t} \|Q(t)x\| \, \mathrm{d}t \leq \int_0^\infty \mathrm{e}^{-\lambda t} \|x\| \, \mathrm{d}t = \frac{1}{\lambda}\|x\|$.
- $R(\lambda) = (\lambda - A)^{-1}$, since for any $\varepsilon > 0$, $x \in X$:

$$\begin{aligned}
\frac{1}{\varepsilon}\big(Q(\varepsilon) - \mathrm{id}\big)R(\lambda)x &= \frac{1}{\varepsilon}\int_0^\infty \mathrm{e}^{-\lambda t}\big(Q(t+\varepsilon) - Q(t)\big)x \, \mathrm{d}t \\
&= \frac{1}{\varepsilon}\int_\varepsilon^\infty \mathrm{e}^{-\lambda(t-\varepsilon)} Q(t)x \, \mathrm{d}t - \frac{1}{\varepsilon}\int_0^\infty \mathrm{e}^{-\lambda t} Q(t)x \, \mathrm{d}t \\
&= \frac{1}{\varepsilon}\big(\mathrm{e}^{\lambda\varepsilon} - 1\big)\int_0^\infty \mathrm{e}^{-\lambda t} Q(t)x \, \mathrm{d}t - \frac{1}{\varepsilon}\mathrm{e}^{\lambda\varepsilon}\int_0^\varepsilon \mathrm{e}^{-\lambda t} Q(t)x \, \mathrm{d}t.
\end{aligned}$$

Hence $\lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon}\big(Q(\varepsilon) - \mathrm{id}\big)R(\lambda)x = \lambda R(\lambda)x - x$, i.e., $R(\lambda)x \in \mathcal{D}(A)$ and $AR(\lambda)x = \lambda R(\lambda)x - x$, so

$$(\lambda - A)R(\lambda) = \mathrm{id}. \tag{$*$}$$

Moreover, for $x \in \mathcal{D}(A)$: $R(\lambda)Ax = \int_0^\infty \mathrm{e}^{-\lambda t} Q(t)Ax \, \mathrm{d}t = \int_0^\infty \mathrm{e}^{-\lambda t} AQ(t)x \, \mathrm{d}t = AR(\lambda)x$. Thus $R(\lambda)(\lambda - A)x \overset{(*)}{=} x$ for all $x \in \mathcal{D}(A)$.

In summary this shows $\lambda \in \varrho(A)$ and $\left\|(\lambda - A)^{-1}\right\| \leq \frac{1}{\lambda}$.

"⇐": This immediately follows from the next lemma, which deals with the Yosida approximation of $A : \mathcal{D}(A) \to X$ satisfying (1) and (2) in the Hille–Yosida theorem. □

Lemma 10.2.5 (Yosida approximation): Let $A : \mathcal{D}(A) \to X$ be the infinitesimal generator of a C_0-semigroup $Q(t)$, $t \geq 0$. Assume that $(0, \infty) \subset \varrho(A)$ and $\left\|(A - \lambda)^{-1}\right\| \leq \frac{1}{\lambda}$ for all $\lambda > 0$. Then $A_\lambda := \lambda A(\lambda - A)^{-1}$, $\lambda > 0$, is a bounded operator with

(1) $\lim_{\lambda\to\infty} A_\lambda x = Ax$ for all $x \in \mathcal{D}(A)$,

(2) $\lim_{\lambda\to\infty} \exp(tA_\lambda)x = Q(t)x$ for all $x \in X$, $t \geq 0$,

(3) $\|Q(t)\| \leq 1$ for all $t \geq 0$.

Proof. Alternative representation of A_λ:

$$A_\lambda = \lambda(A-\lambda)(\lambda-A)^{-1} + \lambda^2(\lambda-A)^{-1} = -\lambda + \lambda^2(\lambda-A)^{-1} \in \mathcal{L}(X).$$

Claim: $\lim_{\lambda\to\infty} \lambda(\lambda-A)^{-1}x = x$ for all $x \in X$. Indeed, for $x \in \mathcal{D}(A)$:

$$\big\|\lambda(\lambda-A)^{-1}x - x\big\| = \big\|A(\lambda-A)^{-1}x\big\| = \big\|(\lambda-A)^{-1}Ax\big\| \le \frac{1}{\lambda}\,\|Ax\| \xrightarrow{\lambda\to\infty} 0.$$

This extends to $x \in X$, since $\mathcal{D}(A)$ is dense in X and $\big\|\lambda(\lambda-A)^{-1}\big\| \le 1$.

(1) $\lim_{\lambda\to\infty} A_\lambda x = \lim_{\lambda\to\infty} \lambda(\lambda-A)^{-1}Ax = Ax$ by the claim.

(2) Since $A_\lambda \in \mathcal{L}(X)$, it defines a uniformly continuous semigroup via $Q_\lambda(t) = \exp(tA_\lambda)$, which is a contraction:

$$\|Q_\lambda(t)\| = \Big\|\exp\big(\lambda^2(\lambda-A)^{-1}t\big)\mathrm{e}^{-\lambda t}\Big\| \le \mathrm{e}^{-\lambda t}\exp\big(t\lambda^2\,\big\|(\lambda-A)^{-1}\big\|\,\big) \le \mathrm{e}^{-\lambda t}\mathrm{e}^{\lambda t} = 1.$$

For $\mu, \lambda > 0$, $x \in X$ (note that A_λ, A_μ and the semigroups commute):

$$\begin{aligned}
\|Q_\lambda(t)x - Q_\mu(t)x\| &= \Big\| \int_0^1 \frac{\mathrm{d}}{\mathrm{d}s} Q_\lambda(ts)Q_\mu(t(1-s))x\,\mathrm{d}s\Big\| \\
&\le \int_0^1 \big\|Q_\lambda(ts)\,t\,Q_\mu(t(1-s))\big[A_\lambda - A_\mu\big]x\big\|\,\mathrm{d}s \\
&\le t \sup_{s\in[0,1]} \|Q_\lambda(ts)\|\,\|Q_\mu(t(1-s))\|\,\|(A_\lambda - A_\mu)x\| \\
&\le t\,\|(A_\lambda - A)x\| + t\,\|(A_\mu - A)x\| \quad (x \in \mathcal{D}(A)).
\end{aligned}$$

Using (1), this shows that $Q_\lambda(t)x$ is a Cauchy sequence and hence converges as $\lambda \to \infty$ for any $x \in \mathcal{D}(A)$, $t \ge 0$. This convergence is uniform on bounded intervals w.r.t. $t \in [0,\infty)$. Since $\mathcal{D}(A)$ is dense in X and $\|Q_\lambda(t)\| \le 1$, $Q_\lambda(t)x$ converges as $\lambda \to \infty$ for any $x \in X$, uniformly for t in any bounded interval. Set

$$S(t)x := \lim_{\lambda\to\infty} Q_\lambda(t)x, \qquad x \in X.$$

Then:

(i) $S(t)$, $t \ge 0$, is a semigroup, since the $Q_\lambda(t)$ satisfy the semigroup property;

(ii) $S(t)$, $t \ge 0$, is a contraction, since $\|Q_\lambda(t)\| \le 1$;

(iii) $\lim_{t\to 0} \|S(t)x - x\| = 0$ for any $x \in X$, since $Q_\lambda(t)$ is strongly continuous and the convergence $\lambda \to \infty$ is uniform w.r.t. t on a bounded interval;

(iv) A is the infinitesimal generator of $S(t)$, $t \ge 0$, see below.

The uniqueness of the semigroup (corollary above) then implies $S(t) = Q(t)$ and hence also (3).

Proof of (iv): For $x \in \mathcal{D}(A)$, using $\frac{\mathrm{d}}{\mathrm{d}s}\mathrm{e}^{sA_\lambda}x = \mathrm{e}^{sA_\lambda}A_\lambda x$, (1), the definition of S and uniformity of the limit w.r.t. λ:

$$S(t)x - x = \lim_{\lambda\to\infty} \exp(tA_\lambda)x - x = \lim_{\lambda\to\infty} \int_0^t \mathrm{e}^{sA_\lambda}A_\lambda x\,\mathrm{d}s = \int_0^t S(s)Ax\,\mathrm{d}s.$$

Hence $\lim_{t\to 0} \frac{1}{t}\big(S(t)x - x\big) = \lim_{t\to 0} \frac{1}{t}\int_0^t S(s)Ax\,\mathrm{d}s = S(0)Ax = Ax$. If $B : \mathcal{D}(B) \to X$, $\mathcal{D}(B) \subset X$, denotes the generator of S, then we have shown that $A \subset B$. By assumption $1 \in \varrho(A)$ and hence

$$(\mathrm{id} - B)\mathcal{D}(A) = (\mathrm{id} - A)\mathcal{D}(A) = X. \qquad (*)$$

Since also $1 \in \varrho(B)$ by the "$\Rightarrow$"-part of the Hille–Yosida theorem, we have $\mathcal{D}(B) = (\mathrm{id}-B)^{-1}X \overset{(*)}{=} \mathcal{D}(A)$, i.e., $A = B$. □

Example 10.2.6 (Heat flow in $L^1(\mathbb{R}^d)$): The kernel $k_t : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$,

$$k_t(x,y) = \frac{1}{(4\pi t)^{d/2}} \mathrm{e}^{-\frac{|x-y|^2}{4t}}, \qquad t > 0,$$

defines via $K_t f := \int k_t(\cdot, y) f(y)\,\mathrm{d}y$, $K_0 f := f$, a C_0-semigroup of contractions on $L^1(\mathbb{R}^d)$. Its infinitesimal generator is the Laplacian, and one has

$$\partial_t K_t f = \Delta K_t f, \qquad t > 0.$$

10.3 Dissipative semigroups and the Lumer–Phillips theorem

Definition 10.3.1: A densely defined operator $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$, is called *dissipative* if for all $\lambda > 0$, $x \in \mathcal{D}(A)$:

$$\|(\lambda - A)x\| \geq \lambda \|x\|.$$

Lemma 10.3.2: Let $A : \mathcal{D}(A) \to X$ be dissipative.

(1) For any $\lambda > 0$, $(\lambda - A) : \mathcal{D}(A) \to X$ is injective and for any $y \in (\lambda - A)\mathcal{D}(A)$: $\left\|(\lambda - A)^{-1} y\right\| \leq \frac{1}{\lambda} \|y\|$.

(2) $\lambda - A : \mathcal{D}(A) \to X$ is surjective for all $\lambda > 0$ if and only if it is surjective for some $\lambda > 0$. In this case, A is closed and $(0, \infty) \subset \varrho(A)$.

Proof. (1) Immediate from the definition.

(2) Let $\lambda_0 > 0$ be such that $(\lambda_0 - A)\mathcal{D}(A) = X$. Then $\lambda_0 - A : \mathcal{D}(A) \to X$ is bijective by (1) and for all $y \in X$: $\left\|(\lambda_0 - A)^{-1} y\right\| \leq \frac{1}{\lambda_0} \|y\|$. If $(x_n) \subset \mathcal{D}(A)$ with $x_n \to x$ and $Ax_n \to y$, then

$$x_n = (\lambda_0 - A)^{-1}(\lambda_0 - A)x_n \to (\lambda_0 - A)^{-1}(\lambda_0 x - y) \in \mathcal{D}(A)$$

and $(\lambda_0 - A)x = \lambda_0 x - y$, i.e., $Ax = y$. Thus A is closed by Theorem 8.1.1.

Consider $\Lambda := \{\lambda \in (0,\infty) \mid (\lambda - A)\mathcal{D}(A) = X\}$. By assumption $\lambda_0 \in \Lambda$ and by the above $\Lambda \subset \varrho(A)$. We claim $\Lambda = (0,\infty)$:

(i) $\Lambda = \varrho(A) \cap (0,\infty)$ is an open subset of $(0,\infty)$, since $\varrho(A)$ is open.

(ii) Λ is also a closed subset of $(0,\infty)$, since for any $(\lambda_n) \subset \Lambda$ with $\lambda_n \to \lambda > 0$ we conclude that for all $y \in X$ there is $(x_n) \subset \mathcal{D}(A)$ with $(\lambda_n - A)x_n = y$, i.e., $x_n = (\lambda_n - A)^{-1} y$, and

- $\|x_n\| = \left\|(\lambda_n - A)^{-1} y\right\| \leq \frac{1}{\lambda_n} \|y\| \leq C \|y\|$ for some $C \in (0,\infty)$;
- $\lambda_n \|x_n - x_m\| \leq \|(\lambda_n - A)(x_n - x_m)\| = \|y - (\lambda_n - A)x_m\| = |\lambda_n - \lambda_m| \, \|x_m\| \leq |\lambda_n - \lambda_m| \, C \|y\|$, using $(\lambda_n - A)x_m = (\lambda_m - A)x_m + (\lambda_n - \lambda_m)x_m$.

Thus $(x_n) \subset \mathcal{D}(A)$ is a Cauchy sequence and hence there is $x \in X$ with $x_n \to x$. Also $Ax_n = -y + \lambda_n x_n \to -y + \lambda x$. Since A is closed, by Theorem 8.1.1: $x \in \mathcal{D}(A)$ and $Ax = -y + \lambda x$. Thus $(\lambda - A)\mathcal{D}(A) = X$.

Since $(0,\infty)$ is connected, $\Lambda = (0,\infty)$. □

Theorem 10.3.1: Lumer–Phillips

A densely defined $A : \mathcal{D}(A) \to X$, $\mathcal{D}(A) \subset X$, is the generator of a C_0-semigroup of contractions if and only if

(1) A is dissipative and

(2) $(\lambda - A)\mathcal{D}(A) = X$ for some $\lambda > 0$.

Proof. "⇐": We aim to apply the Hille–Yosida theorem. By Lemma 10.3.2: A is closed, $(0,\infty) \subset \varrho(A)$ and $\left\|(\lambda - A)^{-1} y\right\| \leq \frac{1}{\lambda} \|y\|$ for all $y \in X$, $\lambda > 0$.

"⇒": By Hille–Yosida: $(0,\infty) \subset \varrho(A)$, $\left\|(\lambda - A)^{-1}\right\| \leq \frac{1}{\lambda}$. In particular $(\lambda - A)\mathcal{D}(A) = X$ for all $\lambda > 0$ and for all $x \in \mathcal{D}(A)$:

$$\|x\| = \left\|(\lambda - A)^{-1}(\lambda - A)x\right\| \leq \left\|(\lambda - A)^{-1}\right\| \|(\lambda - A)x\| \leq \frac{1}{\lambda} \|(\lambda - A)x\|,$$

i.e., A is dissipative. □

Remark 10.3.3: Operators satisfying (1) and (2) are called *maximal dissipative*.

Corollary 10.3.4: Let $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, be densely defined and closed on some Hilbert space $\mathcal{H}$. If A and A^* are dissipative, then A is the generator of a C_0-semigroup of contractions.

Proof. It suffices to show $(\not< - A)\mathcal{D}(A) = \mathcal{H}$.

(i) $(\not< - A)\mathcal{D}(A)$ is closed, since for any $(x_n) \subset \mathcal{D}(A)$ with $(\not< - A)x_n \to y$ we have, as A is dissipative, $\|x_n - x_m\| \leq \|(\not< - A)(x_n - x_m)\|$. Thus (x_n) is Cauchy and $x_n \to x$ for some $x \in \mathcal{H}$. Since A is closed, we have $(\not< - A)x = y$.

(ii) Suppose $(\not< - A)\mathcal{D}(A) \neq \mathcal{H}$, i.e., there is $\|y\| = 1$ with $y \in \big((\not< - A)\mathcal{D}(A)\big)^{\perp}$. By Lemma 8.4.7: $y \in \ker(\not< - A^*)$, i.e., $y = A^* y$. This contradicts $\|(\not< - A^*)y\| \geq \|y\| = 1$, as A^* is dissipative. □

Lemma 10.3.5: $A : \mathcal{D}(A) \to \mathcal{H}$ densely defined on some Hilbert space $\mathcal{H}$ is dissipative if and only if for all $x \in \mathcal{D}(A)$:

$$\operatorname{Re} \langle Ax, x \rangle \leq 0.$$

Proof. Let $x \in \mathcal{D}(A)$, $\lambda > 0$.

"⇐": $\|(\lambda - A)x\| \, \|x\| \geq |\langle (\lambda - A)x, x \rangle| \geq \operatorname{Re} \langle (\lambda - A)x, x \rangle \geq \lambda \|x\|^2$.

"⇒": $\lambda^2 \|x\|^2 \leq \|(\lambda - A)x\|^2 = \langle (\lambda - A)x, (\lambda - A)x \rangle = \lambda^2 \|x\|^2 + \|Ax\|^2 - 2\lambda \operatorname{Re} \langle Ax, x \rangle$, hence $2\lambda \operatorname{Re} \langle Ax, x \rangle \leq \|Ax\|^2$. Since this holds for all $\lambda > 0$, we conclude $\operatorname{Re} \langle Ax, x \rangle \leq 0$. □

Example 10.3.6: $\mathcal{H} = L^2(\Lambda)$, $\Lambda = (a_1, b_1) \times \cdots \times (a_d, b_d) \subset \mathbb{R}^d$. The Dirichlet Laplacian $\Delta_D : \mathcal{D}(\Delta_D) \to L^2(\Lambda)$ from Section 9.3 is self-adjoint and dissipative, i.e., for $\psi \in \mathcal{D}(\Delta_D)$: $\langle \Delta_D \psi, \psi \rangle = -q(\psi, \psi) \leq 0$. It hence generates a C_0-semigroup of contractions Q. Solutions to the heat equation with Dirichlet boundary conditions,

$$\partial_t \psi_t = \Delta_D \psi_t \quad \text{with initial condition } \psi_0 \in L^2(\Lambda),$$

are given by $\psi_t = Q(t)\psi_0$.

11 A summary of spectral theory for unbounded self-adjoint operators (without proofs)

11.1 Integration with respect to a PV measure $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$

Let $\Sigma_{\mathbb{R}}$ be the Borel σ-algebra on $\mathbb{R}$. Any PV measure $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$ with underlying Hilbert space $\mathcal{H}$ defines for $x \in \mathcal{H}$ a Borel measure

$$m_x(I) := \langle E(I)x, x \rangle, \qquad I \in \Sigma_{\mathbb{R}},$$

which is bounded, $m_x(\mathbb{R}) \leq \|x\|^2$. For elementary functions $f : \mathbb{R} \to \mathbb{C}$ we already defined $\int f(t)\,\mathrm{d}E(t)x$, $x \in \mathcal{H}$ (cf. Section 4).

The goal is to extend this definition to all measurable $f : \mathbb{R} \to \mathbb{C}$ – however not in the sense of bounded operators on all $x \in \mathcal{H}$, but rather on a dense domain:

(1) For any $x \in \mathcal{H}$ and any $f \in L^2(\mathbb{R}, m_x)$ there is a sequence (f_n) of elementary functions such that
$$\lim_{n\to\infty} \int |f - f_n|^2 \,\mathrm{d}m_x = 0, \qquad \text{i.e., } f_n \to f \text{ in } L^2(\mathbb{R}, m_x).$$

(2) For any Cauchy sequence (f_n) of elementary functions in $L^2(\mathbb{R}, m_x)$ (cf. the proof of Lemma 4.2.1):
$$\Big\| \int f_n \,\mathrm{d}Ex - \int f_m \,\mathrm{d}Ex \Big\|^2 = \Big\| \int (f_n - f_m)\,\mathrm{d}Ex \Big\|^2 = \int |f_n - f_m|^2 \,\mathrm{d}m_x. \qquad (*)$$
Thus $\big(\int f_n \,\mathrm{d}Ex\big)$ is a Cauchy sequence in $\mathcal{H}$.

Definition 11.1.1: For $f : \mathbb{R} \to \mathbb{C}$ measurable and $x \in \mathcal{H}$ with $f \in L^2(\mathbb{R}, m_x)$ we define
$$f_E x := \int f \,\mathrm{d}Ex := \lim_{n\to\infty} \int f_n \,\mathrm{d}Ex,$$
where (f_n) is any sequence of elementary functions with $f_n \to f$ in $L^2(\mathbb{R}, m_x)$.

Remark 11.1.2 (Elementary properties of the integral, $x \in \mathcal{H}$):

(1) Independence of the approximating sequence (f_n).

(2) L^2-property: for any $f \in L^2(\mathbb{R}, m_x)$: $\big\|\int f \,\mathrm{d}Ex\big\|^2 = \int |f|^2 \,\mathrm{d}m_x$.

(3) Linearity: for any $f, g \in L^2(\mathbb{R}, m_x)$, $\lambda \in \mathbb{C}$: $\int (f + \lambda g)\,\mathrm{d}Ex = \int f \,\mathrm{d}Ex + \lambda \int g \,\mathrm{d}Ex$.

The next theorem in particular asserts that the integral is also linear in x and is a densely defined linear operator on $\mathcal{H}$.

Theorem 11.1.1

For any PV measure $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$ and any measurable $f : \mathbb{R} \to \mathbb{C}$,

$$\mathcal{D}(f_E) := \{x \in \mathcal{H} \mid f \in L^2(\mathbb{R}, m_x)\}$$

is a dense subspace and $f_E x := \int f \, \mathrm{d}Ex$ defines a normal linear operator with the following properties for $f, g : \mathbb{R} \to \mathbb{C}$ measurable:

(1) For $x \in \mathcal{D}(f_E)$, $y \in \mathcal{D}(g_E)$:

$$\langle f_E x, g_E y\rangle = \lim_{n\to\infty} \int \varphi_n(f(t))\overline{\varphi_n(g(t))}\,\mathrm{d}m_{x,y}(t) =: \int f(t)\overline{g(t)}\,\mathrm{d}m_{x,y}(t)$$

with $\varphi_n : \mathbb{C} \to \mathbb{C}$, $\varphi_n(z) = z$ if $|z| \le n$ and 0 else, and $m_{x,y}(I) = \langle E(I)x, y\rangle$, cf. Section 4.

(2) If $f \in \mathcal{B}_\infty(\mathbb{R})$, then $f_E \in \mathcal{L}(\mathcal{H})$ with $\|f_E\| \le \|f\|_\infty$. If $f = 1$, then $f_E = \mathrm{id}_{\mathcal{H}}$.

(3) For $x \in \mathcal{D}(f_E)$, $y \in \mathcal{H}$: $\langle f_E x, y\rangle = \int f(t)\,\mathrm{d}m_{x,y}(t)$.

(4) (i) $f_E + g_E \subset (f+g)_E$ and $\mathcal{D}(f_E + g_E) = \mathcal{D}\big((|f| + |g|)_E\big)$; (ii) $f_E g_E \subset (fg)_E$ and $\mathcal{D}(f_E g_E) = \mathcal{D}\big((fg)_E\big) \cap \mathcal{D}(g_E)$.

(5) $\mathcal{D}(f_E) = \mathcal{D}(\overline{f}_E)$ and $f_E^* = \overline{f}_E$. In particular, if f is real, then f_E is self-adjoint.

(6) If $I \in \Sigma_{\mathbb{R}}$, then $(1_I)_E = E(I)$.

11.2 Von Neumann's spectral theorem

Theorem 11.2.1: von Neumann

For every self-adjoint operator $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, on a Hilbert space $\mathcal{H}$, there is exactly one PV measure $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$ such that

$$A = \int_{\sigma(A)} t \,\mathrm{d}E(t).$$

For any $x, y \in \mathcal{H}$, $t \in \mathbb{R}$ one has

$$\langle E((-\infty, t])x, y\rangle = \lim_{\delta\downarrow 0} \lim_{\varepsilon\downarrow 0} \frac{1}{2\pi\mathrm{i}} \int_{-\infty}^{t+\delta} \big\langle \big[(A - s - \mathrm{i}\varepsilon)^{-1} - (A - s + \mathrm{i}\varepsilon)^{-1}\big]x, y\big\rangle \,\mathrm{d}s. \qquad (*)$$

Remark 11.2.1:

(1) One-to-one correspondence between self-adjoint operators and PV measures $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$.

(2) Equivalent to $(*)$ (*Stone's formula*): for $-\infty < a < b < \infty$,

$$\frac{1}{2}\big\langle \big(E((a,b)) + E([a,b])\big)x, y\big\rangle = \lim_{\varepsilon\downarrow 0} \frac{1}{2\pi\mathrm{i}} \int_a^b \big\langle \big[(A - s - \mathrm{i}\varepsilon)^{-1} - (A - s + \mathrm{i}\varepsilon)^{-1}\big]x, y\big\rangle \,\mathrm{d}s.$$

Corollary 11.2.2: In the situation of the theorem, let $m_x(\cdot) := \langle E(\cdot)x, x\rangle$. Then $\operatorname{supp} m_x \subset \sigma(A)$ for all $x \in \mathcal{H}$.

The von Neumann theorem allows us to define a functional calculus.

Definition 11.2.3: Let $E : \Sigma_{\mathbb{R}} \to \mathcal{L}(\mathcal{H})$ be the PV measure of a self-adjoint operator $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$. For any measurable $f : \mathbb{R} \to \mathbb{C}$ we set $f(A) := f_E$.

Example 11.2.4:

(1) *Maximal multiplication operator* associated with a Borel measurable $f : \mathbb{R}^d \to \mathbb{R}$ on $L^2(\mathbb{R}^d, \mu)$ with a regular Borel measure μ: $\mathcal{D}(M_f) := \{\psi \in L^2(\mathbb{R}^d, \mu) \mid f\psi \in L^2(\mathbb{R}^d, \mu)\}$, $M_f\psi := f\psi$. Recall from Section 8 that M_f is self-adjoint. For all $a < b$, $\varphi, \psi \in L^2(\mathbb{R}^d, \mu)$, by Fubini and dominated convergence:

$$\begin{aligned}
&\tfrac{1}{2}\big\langle \big(E([a,b]) + E((a,b))\big)\varphi, \psi\big\rangle \\
&= \lim_{\varepsilon\downarrow 0} \tfrac{1}{2\pi \mathrm{i}} \int_a^b \big\langle \big[(M_f - s - \mathrm{i}\varepsilon)^{-1} - (M_f - s + \mathrm{i}\varepsilon)^{-1}\big]\varphi, \psi\big\rangle \, \mathrm{d}s \\
&= \lim_{\varepsilon\downarrow 0} \tfrac{1}{\pi} \int_a^b \int \tfrac{\varepsilon}{(f(x)-s)^2+\varepsilon^2} \varphi(x)\overline{\psi(x)} \, \mu(\mathrm{d}x) \, \mathrm{d}s \\
&= \lim_{\varepsilon\downarrow 0} \int \varphi(x)\overline{\psi(x)} \, \tfrac{1}{\pi}\Big[\arctan\Big(\tfrac{s-f(x)}{\varepsilon}\Big)\Big]_a^b \mu(\mathrm{d}x) \\
&= \int \varphi(x)\overline{\psi(x)} \, h(x) \, \mu(\mathrm{d}x),
\end{aligned}$$

where

$$h(x) := \lim_{\varepsilon\downarrow 0} \frac{1}{\pi}\Big[\arctan\Big(\frac{b-f(x)}{\varepsilon}\Big) - \arctan\Big(\frac{a-f(x)}{\varepsilon}\Big)\Big] = \begin{cases} 1, & a < f(x) < b, \\ 1/2, & f(x) \in \{a, b\}, \\ 0, & f(x) < a \text{ or } f(x) > b. \end{cases}$$

Hence $E((-\infty, t]) = M_{1_{\{f \le t\}}}$ for all $t \in \mathbb{R}$.

(2) Let $A : \mathcal{D}(A) \to \mathcal{H}$ be self-adjoint on some Hilbert space $\mathcal{H}$ and suppose $A \ge 0$. Then $-A$ is maximally dissipative and hence $Q(t) = g_t(A)$ with $g_t : [0, \infty) \to \mathbb{R}$, $g_t(x) = \mathrm{e}^{-tx}$, $t \ge 0$, is a C_0-semigroup of contractions.

(3) Let $t \in \mathbb{R}$ and $f_t : \mathbb{R} \to \mathbb{C}$, $f_t(s) := \mathrm{e}^{-\mathrm{i}ts}$. For any self-adjoint $A : \mathcal{D}(A) \to \mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, the operators $U_t := f_t(A) \in \mathcal{L}(\mathcal{H})$ satisfy:

   (i) group property: $U_{t_1}U_{t_2} = U_{t_1+t_2}$ for all $t_1, t_2 \in \mathbb{R}$, $U_0 = \mathbb{1}$. *Proof:* functional calculus and $f_{t_1}f_{t_2} = f_{t_1+t_2}$, $f_0 = 1$.

   (ii) isometry: $\|U_t x\| = \|x\|$ for all $t \in \mathbb{R}$, $x \in \mathcal{H}$. *Proof:* $\|U_t x\|^2 = \int \big|\mathrm{e}^{-\mathrm{i}ts}\big|^2 m_x(\mathrm{d}s) = m_x(\mathbb{R}) = \|x\|^2$.

   Since $\mathcal{H} = \operatorname{ran}\mathbb{1} = \operatorname{ran} U_t U_{-t} \subset \operatorname{ran} U_t$, this implies that the U_t form a group of unitary operators.

12 Unitary groups on Hilbert space

12.1 Strongly continuous unitary groups

Motivation: requirements on quantum dynamics $U_t : \mathcal{H} \to \mathcal{H}$ on some Hilbert space $\mathcal{H}$:

(a) Linearity: for all times $t \in \mathbb{R}$: $U_t \in \mathcal{L}(\mathcal{H})$.

(b) Isometry: for all $t \in \mathbb{R}$, $x \in \mathcal{H}$: $\|U_t x\| = \|x\|$.

(c) Group property: $U_0 = \mathbb{1}$ and for all $s, t \in \mathbb{R}$: $U_{t+s} = U_t U_s$.

Remark 12.1.1: (c) implies $\operatorname{ran} U_t \supset \operatorname{ran} U_t U_{-t} \overset{\text{(c)}}{=} \mathcal{H}$ and hence with (b): U_t is unitary.

Definition 12.1.2: One calls $(U_t)_{t\in\mathbb{R}}$ with properties (a)–(c) a (one-parameter) *group of unitary operators* on $\mathcal{H}$. It is *strongly continuous* iff for all $t \in \mathbb{R}$, $x \in \mathcal{H}$:

$$\lim_{s\to t} U_s x = U_t x.$$

Its *infinitesimal generator* $A : \mathcal{D}(A) \to \mathcal{H}$ is defined on $\mathcal{D}(A) := \{x \in \mathcal{H} \mid \lim_{t\to 0} \frac{1}{t}(U_t - \not{1})x \text{ exists}\}$ by $Ax := \lim_{t\to 0} \frac{1}{t}(U_t - \not{1})x$.

Clearly, any strongly continuous group of unitaries U_t on $\mathcal{H}$ defines two C_0-semigroups with generators $\pm A : \mathcal{D}(A) \to \mathcal{H}$. Consequently, A is densely defined and closed. Since $\|U_t\| \le 1$, the semigroups are contractions and the generators $A, -A$ are maximally dissipative.

Conversely, if $S : \mathcal{D}(S) \to \mathcal{H}$ is self-adjoint, then $A = \pm \mathrm{i}S : \mathcal{D}(S) \to \mathcal{H}$ is maximally dissipative since $\sigma(A) = \pm\mathrm{i}\sigma(S) \subset \pm\mathrm{i}\mathbb{R} \not\ni \lambda$ for $\lambda \in (0,\infty)$, and for all $\lambda > 0$, by Lemma 8.4.10,

$$\left\|(\lambda - A)^{-1}\right\| = \left\|(\lambda \pm \mathrm{i}S)^{-1}\right\| = \left\|(S \mp \mathrm{i}\lambda)^{-1}\right\| \le \frac{1}{\lambda}.$$

Thus A generates a C_0-semigroup of contractions.

Theorem 12.1.1

Let $S : \mathcal{D}(S) \to \mathcal{H}$, $\mathcal{D}(S) \subset \mathcal{H}$, be self-adjoint on some Hilbert space. Then the operators $U_t = \mathrm{e}^{-\mathrm{i}tS}$, $t \in \mathbb{R}$, form a strongly continuous group of unitary operators with generator $A = -\mathrm{i}S$.

Proof. By Example 3 in Section 11 and the above, it remains to establish the form of the generator. For any $x \in \mathcal{D}(S)$, by von Neumann's spectral theorem, $S = \int \lambda\, E(\mathrm{d}\lambda)$, $m_x(\cdot) := \langle E(\cdot)x, x\rangle$, and

$$\left\|\Big(\frac{1}{t}(U_t - \not{1}) + \mathrm{i}S\Big)x\right\|^2 = \int \left|\frac{1}{t}\big(\mathrm{e}^{-\mathrm{i}t\lambda} - 1\big) + \mathrm{i}\lambda\right|^2 m_x(\mathrm{d}\lambda) =: (*).$$

Note $\left|\frac{1}{t}(\mathrm{e}^{-\mathrm{i}t\lambda} - 1) + \mathrm{i}\lambda\right| \le 2\,|\lambda|$ if $|t| \le 1$. Hence by dominated convergence $(*) \xrightarrow{t\to 0} 0$. Thus $-\mathrm{i}S \subset A$ with A the infinitesimal generator of U_t. Since $\mathrm{i}A$ is self-adjoint by Stone's theorem below, we also conclude $\mathrm{i}A = (\mathrm{i}A)^* \subset S^* = S$. □

Theorem 12.1.2: Stone

Let U_t, $t \in \mathbb{R}$, be a strongly continuous group of unitary operators on some Hilbert space $\mathcal{H}$ and $A =: -\mathrm{i}S$ its infinitesimal generator. Then:

(1) $\mathcal{D}(A) = \mathcal{D}(S)$ is dense in $\mathcal{H}$,

(2) S is self-adjoint.

Proof. By the remark above, it remains to prove that S is self-adjoint. Since $\pm\mathrm{i}S$ is maximally dissipative, we also have $(\not{1} \pm \mathrm{i}S)\mathcal{D}(S) = \mathcal{H}$, i.e., $(S \mp \mathrm{i})\mathcal{D}(S) = \mathcal{H}$. By Theorem 9.1.1 it remains to prove that S is symmetric, i.e., $S \subset S^*$. Let $x, y \in \mathcal{D}(S) = \mathcal{D}(A)$:

$$\langle Sx, y\rangle = \langle \mathrm{i}Ax, y\rangle = \lim_{t\to 0} \frac{\mathrm{i}}{t} \langle (U_t - \not{1})x, y\rangle = \lim_{t\to 0} \frac{\mathrm{i}}{t} \Big\langle x, (\underbrace{U_t^*}_{=U_{-t}} - \not{1})y \Big\rangle = \lim_{s\to 0} \frac{-\mathrm{i}}{s} \langle x, (U_s - \not{1})y\rangle$$

$$= \lim_{s\to 0} \big\langle x, \tfrac{\mathrm{i}}{s}(U_s - \not{1})y\big\rangle = \langle x, \mathrm{i}Ay\rangle = \langle x, Sy\rangle\,.$$

□

Corollary 12.1.3: Let U_t, $t \in \mathbb{R}$, be a strongly continuous group of unitary operators on some Hilbert space $\mathcal{H}$ and $A =: -\mathrm{i}S$ its infinitesimal generator. Then for any $x \in \mathcal{D}(S)$:

(1) $x_t := U_t x \in \mathcal{D}(S)$ for all $t \in \mathbb{R}$;

(2) x_t is the unique solution of the initial-value problem $\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}x_t = Sx_t$, $x_0 = x$.

Proof. (1) Using $U_{t+s} = U_s U_t$ and $x \in \mathcal{D}(A)$: $\lim_{s\to 0} \frac{1}{s}(U_s - \mathbb{1})x_t = \lim_{s\to 0} \frac{1}{s}(U_{t+s} - U_t)x = \lim_{s\to 0} U_t \frac{1}{s}(U_s - \mathbb{1})x = U_t A x$, i.e., $x_t \in \mathcal{D}(A)$.

(2) Evidently $x_0 = x$, and $\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}x_t = \mathrm{i}\lim_{s\to 0} \frac{1}{s}(x_{t+s} - x_t) = \mathrm{i}\lim_{s\to 0} \frac{1}{s}(U_s - \mathbb{1})x_t \overset{(1)}{=} \mathrm{i}Ax_t = Sx_t$. The uniqueness of the solution follows by linearity: let x_t, y_t be solutions; then $z_t := y_t - x_t$ is a solution of $\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}z_t = Sz_t$ with $z_0 = 0$. For any $t \in \mathbb{R}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\|z_t\|^2 = \left\langle \tfrac{\mathrm{d}}{\mathrm{d}t}z_t, z_t\right\rangle + \left\langle z_t, \tfrac{\mathrm{d}}{\mathrm{d}t}z_t\right\rangle = \langle -\mathrm{i}Sz_t, z_t\rangle + \langle z_t, -\mathrm{i}Sz_t\rangle = -\mathrm{i}\langle Sz_t, z_t\rangle + \mathrm{i}\underbrace{\langle z_t, Sz_t\rangle}_{=\langle Sz_t, z_t\rangle} = 0.$$

Since $\|z_0\|^2 = 0$, we have $z_t = 0$ for all $t \in \mathbb{R}$. □

Example 12.1.4 (Translation group on $L^2(\mathbb{R})$): $(T_a\psi)(x) := \psi(x-a)$, $a \in \mathbb{R}$, $T_a \in \mathcal{L}(L^2(\mathbb{R}))$. Evidently, T_a is a group of unitary operators. Strong continuity follows using the Fourier transform $\mathcal{F}$:

$$(\mathcal{F}T_a\psi)(k) = \int \mathrm{e}^{-\mathrm{i}kx}\psi(x-a)\frac{\mathrm{d}x}{\sqrt{2\pi}} = \mathrm{e}^{-\mathrm{i}ka}(\mathcal{F}\psi)(k)$$

$$\Rightarrow \quad \|T_a\psi - \psi\|^2 \overset{\text{Plancherel}}{=} \int \underbrace{\left|\mathrm{e}^{-\mathrm{i}ka} - 1\right|^2}_{\to 0 \ (a\to 0),\ \le 4} |(\mathcal{F}\psi)(k)|^2 \ \mathrm{d}k.$$

By dominated convergence: $\lim_{a\to 0}\|T_a\psi - \psi\| = 0$. The infinitesimal generator $A = -\mathrm{i}S$ is given by

$$\mathcal{D}(S) = W^{1,2}(\mathbb{R}), \qquad S\psi = -\mathrm{i}\psi'.$$

Proof. For all $\psi \in W^{1,2}(\mathbb{R})$: $\mathcal{F}\psi, \mathcal{F}\psi' \in L^2(\mathbb{R})$ and $(\mathcal{F}\psi')(k) = \mathrm{i}k(\mathcal{F}\psi)(k)$. By Plancherel,

$$\left\|\left[\frac{1}{a}(T_a - \mathbb{1}) + \mathrm{i}S\right]\psi\right\|^2 = \int \underbrace{\left|\frac{1}{a}\left(\mathrm{e}^{-\mathrm{i}ka} - 1\right) + \mathrm{i}k\right|^2}_{\to 0 \ (a\to 0),\ \le 4|k|^2 \text{ for } |a|\le 1} |(\mathcal{F}\psi)(k)|^2 \ \mathrm{d}k.$$

By the dominated convergence theorem we conclude $\psi \in \mathcal{D}(A)$ and $S \subset \mathrm{i}A$. By self-adjointness, this implies $S = \mathrm{i}A$. □

Remark 12.1.5: This generalizes to $L^2(\mathbb{R}^d)$.

12.2 Dynamical systems through Koopmanism

Let $(M, \mathcal{A}, \mu)$ be some finite measure space, which serves as the phase space of some group $T_t : M \to M$, $t \in \mathbb{R}$, of measure preserving transformations, i.e., for all $s, t \in \mathbb{R}$: $T_t T_s = T_{t+s}$ and $T_0 = \mathbb{1}$, and for all $S \in \mathcal{A}$: $\mu(T_t^{-1}(S)) = \mu(S)$.

Definition 12.2.1 (Koopman operator):

$$U_t : L^2(M,\mu) \to L^2(M,\mu), \qquad (U_t f)(x) := f(T_t x).$$

Lemma 12.2.2: U_t, $t \in \mathbb{R}$, form a group of unitaries.

Proof. The group property is immediate from that of T_t. Isometry: for $f \in L^2(M,\mu)$, $t \in \mathbb{R}$, since T_t is measure preserving,

$$\|U_t f\|^2 = \int_M |f(T_t x)|^2 \,\mu(\mathrm{d}x) = \int_M |f(x)|^2 \,\mu(\mathrm{d}x) = \|f\|^2 . \qquad \square$$

In this situation,

$$F := \bigcap_{t\in\mathbb{R}} \ker(U_t - \mathbb{1})$$

is the closed subspace of $\mathcal{H}$ of invariant vectors: $f \in F \iff U_t f = f$ for all $t \in \mathbb{R}$.

Theorem 12.2.1: von Neumann's mean ergodic theorem

Let U_t, $t \in \mathbb{R}$, be a strongly continuous group of unitary operators on some Hilbert space $\mathcal{H}$. Then for all $x \in \mathcal{H}$:

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T U_t x \,\mathrm{d}t = Px,$$

where P is the orthogonal projection onto the closed subspace $F = \bigcap_{t\in\mathbb{R}} \ker(U_t - \mathbb{1})$.

Remark 12.2.3: Similarly, for any unitary U and $x \in \mathcal{H}$: $\lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^N U^k x = Px$ with P the orthogonal projection onto $\ker(U - \mathbb{1})$.

Proof. Let $r \in \mathbb{R}$ and set $F_r := \ker(U_r - \mathbb{1})$. Then, by Lemma 8.4.7,

$$\operatorname{ran}(U_r - \mathbb{1})^\perp = \ker(U_r^* - \mathbb{1}) = \ker(U_{-r} - \mathbb{1}) = \ker\big(U_{-r}(\mathbb{1} - U_r)\big) = \ker(\mathbb{1} - U_r) = F_r,$$

$$\Rightarrow \quad \mathcal{H} = F_r \oplus \overline{\operatorname{ran}(U_r - \mathbb{1})}.$$

Let $M(T)x := \frac{1}{T}\int_0^T U_t x \,\mathrm{d}t$, $T \in \mathbb{R}$, $x \in \mathcal{H}$.

Step 1: $\lim_{T\to\infty} M(T)x = 0$ *for all* $x \in \overline{\operatorname{ran}(U_r - \mathbb{1})}$. Let $\varepsilon > 0$ and pick $x_\varepsilon \in \operatorname{ran}(U_r - \mathbb{1})$ such that $\|x - x_\varepsilon\| < \varepsilon$. Then $\|M(T)x - M(T)x_\varepsilon\| \le \underbrace{\|M(T)\|}_{\le 1} \|x - x_\varepsilon\| \le \varepsilon$. Let $x_\varepsilon = (U_r - \mathbb{1})y$, $y \in \mathcal{H}$. Then for $T > r$

$$M(T)x_\varepsilon = M(T)(U_r - \mathbb{1})y = \frac{1}{T}\int_0^T U_{t+r} y \,\mathrm{d}t - \frac{1}{T}\int_0^T U_t y \,\mathrm{d}t = \frac{1}{T}\int_T^{T+r} U_t y \,\mathrm{d}t - \frac{1}{T}\int_0^r U_t y \,\mathrm{d}t$$

$\Rightarrow \|M(T)x_\varepsilon\| \le \frac{2r}{T}\|y\| \xrightarrow{T\to\infty} 0$.

Step 2: $\lim_{T\to\infty} M(T)x \in F_r$ *exists for all* $x \in F_r$. The function $\mathbb{R} \ni t \mapsto U_t x$ is r-periodic for $x \in F_r$, since $U_{t+r}x = U_t U_r x = U_t x$. Write $T = nr + \tilde{r}$ with some $n \in \mathbb{Z}$ and $|\tilde{r}| < r$. Then

$$M(T)x = \frac{1}{T}\int_0^{nr+\tilde{r}} U_t x \,\mathrm{d}t = \frac{n}{T}\int_0^r U_t x \,\mathrm{d}t + \frac{1}{T}\int_0^{\tilde{r}} U_t x \,\mathrm{d}t,$$

$$\Rightarrow \quad \lim_{T\to\infty} M(T)x = \frac{1}{r}\int_0^r U_t x \,\mathrm{d}t \in F_r.$$

Step 3: Every $x \in \mathcal{H}$ can be uniquely decomposed into $x = x_r + z_r$ with $x_r \in F_r$, $z_r \in \overline{\operatorname{ran}(U_r - \mathbb{1})} = F_r^\perp$. Thus the limit $\lim_{T\to\infty} M(T)x \in F_r$ exists. Since $r \in \mathbb{R}$ was arbitrary and $F = \bigcap_{r\in\mathbb{R}} F_r$, it follows that $\lim_{T\to\infty} M(T)x \in F$. $\square$

Definition 12.2.4: The group of measure preserving transformations T_t, $t \in \mathbb{R}$, on $(M, \mathcal{A}, \mu)$ is *ergodic* if and only if the constant functions $f \in L^2(M,\mu)$ are the only ones satisfying $f = f \circ T_t$ for all $t \in \mathbb{R}$.

Remark 12.2.5: (T_t) ergodic $\iff$ $F = \operatorname{span}\{1\}$.

Corollary 12.2.6: If additionally T_t, $t \in \mathbb{R}$, is ergodic, then for any $f \in L^2(M, \mu)$:

$$\underbrace{\lim_{T\to\infty} \frac{1}{T} \int_0^T U_t f \, \mathrm{d}t}_{\text{time average}} = \underbrace{\frac{1}{\mu(M)} \int_M f \, \mathrm{d}\mu}_{\text{phase-space average}} .$$

Proof. The L^2-normalized constant function is $1(x) = \frac{1}{\sqrt{\mu(M)}}$, $x \in M$. The orthogonal projection onto $\operatorname{span}\{1\} = F$ is $Pf = \frac{1}{\mu(M)} \int_M f \, \mathrm{d}\mu$. Thus the claim follows from von Neumann's mean ergodic theorem. □

Further reading: M. Reed, B. Simon, *Methods of modern mathematical physics*, Vol. 1, Academic Press 1978.